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Artificial Intelligence and Data Science

Auditing the Calibration of Eligibility Scores

Produced by Large Language Models in Patient–Clinical Trial Matching

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Assessing the Effect of RLHF and
Uncovering Mode Collapse in Open-Source Medical LLMs

Author:

Regis LIKASSI

PGE5 AI & Data Science

Academic Supervisor:

Dr. Abdelhamid Mellouk

aivancity, School for Technology, Business & Society

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Abstract

Patient matching to clinical trials is a bottleneck in translational medicine, and recent large language model (LLM) systems such as TrialGPT aim to automate the assessment of patient eligibility against inclusion and exclusion criteria. Beyond raw accuracy, however, their probabilistic reliability has not been systematically audited: a recent scoping review of twenty-four LLM-based matching systems found none that report calibration metrics. Yet an overconfident model that misclassifies with high confidence is clinically worse than an accurate but honestly uncertain one.

We conducted the first systematic calibration audit of open-source LLMs in this domain: twelve configurations (six models $\times$ two prompts) evaluated on the 1,015 patient–criterion pairs of TrialGPT-Criterion-Annotations, using direct log-probability scoring and four pre-registered hypotheses with Bonferroni-corrected permutation tests.

All twelve configurations exceed the clinical threshold of 0.05 (Expected Calibration Error from 0.124 to 0.602), and RLHF significantly degrades calibration on both model families ($p < 0.003$). Five configurations exhibit mode collapse, and both medical LLMs achieve accuracy at or below 0.4% on the 238 discriminative cases. We then show this collapse to be largely a decoding artefact rather than a knowledge loss: the discriminative signal survives in the log-probabilities (area under the curve up to 0.91) but is masked by a response prior. A single scalar, the signal-to-prior ratio, predicts collapse and explains why monotone recalibration cannot repair it while a simple re-thresholding rule restores up to +0.79 macro-F1. The analysis separates recoverable collapse from irrecoverable corruption, and both replicate on the independent TREC Clinical Trials corpus.

Aggregate metrics can thus mask complete failure on clinically decisive cases, and can equally mask a recoverable signal beneath an apparent failure. A responsible evaluation protocol must combine calibration, prediction diversity, stratified analysis by case difficulty, and a diagnostic distinguishing a collapsed decision rule from a corrupted signal.

Keywords: large language models, calibration, clinical trial matching, RLHF, medical AI, mode collapse, expected calibration error, re-thresholding.

Résumé

L'appariement patient–essai clinique est un goulot d'étranglement de la recherche médicale, et les systèmes récents fondés sur les grands modèles de langage (LLM) tels que TrialGPT visent à automatiser l'évaluation de l'éligibilité des patients face aux critères d'inclusion et d'exclusion. Leur fiabilité probabiliste n'a cependant pas fait l'objet d'un audit systématique : une revue récente de vingt-quatre systèmes n'en identifie aucun qui rapporte des métriques de calibration — alors qu'un modèle sur-confiant qui se trompe avec certitude est cliniquement dangereux.

Nous avons mené le premier audit systématique de la calibration des LLM open source dans ce domaine : douze configurations (six modèles × deux prompts) évaluées sur les 1,015 paires patient–critère de TrialGPT-Criterion-Annotations, par scoring direct des log-probabilités et quatre hypothèses pré-enregistrées (permutation, Bonferroni).

Les douze configurations dépassent le seuil clinique de 0,05 (ECE de 0,124 à 0,602), et le RLHF dégrade significativement la calibration sur les deux familles ($p < 0{,}003$). Cinq configurations présentent un effondrement de mode, et les deux LLM médicaux atteignent une exactitude $\leq 0{,}4\%$ sur les 238 cas discriminants. Cet effondrement est cependant largement un artefact de décodage : le signal discriminant subsiste dans les log-probabilités (AUC jusqu'à 0,91) mais est masqué par un a priori de réponse. Un scalaire unique, le ratio signal/a priori, prédit l'effondrement, explique pourquoi une recalibration monotone ne peut le corriger, et justifie un re-seuillage qui restaure jusqu'à +0,79 de macro-F1. L'analyse distingue enfin l'effondrement récupérable de la corruption irrécupérable, distinction qui se réplique sur le corpus indépendant TREC Clinical Trials.

Les métriques agrégées peuvent ainsi masquer un échec complet sur les cas décisifs comme un signal récupérable sous une défaillance apparente. Un protocole d'évaluation responsable doit combiner calibration, diversité prédictive, analyse stratifiée par difficulté, et un diagnostic distinguant une règle de décision effondrée d'un signal corrompu.

Mots-clés : grands modèles de langage, calibration, appariement d'essais cliniques, RLHF, IA médicale, mode collapse, expected calibration error, re-seuillage.

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List of Abbreviations

Acronym	Definition
AUC	Area Under the Curve
BM25	Best Matching 25 (ranking function)
CI	Confidence Interval
DARE	Drop And REscale (model merging)
DPO	Direct Preference Optimization
ECE	Expected Calibration Error
EHR	Electronic Health Record
GPU	Graphics Processing Unit
IC	Intervalle de Confiance (French)
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MC	Mode Collapse
MCC	Matthews Correlation Coefficient
MedCPT	Medical Contrastive Pre-trained Transformer
ML4H	Machine Learning for Health
NA	Not Applicable
nDCG	Normalized Discounted Cumulative Gain
NEI	Not Enough Information
NF4	Normal Float 4-bit (quantization)
NLL	Negative Log-Likelihood
NLP	Natural Language Processing
OMOP	Observational Medical Outcomes Partnership
PPO	Proximal Policy Optimization
PR-AUC	Precision–Recall Area Under the Curve
QA	Question Answering
RCT	Randomized Controlled Trial
RLHF	Reinforcement Learning from Human Feedback
ROC	Receiver Operating Characteristic
SFT	Supervised Fine-Tuning
SLERP	Spherical Linear Interpolation (model merging)
TIES	Trim, Elect Sign & Merge (model merging)
TREC	Text REtrieval Conference
TRIPOD	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

Chapter 1

Introduction

1.1 Macro Context: AI in Healthcare

Artificial intelligence has moved from a peripheral tool to a central component of many medical workflows over the past decade. The transformation has been particularly rapid since 2022, driven by the release of general-purpose large language models capable of processing free-text clinical documentation and by the emergence of medically specialized variants trained on biomedical corpora. Applications now range from automated summarization of patient records to differential diagnosis, from radiology report generation to the identification of adverse drug interactions.

This expansion carries both promise and risk. On the one hand, LLM-based tools address longstanding bottlenecks in clinical practice, where the volume of unstructured text and the complexity of medical reasoning have historically limited the reach of algorithmic support. On the other hand, the confidence that clinicians and patients place in these tools depends critically on the reliability of the underlying probabilistic outputs. A model that is accurate on average but overconfident on the cases where it is wrong is worse than a model that is less accurate but honestly uncertain: the former induces misplaced trust, while the latter enables appropriate deference to human judgment. This asymmetry between accuracy and reliability motivates the present thesis.

1.2 Micro Context: Clinical Trial Matching

Among the many applications of AI in healthcare, patient matching to clinical trials occupies a particularly consequential position. Clinical trials are the mechanism through which new therapies are evaluated and made available to patients, and their success depends on the timely recruitment of suitable participants. Despite the substantial resources invested in recruitment, less than 5% of adult cancer patients enroll in a clinical trial, and 80% of trials fail to meet their recruitment targets [1]. The bottleneck lies partly in the sheer volume of open studies — over 400,000 trials are currently registered on ClinicalTrials.gov — and partly in the complexity of their eligibility criteria, which combine numeric thresholds, medical history requirements, and drug exclusions in dozens of criteria per trial. Manual screening at this scale is labor-intensive and error-prone.

Since 2023, the release of GPT-4 and the proliferation of open-source LLMs have prompted a wave of LLM-based matching systems, of which TrialGPT [1] in *Nature Communications* is the most influential example. These systems report high accuracy on per-criterion eligibility classification and are beginning to be deployed in real recruitment workflows. A scoping review published in 2025 [2] identified twenty-four such systems published in 2024 alone. Yet the same review noted a striking common feature: none of these systems reports calibration metrics. No Expected Calibration Error, no Brier score, no reliability diagram, no calibration-in-the-large. The probabilistic reliability of LLM-based trial matching thus remains empirically undocumented at a moment when the technology is transitioning from research prototype to clinical deployment.

1.3 Problem Statement

The absence of calibration reporting in LLM-based clinical trial matching is not a minor omission. Van Calster et al. [3], in their influential BMC Medicine article, describe calibration as "the Achilles heel of predictive analytics" and argue that calibration is a prerequisite for the clinical deployment of any predictive model. The 2024 update of the TRIPOD guidelines, TRIPOD+AI [4], formalizes this requirement: any AI-based clinical prediction model must report calibration curves, calibration-in-the-large, and calibration slopes. Failing to do so is grounds for methodological rejection by several leading medical journals.

The problem addressed in this thesis is therefore the following: LLM-based patient–trial matching systems are being deployed in clinical settings without any systematic audit of the probabilistic reliability of their outputs. This deployment carries three specific risks. First, overconfident predictions on wrong labels may direct patients toward inappropriate trials, or exclude patients from trials in which they would have been eligible. Second, the presence of systematic biases in evasion labels — when a model reports "not enough information" rather than committing to a decision — may be undetectable through the standard accuracy metrics used to evaluate these systems. Third, the effect of common LLM post-training procedures, in particular reinforcement learning from human feedback [5], on the calibration of medical predictions remains empirically undocumented. The present thesis addresses these risks through a systematic audit of twelve open-source LLM configurations.

1.4 Research Question and Hypotheses

The main research question addressed in this work is stated as follows:

> *Are the eligibility scores produced by large language models in patient–clinical trial matching probabilistically reliable, and does reinforcement learning from human feedback affect this reliability?*

This question is decomposed into four falsifiable hypotheses, each associated with a numerical rejection threshold defined prior to the experimental phase:

- **H1** predicts that current LLM pipelines are systematically miscalibrated, with Expected Calibration Error exceeding the clinical threshold of 0.05.
- **H2** predicts that reinforcement learning from human feedback degrades calibration, extending a finding of Kadavath et al. [6] from general question answering to the clinical domain. This is the central pre-registered contribution of the thesis.
- **H3** predicts that model scores respect the ordinal structure of the eligibility labels, with a concordance index above 0.70.
- **H4** predicts that confidence-based selective classification improves macro-F1 by at least 0.02 when coverage is restricted to the top 80% most confident cases.

The pre-registration of these hypotheses, together with the fixed rejection thresholds, aligns the study with the reporting standards of TRIPOD+AI [4] and protects the analysis against post-hoc adjustment of the criteria to fit the observed results.

1.5 Contributions

The thesis contributes to the calibration and clinical AI communities along four complementary directions.

The first contribution is methodological: the study provides a reusable protocol for auditing the calibration of LLM-based clinical prediction pipelines. The protocol combines a pre-registered hypothesis-testing framework, a log-probability scoring pipeline that produces well-defined probability distributions, a double-prompt design that treats prompt engineering as an experimental variable, and a composite utility score that integrates calibration, discrimination, and prediction diversity. Every design element is documented in Chapter 4 and archived for independent reproduction (Appendix D).

The second contribution is confirmatory: the study tests the pre-registered hypothesis that RLHF degrades calibration on two independent model families (Mistral-7B and

Llama-3.1-8B) and two independent prompt designs. All four permutation tests reject the null hypothesis at Bonferroni-corrected significance ($p < 0.003$), extending to the clinical domain a finding previously documented only in general question answering.

The third contribution is exploratory: the study documents twelve original findings that emerged from the deeper analysis. Three of these deserve particular emphasis. The enriched-prompt paradox (F3) shows that expert prompt engineering degrades calibration in five of six models. Mode collapse (F7 and F8) affects five of the twelve configurations, which concentrate more than 97% of their predictions on a single label, and this collapse persists under constrained decoding. Complete failure on truly discriminative cases (F9) reveals that the two medical LLMs tested achieve accuracy at or below 0.4% on the 238 pairs requiring a genuine clinical decision — despite their apparent superiority on global accuracy. These three findings jointly demonstrate that global accuracy and ECE, taken in isolation, are insufficient to characterize the clinical usability of an LLM classifier.

The fourth contribution is mechanistic: a deeper analysis of the log-probabilities reinterprets mode collapse as a property of the decision rule rather than a loss of knowledge. The study shows that the discriminative signal survives collapse (area under the curve up to 0.91) but is masked by a response prior, identifies a scalar signal-to-prior ratio that predicts collapse across configurations (F11), and demonstrates that a simple re-thresholding rule recovers up to +0.79 macro-F1 where monotone recalibration provably cannot (F10). The same analysis separates recoverable collapse from an irrecoverable corruption of the signal, and both the collapse and its mechanism replicate on the independent TREC Clinical Trials corpus (F12). This contribution turns a descriptive failure mode into a diagnosable and often correctable property of the decoding step.

1.6 Structure of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 reviews the state of the art at the intersection of clinical trial matching, probabilistic calibration, and medical large language models, and identifies the specific gap that motivates this work.

Chapter 3 describes the academic framework of the project, the origin and evolution of the research question, the TrialGPT-Criterion-Annotations benchmark used as the empirical foundation, and the ethical and computational constraints that shape the study.

Chapter 4 presents the experimental design in full detail: the four pre-registered hypotheses, the data preparation, the model and prompt selection, the log-probability scoring pipeline, the evaluation metrics, and the statistical analysis plan.

Chapter 5 reports the empirical findings organized around the four pre-registered hypotheses and the twelve original findings. The chapter begins with the pre-LLM audit of the dataset, proceeds through the calibration audit and the RLHF effect, documents mode collapse and its reinterpretation as a decoding artefact, and closes with an external replication on the TREC Clinical Trials corpus.

Chapter 6 summarizes the outcomes, answers the research question, discusses the limitations of the study, and proposes future directions, including broader external validation beyond the TREC replication reported here, multi-agent adversarial debate as a mitigation for mode collapse, and mechanistic investigation of the origin of the response prior in medical LLMs.

Chapter 2

State of the Art

This chapter reviews the state of the art at the intersection of three research areas that jointly frame the present study: patient–clinical trial matching, probabilistic calibration of classifiers, and the evaluation of medically specialized large language models. Rather than surveying each area in isolation, we organize the review around the specific gap that motivates this thesis — the absence of systematic calibration audits in LLM-based clinical trial matching — and use each section to build the position of this work with respect to the closest prior contributions. Section 2.5 closes the chapter with a comparative table and an explicit statement of the contribution.

2.1 Patient–Clinical Trial Matching

2.1.1 Clinical Context and Recruitment Bottlenecks

Clinical trial recruitment is one of the most persistent bottlenecks in translational medicine. Despite widespread interest from patients and physicians, less than 5% of adult cancer patients enroll in a clinical trial, and 80% of trials fail to meet their recruitment targets [1]. The mismatch between patients and appropriate trials is caused, in part, by the sheer volume of open studies (over 400,000 registered on ClinicalTrials.gov at the time of writing) and the complexity of their eligibility criteria, which combine numeric thresholds, medical history requirements, and drug exclusions in dozens of criteria per trial. Manual screening of a patient against this space of criteria is labor-intensive and error-prone, and automating even a fraction of this process would substantially accelerate recruitment.

Automated matching systems have thus been a topic of active research for over a decade. The field has passed through three distinct waves, each corresponding to a different technological substrate.

2.1.2 Pre-LLM Approaches: Rule-Based and Embedding Systems

The first wave of matching systems relied on structured information extraction. EliXR [7], EliIE [8], and Criteria2Query [9] transformed free-text eligibility criteria into structured OMOP or SQL queries, matched against structured electronic health records. These systems produce deterministic Boolean outputs (match or no match) and provide no probability estimates or uncertainty quantification. They perform well on criteria that map cleanly to structured concepts (age, diagnoses, laboratory values) but struggle with criteria requiring nuanced interpretation.

The second wave introduced end-to-end neural matching. DeepEnroll [10] encoded criteria with BERT and patient EHRs with hierarchical embeddings, framing matching as textual entailment and reporting PR-AUC and F1 scores. COMPOSE [11] added a cross-modal pseudo-siamese network with a composite inclusion/exclusion loss, achieving 98.0% patient–criterion AUC and 83.7% patient–trial accuracy. Neither system reported calibration metrics. This omission is not accidental: the primary evaluation targets of these systems were discrimination metrics (AUC, F1) rather than the reliability of the probabilistic outputs.

2.1.3 The LLM Wave: 2023–2026

The third and current wave dates from the release of GPT-4 and the subsequent proliferation of open-source large language models. The most influential system in this wave is TrialGPT [1], published in *Nature Communications* in 2024. TrialGPT is a three-module pipeline: a retrieval stage recovers more than 90% of relevant trials in less than 6% of the collection using a hybrid BM25 and MedCPT retriever; a matching stage emits a per-criterion ordinal label ("included / not included / no relevant information"); and a ranking stage aggregates criterion labels into trial-level scores. TrialGPT reports a criterion-level accuracy of 87.3% and correlation patterns against the ground truth, but does not report any calibration metric — no ECE, no Brier score, no reliability diagram, and no monotonicity check across the ordinal scale.

TrialGPT is not an isolated case. RECTIFIER [12], published in *NEJM AI*, applies a retrieval-augmented GPT-4 to the COPILOT-HF heart-failure trial and reports 97.9–100% accuracy with Matthews correlation coefficients between 0.837 and 1.00. The system has been deployed clinically at Mass General Brigham, yet its outputs are reported only as accuracy and MCC, without probabilistic reliability assessment. PRISM and its variant OncoLLM [13], published in *npj Digital Medicine*, introduce a weighted-tier scoring method with concept-level importance and report top-k accuracy and nDCG (68% vs. 62.6% for GPT-3.5) but again omit calibration. Trial-LLaMA distills open-source Llama on synthetic criterion labels [14] and reports accuracy alone. Zero-shot matching approaches by Wornow et al. [15] achieve 0.93 micro-F1 on the n2c2 2018 dataset and elicit a self-declared confidence per criterion — the closest any published system comes to uncertainty quantification — but the self-declared confidence is never calibrated against ground truth. More recent systems including TrialMatchAI [16], multimodal pipelines by Callies et al. [17], and cross-modal trial-site selectors extend the scope of matching but continue the pattern of accuracy-only reporting.

The definitive scoping review of the field, published by Chen et al. [2] in *JCO Clinical Cancer Informatics*, screened 2,357 papers and included 24 LLM-based matching systems, of which 21 were published in 2024. The review confirms that *none* of these systems reports calibration metrics: no ECE, no Brier score, no reliability diagram, no calibration-in-the-large, and no slope. This gap defines the primary methodological opportunity addressed by the present thesis.

2.2 Probabilistic Calibration of Classifiers

2.2.1 Foundations and the Modern Miscalibration Problem

Probability calibration — the requirement that predicted probabilities match empirical accuracies — has a long history in statistical prediction. Historical foundations include Platt scaling [18] for logistic regression outputs, isotonic regression [19] for non-parametric monotone calibration, and the systematic comparisons of Niculescu-Mizil and Caruana [20]. For decades, calibration was treated as a solved problem: shallow classifiers such as logistic regression, naïve Bayes, and calibrated support vector machines were well calibrated after appropriate post-hoc correction.

This picture changed with the rise of deep learning. Guo, Pleiss, Sun, and Weinberger [21] showed in 2017 that modern deep neural networks, despite their superior accuracy, are *systematically overconfident* relative to shallower architectures: a network with 85% accuracy may assign 99% confidence to its predictions, including its errors. This paper introduced the Expected Calibration Error (ECE) as the standard metric for measuring miscalibration in modern classifiers, and demonstrated the effectiveness of temperature scaling, a single-parameter post-hoc method that does not modify argmax predictions.

Since 2017, the calibration literature has expanded rapidly. Alternative metrics have been proposed to address specific limitations of ECE: adaptive-ECE, class-wise ECE, and the more refined estimators of Nixon et al. [22] and Kumar, Liang, and Ma [23].

The Brier score [24] provides a joint measure of calibration and sharpness. Beyond ECE, distribution-free confidence intervals for calibration have been studied by Gupta and Ramdas [25].

2.2.2 Clinical Importance of Calibration

The clinical importance of calibration has been argued particularly forcefully by Van Calster et al. [3], whose 2019 *BMC Medicine* article titled "Calibration: the Achilles heel of predictive analytics" establishes calibration as a prerequisite for clinical deployment of any predictive model. Van Calster and colleagues distinguish four progressive levels of calibration — calibration in the large, weak calibration, moderate calibration, and strong calibration — and argue that most published clinical prediction models fail even the weakest level. They also introduce the threshold ECE < 0.05 as a practical benchmark for clinically usable calibration, which the present thesis adopts as the rejection criterion for Hypothesis H1 (Section 4.2).

The 2024 update of the TRIPOD guidelines, known as TRIPOD+AI [4], formalizes calibration as a required reporting element for any AI-based clinical prediction model. Under TRIPOD+AI, a paper introducing a clinical AI model must report calibration curves, calibration-in-the-large, and calibration slopes; failure to do so is now grounds for methodological rejection by several leading medical journals. This regulatory shift makes the absence of calibration reporting in LLM-based trial matching systems increasingly untenable.

2.2.3 Recalibration Methods

Post-hoc recalibration methods apply a monotone transformation to uncorrected confidences after the classifier has been trained. The three methods evaluated in the present thesis — temperature scaling, isotonic regression, and Platt scaling — represent the classical choices; more elaborate methods include beta calibration [26], Dirichlet calibration [27], and the matrix/vector scaling family. Intrinsic methods, which modify the training objective itself, include label smoothing [28] and focal loss [29]. Distribution-free calibration approaches combine post-hoc scaling with conformal guarantees [25].

Complementary to point-estimate calibration, split conformal prediction [30], [31] produces prediction sets with a guaranteed marginal coverage of $1 - \alpha$ without any distributional assumption on the underlying model. The guarantees hold under exchangeability of calibration and test data, which is satisfied in the present study by the split protocol described in Section 4.3.4. Selective classification — the option to abstain from prediction — has been formalized by Geifman and El-Yaniv [32], [33] through risk–coverage curves, which are used in Section 5.10 to test Hypothesis H4.

2.3 Large Language Models: Alignment and Calibration

The calibration of large language models raises questions that differ qualitatively from those of traditional classifiers. LLMs are not trained end-to-end for classification but for next-token prediction, and their class probabilities in downstream tasks are extracted through prompting strategies rather than through a dedicated output layer. This structural difference has motivated a dedicated research thread on the calibration of LLMs and, in particular, on the effect of post-training alignment procedures.

2.3.1 RLHF and Its Effect on Calibration

Reinforcement learning from human feedback (RLHF), introduced at scale by Ouyang et al. [5] in the development of InstructGPT, has become the dominant paradigm for aligning large language models with human preferences. The procedure consists of three stages: supervised fine-tuning on demonstration data, training of a reward model from human preference judgments, and reinforcement learning of the base model to maximize the reward. RLHF has demonstrably improved user-facing metrics such as helpfulness, truthfulness, and adherence to safety guidelines.

The calibration cost of this alignment procedure was first documented by Kadavath et al. [6] in a systematic study of Anthropic's Claude family. Using general question-answering benchmarks (TriviaQA, Lambada, TruthfulQA), the authors showed that base models produce well-calibrated confidence estimates on multiple-choice questions, while their RLHF-aligned counterparts systematically shift probability mass toward the model's top prediction, degrading calibration. Kadavath and colleagues attribute this effect to the alignment reward signal, which incentivizes confident, assertive answers regardless of their actual reliability — because human raters, on average, prefer confident responses.

This finding has been replicated in several follow-up studies and is now considered a robust property of RLHF-based alignment. However, no prior work has systematically extended this observation to the clinical domain, where the trade-off between assertiveness and reliability carries substantially higher stakes. The central pre-registered contribution of the present thesis (Hypothesis H2, tested in Section 5.4) is precisely this extension: we replicate Kadavath's finding on two independent open-source model families (Mistral-7B and Llama-3.1-8B) applied to clinical trial matching, and show that the effect not only persists but is quantitatively amplified in this domain.

2.3.2 Uncertainty Elicitation and Verbalized Confidence

An alternative to log-probability-based calibration is verbalized confidence elicitation, where the model is prompted to explicitly report its confidence in a natural-language format (e.g., "I am 80% sure that..."). This approach has been studied in general LLMs and was applied in the clinical domain by Wornow et al. [15], whose zero-shot clinical trial matching system asks GPT-4 to output a confidence score alongside each criterion label. Verbalized confidence is attractive for its interpretability, but it introduces two complications: the elicited scores are not necessarily connected to the model's internal probability estimates, and they are typically not calibrated against ground truth. The present thesis avoids verbalized confidence in favor of direct log-probability scoring (Section 4.6), which produces proper probability distributions suitable for standard calibration analysis.

2.4 Medical Large Language Models

The rise of general-purpose LLMs has been paralleled by a growing ecosystem of medically specialized models. These models are typically built by continued pretraining, instruction tuning, or model merging on top of a general-purpose base, using biomedical corpora such as PubMed abstracts, clinical guidelines, or medical textbooks. This section reviews the four open-source medical LLMs most relevant to the present study.

2.4.1 Meditron

Meditron [34], released by EPFL in late 2023, represents the first open-source medical LLM to be developed at scale. The 7B and 70B versions are built by continued pretraining of Llama-2 on the GAP-Replay corpus, which consists of 48,000 clinical practice guidelines and a curated subset of PubMed abstracts totaling approximately 46 billion tokens. Meditron reports strong performance on established medical QA benchmarks (MedQA, PubMedQA, MedMCQA) but does not report calibration metrics. Notably, no official instruction-tuned or RLHF-aligned version of Meditron-7B has been released, which makes it the ideal candidate to isolate the effect of medical pretraining without any confound from post-training alignment (Section 4.4).

2.4.2 BioMistral

BioMistral [35], released in early 2024, takes a different construction path: it is built by continued pretraining of Mistral-7B-Instruct-v0.1 on the PubMed Central Open Access subset (3 billion tokens), and then combined with the original Mistral through several merging strategies (DARE, TIES, SLERP). BioMistral is the only open-source medical LLM to date whose original publication reports ECE values on medical QA tasks, providing an external validation point for calibration measurement methodology. Because BioMistral inherits its alignment from Mistral-Instruct, it represents the "medical + RLHF" corner of the two-by-two matrix used in the present study.

2.4.3 Med42, HuatuoGPT-o1, and the Broader Ecosystem

The medical LLM ecosystem has expanded rapidly since 2024. Med42-v2 [36], released by M42 Health, is available in 8B and 70B parameter versions and uses a full supervised-fine-tuning-plus-DPO pipeline on top of Llama-3. However, the base variant of Med42-v2 is not publicly released, which precludes a Base-versus-Instruct comparison of the form required for testing Hypothesis H2. HuatuoGPT-o1 [37], released in late 2024, extends the reasoning capabilities of medical LLMs through verification-based multi-step training, but its primary training corpus is Chinese, which introduces a language confound with the English-only benchmark used here. Other medical LLMs considered but excluded from the final selection include OpenBioLLM (redundant architecture with BioMistral), Aloe-70B (memory budget), and various proprietary systems (GPT-4-Medical, Med-PaLM) for which the underlying weights are not available for calibration audits.

The comparative characteristics of the six models selected for the present study are summarized in Table 4.3 of Chapter 4. Two properties are shared across all current open-source medical LLMs: they are evaluated primarily on multiple-choice medical QA benchmarks, and calibration is not reported. The present thesis contributes to filling this gap by applying standard calibration protocols to four representative medical and generalist configurations on a common downstream task.

2.5 Positioning of This Work

The three research threads reviewed above — clinical trial matching, probabilistic calibration, and medical LLMs — have developed largely in parallel. Table 2.1 places the present thesis in the intersection of these threads, alongside the closest prior works.

Three specific gaps in the literature are addressed by this thesis. First, no prior LLM-based clinical trial matching system reports calibration metrics, as documented by the Chen et al. [2] scoping review across 24 published systems. The present study is, to our knowledge, the first to conduct a systematic calibration audit in this domain. Second, no prior work has tested the RLHF-degrades-calibration hypothesis of Kadavath et al. [6] outside general-purpose question answering. The present study extends this finding to the clinical domain across two independent model families. Third, the standard

Table 2.1: Comparative positioning of the closest prior works. A checkmark indicates that the corresponding aspect is systematically addressed; a dash indicates that it is not.

Work	**Domain**	**Models**	**RLHF**	**Calib.**	**Recalib.**
Jin et al. 2024 (TrialGPT)	Trial matching	GPT-4	—	—	—
Kadavath et al. 2022	General QA	Anthropic	✓	✓	—
Labrak et al. 2024 (BioMistral)	Medical QA	BioMistral	—	✓	—
Unlu et al. 2024 (RECTIFIER)	Trial matching	GPT-4	—	—	—
Nievas et al. 2024 (Trial-LLaMA)	Trial matching	Llama	—	—	—
Chen et al. 2025 (scoping review)	Trial matching	24 LLMs	—	—	—
This work	**Trial matching**	**6 LLMs**	✓	✓	✓

evaluation protocol for medical LLMs — multiple-choice medical QA — masks two failure modes that are exposed here: mode collapse in evasion labels and complete failure on truly discriminative cases. The methodology and findings of this thesis are intended to make these failure modes detectable in future medical LLM evaluations.

The specific contributions of this work, summarized as four pre-registered hypotheses and twelve original findings, are presented in Chapter 5 and synthesized in Chapter 6.

Chapter 3

Project Context and Scope

3.1 Academic Research Framework

This thesis was conducted within the Final Year Project (*Projet de Fin d'Études*, PFE) of the PGE5 program at aivancity, from March to September 2026. Unlike most PFE projects at aivancity, which are structured around a six-month industry internship in a host company, the present project is a purely academic research contribution. This choice was made in agreement with the academic supervisor, Dr. Abdelhamid Mellouk, after a preliminary review of the literature identified a specific methodological gap — the systematic auditing of large language model calibration in clinical trial matching — that could be addressed with academic resources alone, without requiring hospital or industry partners.

The academic framing has three practical consequences for the study. First, the research question and the four pre-registered hypotheses were formulated to be testable with publicly available datasets and open-source models, avoiding any dependency on proprietary systems or restricted patient data. Second, the study operates on a purely retrospective basis: no prospective clinical validation is performed, and no recommendations for real patient enrollment are made. Third, the computational budget is limited to what a single graduate student can access on free-tier cloud platforms, which caps the model sizes at seven to nine billion parameters and shapes several of the design choices described in Chapter 4.

Regular supervision meetings with Dr. Abdelhamid Mellouk were held throughout the project. Progress was formalized through monthly reports covering the scoping phase (March), the state of the art review (April), the experimental pipeline design (May), the execution of the twelve model runs (June), the extended analyses on mode collapse and truly discriminative cases (July), and the writing phase (August–September).

3.2 Origin and Evolution of the Research Question

The initial research question, formulated during the scoping phase, focused on patient-reported uncertainty communication in clinical trial matching: how do LLMs express uncertainty to end users, and does this communication align with what patients expect? A month of preliminary literature review, however, revealed that this framing conflated two distinct research questions — the human factors of uncertainty communication, and the underlying probabilistic reliability of the model outputs — and that only the second could be rigorously addressed within the scope of a Master's thesis.

The question was therefore reformulated to focus exclusively on the probabilistic reliability of LLM outputs. This reformulation offered three advantages. First, it aligned the study with an established research tradition on classifier calibration, providing well-defined metrics (ECE, Brier score, reliability diagrams) and clear evaluation thresholds (Van Calster's 0.05). Second, it made the hypotheses testable with pre-registered statistical procedures, protecting the analysis against confirmation bias. Third, it identified a specific methodological gap in the medical LLM literature: while several papers had evaluated the accuracy of medical LLMs on clinical benchmarks, no systematic audit of their calibration had been performed for the clinical trial matching task. This gap became the specific target of the study.

The final research question — "Are the eligibility scores produced by LLM-based patient–trial matching systems probabilistically reliable, and does RLHF post-training affect this reliability?" — is a narrower, more falsifiable version of the initial intent, and the tradeoff is a deliberate methodological choice.

3.3 Dataset: TrialGPT-Criterion-Annotations

3.3.1 Dataset Description

The empirical foundation of this study is the TrialGPT-Criterion-Annotations dataset, released by Jin et al. [1] alongside their TrialGPT system in *Nature Communications*. The dataset comprises 1,015 patient–criterion pairs, where each pair associates a synthetic patient note with a single eligibility criterion drawn from a real clinical trial listed on ClinicalTrials.gov. Each pair is annotated independently by three physicians (whose consensus label serves as the ground truth) and by GPT-4 (used as an automated second opinion). Both annotation sources apply the same six-level ordinal scale, described in detail in Section 4.3.1.

The dataset is publicly available under a CC BY 4.0 license, hosted on Zenodo alongside the TrialGPT source code. Its release supports the reproducibility of the original TrialGPT evaluation and, by extension, the present audit.

3.3.2 Justification of the Single-Dataset Choice

An important scoping decision of this study is the deliberate reliance on a single benchmark for the *primary* analysis. Three alternatives were considered and rejected as the primary corpus during the design phase. First, the TREC Clinical Trials 2021–2022 [38], [39] tracks provide relevance judgments for patient–trial pairs, but they operate at the trial level rather than the criterion level and use only three ordinal grades (definitely relevant, possibly relevant, irrelevant). Adapting the calibration protocol to this coarser granularity would preclude the fine-grained analysis of inclusion versus exclusion criteria that motivates several of our findings. Second, the SIGIR 2016 clinical trial task shares the same limitations as TREC. Third, hospital-derived cohorts (n2c2, MIMIC-derived subsets) would provide realistic patient data but require institutional partnerships and ethics approval that fall outside the academic scope of the project.

TrialGPT-Criterion-Annotations is, to our knowledge, the only publicly available benchmark that combines per-criterion granularity, a six-level ordinal scale, and an explicit inclusion/exclusion typology. These properties are jointly essential for the calibration analysis described in Chapter 4, which is why it is retained as the primary corpus. The coarser granularity that disqualifies TREC for the calibration hypotheses does not, however, prevent it from serving a distinct and valuable role: because the mechanistic analysis of mode collapse (Chapter 5) operates on the model's probability distributions rather than on the six-level scale, TREC can be used as an independent corpus to test whether the collapse behaviour and its mechanism generalize beyond the primary benchmark. We therefore use TREC Clinical Trials 2021–2022 as a secondary, external validation corpus, described in Section 4.3.5 and reported in Section 5.15.

3.4 Ethical Considerations

The ethical framework of this study is straightforward due to the nature of the dataset. TrialGPT-Criterion-Annotations contains no real patient data: all clinical notes were synthesized by the original authors from publicly available templates on ClinicalTrials.gov, using clinical concept substitution to produce notes that are semantically realistic yet fully non-identifying. No personal health information is processed at any stage of the pipeline, and no institutional review board approval or ethics committee clearance was required.

The absence of real patient data does not eliminate all ethical considerations. The results reported in Chapter 5 concern models that could plausibly be deployed in real clinical screening workflows, and the mode collapse behaviors documented in Section 5.6 would translate into systematic screening failures if left undetected in production. The methodology described in this thesis is therefore intended as a template for pre-deployment audits rather than as a certification of the models themselves for clinical use. Practitioners integrating LLMs into recruitment workflows retain full responsibility for validating the models on their own populations and for maintaining human oversight of eligibility decisions.

Beyond patient data, the study also complies with the licensing terms of the models under evaluation. All six models are released under permissive open-source licenses (Apache 2.0 for Mistral, custom research licenses for Llama-3.1 and Meditron, MIT for BioMistral) that permit non-commercial evaluation and redistribution of inference outputs. The prediction files archived in the project repository (Appendix D) fall within the scope of these licenses.

3.5 Constraints and Resources

3.5.1 Computational Constraints

All experiments were performed on the Kaggle free-tier compute environment, which provides access to a single Tesla T4 GPU with 16 GB of memory and a weekly quota of 30 GPU-hours. This constraint shaped two important design choices. First, the model sizes are capped at seven to nine billion parameters, which excludes larger medical LLMs such as Meditron-70B or Med42-v2-70B from the evaluation. Second, all models are quantized to 4-bit NF4 precision, which reduces the memory footprint from approximately 14 GB per model (in half-precision) to approximately 4 GB. The justification of these choices and their potential impact on the results are discussed in Sections 4.4 and 4.7.

3.5.2 Model Access Constraints

Access to the six selected models is subject to licensing terms that occasionally require prior approval. Llama-3.1 requires acceptance of Meta's community license through the HuggingFace hub, which was obtained during the setup phase. Gemma and Med42-v2, considered but excluded from the final selection, are subject to more restrictive licenses that complicate reproducibility and were partly responsible for their exclusion (Section 4.4.3). All models retained in the final evaluation are freely accessible for research use without additional approval.

3.5.3 Time Constraints

The PFE was conducted over six months, from March to September 2026. The scoping phase (March) established the research question and the literature review. The design phase (April–May) formalized the pre-registered hypotheses and the experimental matrix. The execution phase (June–July) ran the twelve model configurations and the downstream analyses. The writing phase (August–September) produced the present manuscript. The time constraint precludes prospective clinical validation and extension to larger model scales; external validation was limited to a single independent corpus (TREC Clinical Trials, Section 5.15) rather than the broader multi-dataset replication that would be desirable. These directions are discussed in Chapter 6.

3.6 Scope of This Work

The scope of this thesis is delimited both by what it includes and by what it deliberately excludes. Within scope, the study contributes: a systematic calibration audit of twelve LLM configurations on a primary benchmark; a pre-registered test of the effect of RLHF post-training on calibration in the clinical domain; a comparison between generalist and medically specialized open-source models; a documentation of two novel failure modes (mode collapse and complete failure on discriminative cases) that had not been previously reported for these models, together with a mechanistic reinterpretation of mode collapse as a recoverable property of the decoding rule; an evaluation of post-hoc recalibration, re-thresholding, and conformal prediction as practical remedies; and an external replication of the collapse mechanism on the independent TREC Clinical Trials corpus.

Explicitly out of scope, and deferred to future work in Chapter 6, are: broader external validation across multiple independent benchmarks beyond the single-corpus TREC replication reported here; prospective clinical validation involving real patient enrollment; extension to model sizes above nine billion parameters; extension to multi-agent debate architectures as a mitigation for mode collapse; fine-tuning or continued pretraining of the evaluated models; and multilingual evaluation on non-English clinical text.

This delimitation is intended to make the contribution of the thesis falsifiable and specific, rather than exhaustive. The methodology proposed here can be extended in each of the excluded directions, and several such extensions are outlined as concrete research programs in Chapter 6.

Chapter 4

Methodology and Technical Approach

4.1 Overview of the Experimental Design

This chapter describes the methodology used to audit the calibration of large language models in patient–clinical trial matching. The experimental design follows a pre-registered protocol consistent with the TRIPOD+AI reporting guidelines [4], which recommend explicit hypothesis specification, statistical rigor, and reproducibility for clinical prediction models.

The evaluation pipeline is organized around three principles. First, all predictions are generated on a single held-out benchmark, TrialGPT-Criterion-Annotations [1], which contains 1,015 patient–criterion pairs annotated by three physicians and by GPT-4. Second, model probabilities are extracted directly from the log-probabilities of candidate labels rather than from free-form generation, in order to obtain well-defined probability distributions suitable for calibration analysis. Third, every hypothesis test is defined before the analysis begins, and p-values are corrected for multiple comparisons.

The design forms a 2×2 matrix crossing model type and post-training regime. Along one axis, we compare generalist models (Mistral-7B, Llama-3.1-8B) with medically specialized models (Meditron-7B, BioMistral-7B). Along the other axis, we compare base models (without RLHF) with instruction-tuned models (with RLHF). This crossing produces six models, each of which is evaluated with two prompt variants: a generic minimalist prompt (Prompt A) and an adapted prompt with expert guidance (Prompt B). The full factorial yields twelve configurations, all evaluated on the same 1,015 pairs, for a total of 12,180 predictions.

Figure 4.1 summarizes the pipeline. Section 4.2 states the four pre-registered hypotheses. Section 4.3 describes the dataset. Sections 4.4 and 4.5 justify the model and prompt selection. Section 4.6 details the log-probability scoring pipeline. Section 4.8 defines the calibration and validity metrics. Section 4.9 presents the statistical analysis plan, including the recalibration and conformal prediction procedures.

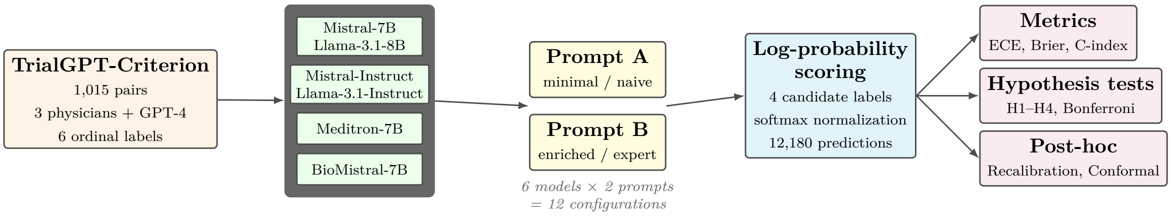

Figure 4.1: Overview of the experimental design. Six open-source LLMs are evaluated with two prompt variants on 1,015 patient–criterion pairs, producing twelve configurations tested against four pre-registered hypotheses.

4.2 Research Question and Pre-registered Hypotheses

The main research question addressed in this work is stated as follows:

> *Are the eligibility scores produced by large language models in patient–clinical trial matching probabilistically reliable, and does reinforcement learning from human feedback affect this reliability?*

This question is decomposed into four falsifiable hypotheses, each associated with a numerical rejection threshold defined prior to the experimental phase. Pre-registration of hypotheses is a standard safeguard against confirmation bias in empirical work [4] and provides a clear criterion for evaluating each finding.

Hypothesis H1 tests whether current LLM pipelines are systematically miscalibrated. The rejection threshold is derived from Van Calster et al. [3], who characterize an Expected Calibration Error (ECE) below 0.05 as evidence of a well-calibrated predictive system in clinical contexts.

Hypothesis 1 (H1: Poor calibration of current LLM pipelines)**.** *The Expected Calibration Error of an LLM-based eligibility scoring pipeline exceeds the well-calibrated threshold of* 0.05*:*

$$ECE(pipeline) > 0.05.$$

Hypothesis H2 extends the finding of Kadavath et al. [6], who observed that RLHF post-training degrades calibration on general question-answering tasks, to the specific domain of clinical trial matching. It is the central original contribution of this thesis. The comparison uses a permutation test on the difference in ECE between paired Base and Instruct versions of the same model family, with Bonferroni correction across families and prompt variants.

Hypothesis 2 (H2: RLHF degrades calibration)**.** *Within a given model family, the*

RLHF-aligned Instruct variant exhibits worse calibration than its Base counterpart:

$$ECE(Instruct) > ECE(Base).$$

Hypothesis H3 tests whether, independently of absolute calibration, the model's scores respect the ordinal structure of the eligibility labels. Ordinal validity is measured by the concordance index (C-index), and the threshold of 0.70 corresponds to conventional discrimination criteria used in clinical prediction modeling [3].

Hypothesis 3 (H3: Ordinal validity)**.** *The C-index between the LLM ordinal score and the expert ordinal label exceeds the standard discrimination threshold:*

$$\text{C-index} > 0.70.$$

Hypothesis H4 addresses the practical value of confidence-based abstention. If confidence scores are meaningful, restricting predictions to the most confident subset should yield higher classification quality. The threshold of +0.02 on macro-F1 gain is chosen to reflect a clinically noticeable improvement without being trivially achievable.

Hypothesis 4 (H4: Selective classification improves quality)**.** *Restricting predictions to the top 80% most confident cases increases macro-F1 by at least* 0.02*:*

$$F_1(coverage = 80\%) > F_1(coverage = 100\%) + 0.02.$$

Together, these four hypotheses probe distinct aspects of probabilistic reliability: absolute calibration (H1), the effect of RLHF (H2), ordinal consistency (H3), and practical usability under abstention (H4). All rejection criteria and statistical tests were fixed prior to running the experiments and are reported without modification in Chapter 5.

4.3 Data Preparation

The experimental evaluation is conducted on the TrialGPT-Criterion-Annotations dataset [1], released alongside the TrialGPT system by Jin et al. To our knowledge, it is the only publicly available benchmark that provides multi-level expert annotations at the level of individual eligibility criteria, with an explicit distinction between inclusion and exclusion criteria. These properties are essential for a fine-grained calibration analysis: aggregating annotations at the trial level, as in TREC Clinical Trials or SIGIR benchmarks, would preclude the per-criterion probability estimation required by the present protocol.

4.3.1 Dataset Composition

The dataset comprises 1,015 patient–criterion pairs. Each pair associates a synthetic patient note with a single eligibility criterion drawn from a real clinical trial. Two independent annotations are provided per pair: a consensus label from three physicians, and a label produced by GPT-4. Both annotations use the same six-level ordinal scale, described in Table 4.1.

Table 4.1: The six eligibility labels used in TrialGPT-Criterion-Annotations, organized by criterion type.

Label	Criterion type	Semantics
included	inclusion	The patient satisfies the inclusion criterion.
not included	inclusion	The patient does not satisfy the criterion.
excluded	exclusion	The patient has the excluding condition.
not excluded	exclusion	The patient does not have the excluding condition (favorable).
not enough information	both	The note does not permit a decision.
not applicable	both	The criterion does not apply to this patient.

Two of the six labels function as evasion or uncertainty markers rather than decisive clinical judgments (“not enough information” and “not applicable”). We deliberately keep them in the label space to preserve the semantics of the original benchmark, while treating their behavior as an object of study in its own right (see Sections 5.9 and 5.6).

4.3.2 Label Distribution

The distribution of expert annotations is markedly imbalanced (Table 4.2). “Not excluded” dominates at 47.7%, reflecting the fact that most patients do not present the specific medical conditions targeted by exclusion criteria. “Not enough information” accounts for a further 28.9%. The remaining 23.4% of pairs, distributed across four labels, represent the cases requiring a truly discriminative clinical judgment. This asymmetry is critical for the interpretation of results in Chapter 5: aggregate metrics computed over the full 1,015 pairs can be dominated by the two majority classes and may mask complete failure on the discriminative subset.

4.3.3 Textual Characteristics

Patient notes are short and structured: their length ranges from 232 to 1,447 characters with a median of 579 characters. This compactness is important for two reasons. First, it means that no aggressive truncation is required to fit the notes within LLM context

Table 4.2: Expert-label distribution in TrialGPT-Criterion-Annotations.

Label	Count	Proportion
not excluded	484	47.7%
not enough information	293	28.9%
included	74	7.3%
excluded	55	5.4%
not applicable	55	5.4%
not included	54	5.3%
Total	1,015	100.0%

windows, avoiding an additional source of variability. Second, short notes place a heavier reasoning burden on the model: information is often implicit, and the model must infer unstated conditions from the surrounding clinical context. Criterion texts are also short, with a median length of 58 characters, ranging from declarative statements (*Age 18 to 75 years*) to more elaborate clauses involving numeric thresholds or drug interactions.

4.3.4 Split for Recalibration Analysis

The dataset is used in three distinct ways depending on the analysis, each governed by a specific splitting strategy.

For the primary calibration audit (Chapter 5, Section 5.3) and the H2 permutation tests, the models are queried on the full set of 1,015 pairs. No split is required at this stage because the models are used off-the-shelf, without any fitting on the target data. Each of the twelve configurations produces one prediction per pair, and calibration metrics are computed on the full set. The permutation tests reshuffle pair-level assignments between Base and Instruct variants, preserving the identity of individual pairs to maintain paired comparisons.

For the post-hoc recalibration analysis (Section 5.11), we adopt a 70/30 random split. The 70% subset (approximately 710 pairs) is used to fit the recalibration function (isotonic regression, temperature scaling, or Platt scaling), while the 30% subset (approximately 305 pairs) is used to evaluate the recalibrated ECE on data unseen during fitting. The split is stratified by criterion type to preserve the inclusion/exclusion ratio in both subsets and prevent imbalance artifacts. The random seed is fixed at 42 for reproducibility, and the same split is applied consistently across all twelve configurations to enable direct comparison.

To assess whether the recalibration gain observed on a single 70/30 split is stable or an artifact of a favorable split, we complement the primary analysis with a 5-fold cross-validation on the same data. Each fold uses 80% of the pairs for fitting and

20% for evaluation, with the folds rotating through the dataset. We report the mean ECE reduction across the five folds together with its standard deviation, providing a direct measure of the stability of the recalibration effect. This cross-validation is applied to isotonic recalibration only, which is the method retained as the primary recommendation in Chapter 5.

For the analysis of performance on truly discriminative cases (Section 5.7), we introduce a further stratification that separates the 238 pairs whose expert label falls outside the two dominant classes ("not excluded" and "not enough information") from the remaining 777 pairs. This stratification is not a random split but a principled selection driven by the class-imbalance analysis reported in Section 4.3.2, and it is used to reveal model behavior on cases requiring genuine clinical decisions rather than majority-class shortcuts.

4.3.5 External Validation Corpus

To test whether the mechanistic findings of Chapter 5 generalize beyond the primary benchmark, we assembled a secondary corpus from the TREC Clinical Trials 2021–2022 tracks [38], [39]. TREC provides patient–trial relevance judgements on a three-grade scale (eligible, excluded, not relevant) rather than the six-level per-criterion annotations of TrialGPT-Criterion; it is therefore not suitable for re-testing the pre-registered calibration hypotheses, but it is well suited to testing whether the collapse behaviour and its signal-to-prior mechanism (Sections 5.15 and 5.6.4) are properties of the models rather than of the primary benchmark.

We sampled 40 topics with an even split between eligible and non-relevant trials, drew five trials per topic, and decomposed each trial's inclusion and exclusion criteria into individual patient–criterion pairs using the same parsing rules as the primary dataset, capping the number of criteria per trial to limit inference cost. This yields 199 patient–trial pairs and 923 patient–criterion pairs. The trial-level relevance grade is used as a coarse gold label for the recoverability analysis, binarized as eligible versus the remainder. The external corpus is scored with an identical pipeline — same Prompt A, same log-probability scoring, same 4-bit quantization — so that the corpus is the only variable that differs from the primary evaluation. The results of this replication are reported in Section 5.15.

4.3.6 Ethical and Reproducibility Considerations

The ethical framework of this study is straightforward due to the nature of the dataset. TrialGPT-Criterion-Annotations contains no real patient data: all clinical notes were synthesized by the original authors [1] to match the structure and semantics of authentic recruitment scenarios without disclosing any personal health information. The synthesis process combined publicly available templates from ClinicalTrials.gov with clinical concept substitution to produce notes that are semantically realistic yet fully non-identifying. No institutional review board approval or ethics committee clearance was therefore required for the present study, since no human subject data is processed.

The dataset is released under a CC BY 4.0 license, permitting redistribution and derivative use for research purposes with attribution. This licensing framework aligns with the reproducibility commitments of this thesis. All prediction files produced during the twelve model runs, comprising the softmax distributions over candidate labels for each of the 1,015 pairs, are preserved as Parquet files and archived in the project repository listed in Appendix D. The complete set of downstream analysis outputs (calibration metrics, permutation test p-values, recalibration results, conformal prediction sets, and mode collapse statistics) is also archived as CSV files to enable independent verification of every numerical result reported in Chapter 5.

Beyond data ethics, the study raises considerations related to the deployment of the audited models in clinical practice. The results reported in Chapter 5 are informative but not directly actionable: no model tested reaches a level of probabilistic reliability sufficient for autonomous eligibility screening, and the mode collapse behaviors documented in Section 5.6 would in a clinical setting translate into systematic screening failures if left undetected. These findings underscore the responsibility of practitioners who consider integrating LLMs into recruitment workflows to conduct calibration audits before deployment and to interpret model outputs jointly with human clinical judgment rather than as replacements for it. The methodology described in this chapter is intended as a practical template for such pre-deployment audits.

4.4 Model Selection

The choice of models under evaluation is the central design decision of this work. Rather than surveying a broad catalogue of open-source LLMs, we prioritize a small selection organized as a controlled experimental matrix. This design enables clean isolation of the effects under study — particularly the effect of RLHF post-training — while remaining feasible on the limited computational resources available.

4.4.1 Selection Criteria

Three criteria guide the selection of models. First, the model must be available in both a base and an instruction-tuned variant with a common pretraining, since testing hypothesis H2 requires paired comparisons within the same model family. Second, the selection must span the two-by-two matrix of interest: generalist versus medically specialized pretraining, crossed with the presence or absence of RLHF post-training. Third, the model must fit into 16 GB of GPU memory when quantized to 4 bits, which is the memory budget of a free-tier T4 GPU on Kaggle. This constraint effectively caps the parameter count at 7–9 billion.

These three criteria eliminate several candidates that would otherwise be attractive. Models larger than 9 B parameters (Meditron-70B, Med42-v2-70B, Llama-3.3-70B) exceed the memory budget. Model families without a clean base–instruct pairing (Med42-v2 releases only the instruction-tuned variant) preclude the H2 comparison. Model families released without a medical counterpart (Gemma, DeepSeek) do not add information for our two-by-two matrix.

4.4.2 The Six Selected Models

The six selected models, together with their scientific role in the experimental matrix, are summarized in Table 4.3.

Table 4.3: Selected models and their scientific role in the two-by-two experimental matrix.

Model	**Type**	**Scientific Role**
Mistral-7B-v0.3	Generalist, no RLHF	H2 control
Mistral-7B-Instruct-v0.3	Generalist, with RLHF	H2 test
Llama-3.1-8B	Generalist, no RLHF	H2 replication
Llama-3.1-8B-Instruct	Generalist, with RLHF	H2 replication
Meditron-7B	Medical, no RLHF	Pure specialization effect
BioMistral-7B	Medical, inherited RLHF	Validation vs. Labrak et al. (2024)

Mistral-7B-v0.3 and its instruction-tuned counterpart Mistral-7B-Instruct-v0.3 form the primary base–instruct pair for testing H2. The two models share the same pretraining corpus and architecture; the Instruct variant differs only through subsequent supervised fine-tuning and RLHF. This clean isolation of the RLHF stage makes them the reference comparison for our central hypothesis.

Llama-3.1-8B and Llama-3.1-8B-Instruct play the same role for a second, independent model family. Testing H2 on two families rather than one rules out the possibility that any observed effect is specific to a single training pipeline. Llama-3.1 also brings a different tokenizer, a larger context window, and a different alignment methodology, which strengthens the generality of the conclusions.

Meditron-7B [34] is a foundation model pretrained on 46,000 clinical practice guidelines and a large corpus of PubMed abstracts, starting from Llama-2 weights. Crucially, no official instruction-tuned variant of Meditron-7B has been released; the base model is the only publicly available version. Meditron therefore isolates the effect of medical pretraining without any confound with RLHF, providing a direct test of whether medical specialization improves calibration in the absence of instruction alignment.

BioMistral-7B [35] occupies the opposite corner of the two-by-two matrix. It is derived from Mistral-7B-Instruct-v0.1 through continued pretraining on PubMed Central Open Access texts. It therefore inherits the RLHF alignment of its Mistral-Instruct parent while adding a medical specialization stage. BioMistral is also the only open-source medical LLM with published ECE values, which provides an external validation point for our measurement methodology.

Together, these six models populate the two-by-two matrix completely (Table 4.4). Each of the three scientific questions associated with the matrix — the effect of RLHF, the effect of medical specialization, and their interaction — can be tested through paired comparisons within this design.

Table 4.4: The two-by-two experimental matrix of selected models.

	Without RLHF	**With RLHF**
Generalist	Mistral-7B Llama-3.1-8B	Mistral-7B-Instruct Llama-3.1-8B-Instruct
Medical	Meditron-7B	BioMistral-7B

4.4.3 Excluded Models

Several candidate models were considered and deliberately excluded from the final selection. Qwen2.5-7B and DeepSeek-V2-Lite were excluded as redundant additions to the generalist axis: they would not provide new evidence for H2 beyond what Mistral and Llama already establish. Gemma-2-9B was excluded for the same reason, and additionally because its license restricts commercial redistribution of derivative model outputs, complicating reproducibility.

Among medical LLMs, Med42-v2 [36] and OpenBioLLM-8B were excluded because their base versions are not publicly released, precluding a within-family H2 comparison. HuatuoGPT-o1 [37] was excluded because its training corpus is primarily Chinese, which introduces a confound with the English-only nature of the TrialGPT-Criterion benchmark. Larger medical models (Meditron-70B, Med42-v2-70B, Aloe-70B) all exceed the memory budget of a free-tier T4 GPU, even at 4-bit quantization.

The rationale for this restrictive selection is that a small number of well-matched models tested on the same protocol yields more interpretable conclusions than a broader benchmark that would necessarily mix confounding factors. Extending the analysis to additional families is a natural direction for future work, discussed in Chapter 6.

4.5 Prompt Design

The prompt used to elicit predictions from a large language model is not a neutral component of an evaluation pipeline: it is a design choice that directly affects the model's outputs. Any calibration audit that fixes a single prompt inevitably conflates two effects — the properties of the model and the properties of the prompt — which cannot be separated post hoc. To address this ambiguity, we evaluate each of the six models under two distinct prompt variants, treating the prompt as an experimental variable rather than a fixed convention.

4.5.1 Rationale for Two Prompt Variants

The two prompts are designed to represent the two extremes of a spectrum of elicitation strategies commonly encountered in practice. The first (Prompt A) is intentionally minimal: it presents the classification task and the label vocabulary without any explanation, in the manner of a naive user querying the model. The second (Prompt B) is enriched with definitions, examples, and explicit reasoning rules, in the manner of an expert operator who invests time in prompt engineering. By evaluating each model with both prompts, we obtain four independent measurement points per model family for hypothesis H2 (Base–Instruct comparison at Prompt A, Base–Instruct comparison at Prompt B), doubling the statistical power of the RLHF test.

This design also enables three additional analyses beyond the pre-registered hypotheses. First, the effect of prompt design itself becomes measurable within each model. Second, the interaction between RLHF and prompt design becomes testable: some models may be more sensitive to prompt enrichment than others. Third, the design places our results in a stronger position with respect to external validity: a finding replicated under two different prompts is more robust than a finding measured under a single one.

4.5.2 Prompt A: Generic Baseline

Prompt A is a minimalist prompt containing only the classification instruction, the input note, the criterion text, and the list of candidate labels. It does not define the labels, does not provide examples, and does not include reasoning rules. Its purpose is to establish a baseline that corresponds to how a first-time user would query the model, and to provide a reference point against which enriched prompts can be compared.

The complete text of Prompt A is given in Appendix B. Its essential structure is:

> *Classify whether this patient meets the eligibility criterion.*
> *Patient: [note]*
> *Criterion ([type]): [criterion]*
> *Choose exactly one label from: included, not included, excluded, not excluded, not enough information, not applicable.*
> *Answer:*

Prompt A is applied identically to all six models. For instruction-tuned models, the prompt is wrapped in the model's native chat template. For base models, the prompt is presented as a plain-text continuation task without special tokens.

4.5.3 Prompt B: Adapted with Expert Guidance

Prompt B is an adapted prompt containing definitions of the six labels, one worked example per label, and a critical inference rule for exclusion criteria that addresses a subtle failure mode observed in preliminary experiments. The full text is reproduced in Appendix B.

Three design choices distinguish Prompt B from Prompt A. First, each of the six labels is defined with an explicit semantic description. For example, "not included" is characterized as "The patient note contains information showing the patient does NOT meet this inclusion criterion," which distinguishes it from "not enough information" (where the note is silent). Second, a one-sentence example is provided for each label, drawn from realistic clinical situations. Third, a specific inference rule is stated for exclusion criteria:

> *If the criterion names a specific disease, condition, or medical history, and the note does not mention it, the default is "not excluded" (the patient most likely does not have it). Reserve "not enough information" only for criteria requiring a specific lab value or measurement that is truly missing.*

This rule directly targets the tendency of language models to produce "not enough information" for any exclusion criterion whose target condition is not explicitly discussed in the patient note — a pattern that mechanically converts the absence of a mention into an expression of epistemic uncertainty, when clinically it should be treated as evidence of absence.

4.5.4 Adaptations for Base vs. Instruct Models

Base models and instruction-tuned models expect prompts in different formats, and applying the same prompt verbatim to both would create an artificial handicap for the base variant. To ensure a fair comparison, we adapt the surface form of each prompt to the target model class while preserving its semantic content.

For instruction-tuned models, prompts are formatted using the model's native chat template, which delimits user and assistant roles with model-specific special tokens (`[INST]` for Mistral, `<|start_header_id|>` for Llama-3). This is the format for which these models were aligned and where they behave as expected.

For base models, no chat template is applied. Instead, the prompt includes one or more few-shot examples that establish the desired input-output pattern by demonstration. Prompt A for base models includes a single example; Prompt B includes three examples covering distinct label types. This asymmetry reflects a genuine limitation of base models — they do not natively follow instructions — and treating it as an evaluation artifact would misrepresent how these models are actually used in practice. The introduction of few-shot examples is nevertheless a potential confounder for cross-family comparisons, and we discuss it explicitly in Chapter 6.

The complete text of all four prompt variants (Prompt A / Prompt B, Instruct / Base) is provided in Appendix B to support independent reproduction of the pipeline.

4.6 Scoring Pipeline: Log-Probability Extraction

Standard practice for querying a large language model on a classification task is to sample a textual output and post-process it to recover a label. This approach has two limitations for a calibration audit. First, the sampled output is a point estimate: it discards the underlying probability distribution over candidate labels, which is precisely the object of interest for calibration analysis. Second, generation is stochastic, introducing an additional source of variability that would complicate the reproducibility of downstream statistics. We therefore adopt a log-probability scoring approach that directly reads out the model's probability distribution over the candidate labels, without any generation step.

4.6.1 Principle of Log-Probability Scoring

For a given patient–criterion pair, we first construct the prompt as described in Section 4.5. Rather than sampling a continuation, we consider each candidate label as a hypothetical completion of the prompt. For each label ℓ, we compute the mean log-probability of its token sequence conditional on the prompt:

$$s_\ell = \frac{1}{|\ell|} \sum_{t=1}^{|\ell|} \log p(\ell_t \mid \text{prompt}, \ell_{<t}) \tag{4.1}$$

where ℓ_t is the t-th token of the label and $|\ell|$ is the total number of tokens. Length normalization by $|\ell|$ is necessary because labels have unequal token lengths in the tokenizer: without it, longer labels such as "not enough information" would be systematically penalized against shorter ones such as "included".

The resulting scores $\{s_\ell\}$ are then normalized through a softmax to obtain a proper probability distribution π over the candidate labels:

$$\pi_\ell = \frac{\exp(s_\ell)}{\sum_{\ell' \in \mathcal{L}} \exp(s_{\ell'})} \tag{4.2}$$

The predicted label $\hat{y}$ is the argmax of π, and the associated confidence $\hat{p}$ is the maximum probability. Crucially, the full distribution π is retained for downstream analyses, including recalibration, conformal prediction, and the simulation of constrained label spaces (Section 5.6).

4.6.2 Restriction of the Candidate Set by Criterion Type

The six labels described in Section 4.3 split into two groups by criterion type. For inclusion criteria, the semantically valid labels are *included*, *not included*, *not enough information*, and *not applicable*. For exclusion criteria, they are *excluded*, *not excluded*, *not enough information*, and *not applicable*. In both cases, four labels are candidates and two would be semantically nonsensical (e.g., predicting "included" for an exclusion criterion). We therefore restrict the softmax in Equation 4.2 to the four candidates appropriate to the criterion type at hand. This restriction preserves the ordinal structure of the task and prevents the model from being penalized for probability mass placed on labels that could not have been correct in the first place.

4.6.3 Algorithmic Summary

Algorithm 1 summarizes the full scoring procedure applied to each patient–criterion pair.

Because no sampling is involved, the procedure is fully deterministic: running the same model on the same input twice yields identical scores. This determinism is important for the reproducibility of the entire downstream analysis, including the bootstrap confidence intervals and permutation tests described in Section 4.9.

4.6.4 Implementation Notes

The scoring loop is implemented using the standard causal language modeling interface of the HuggingFace `transformers` library. For a given prompt, we tokenize the con-

Algorithm 1 Log-probability scoring for eligibility label prediction

Require: Patient note n, criterion c, criterion type t
Ensure: Predicted label $\hat{y}$, confidence $\hat{p}$, distribution π

```
1: Build prompt from n, c, t
2: L ← candidate labels according to t
3: for all ℓ ∈ L do
4:     s_ℓ ← mean log-probability of tokens of ℓ
5: end for
6: π ← softmax(s)
7: ŷ ← arg max_ℓ π_ℓ
8: p̂ ← π_ŷ
9: return ŷ, p̂, π
```

catenation of the prompt and each candidate label separately, run a single forward pass per concatenation, and read out the token log-probabilities from the model's logits after applying a log-softmax over the vocabulary. This approach requires four forward passes per patient–criterion pair (one per candidate label), which is approximately twice as expensive as free-form generation of a short answer. In practice, on a T4 GPU with 4-bit quantization, the four passes complete in approximately three seconds per pair for a 7-billion-parameter model, and the full evaluation of 1,015 pairs takes fifty to seventy minutes depending on the model.

4.7 Quantization and Infrastructure

Running 7-to-9-billion-parameter models on a free-tier T4 GPU (16 GB of memory) requires reducing the memory footprint of the model weights. We apply 4-bit NF4 quantization [40] through the `bitsandbytes` library, which represents each weight as a 4-bit value drawn from a distribution matched to the empirical distribution of neural network weights. This reduces the memory required to store a 7-billion-parameter model from approximately 14 GB (in half-precision) to approximately 4 GB, leaving sufficient headroom for the key-value cache during inference.

Prior work has shown that 4-bit NF4 quantization introduces negligible degradation on downstream evaluation benchmarks compared to full precision [40], including on tasks that involve long-form generation or fine-grained probability estimation. We rely on this evidence to justify the use of NF4 quantization for our calibration audit, while acknowledging that a small residual effect of quantization on absolute ECE values cannot be entirely ruled out. Since Hypothesis H2 and the associated Bonferroni-corrected permutation tests compare *differences* in ECE between paired variants evaluated under the same quantization regime, any systematic bias introduced by quantization cancels out and does not affect the conclusions.

All experiments were run on Kaggle notebooks with a single Tesla T4 GPU. The compute quota (30 GPU-hours per week on the free tier) was sufficient to complete the twelve runs and all downstream analyses within the experimental phase of the project.

4.8 Evaluation Metrics

The evaluation combines calibration metrics, classification metrics, ordinal validity metrics, and behavioral metrics. Each captures a distinct aspect of probabilistic reliability, and no single metric is sufficient on its own. This section defines every metric used in Chapter 5.

4.8.1 Expected Calibration Error (ECE)

The Expected Calibration Error, introduced by Naeini et al. and popularized by Guo et al. [21], measures the average absolute gap between predicted confidence and observed accuracy across confidence bins:

$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{n} \left|\mathrm{acc}(B_k) - \mathrm{conf}(B_k)\right| \tag{4.3}$$

where predictions are partitioned into K equal-width bins $B_1, \ldots, B_K$ based on their confidence, n is the total number of predictions, $\mathrm{acc}(B_k)$ is the empirical accuracy of predictions in bin B_k, and $\mathrm{conf}(B_k)$ is the average confidence of predictions in the same bin. A perfectly calibrated model satisfies $\mathrm{acc}(B_k) = \mathrm{conf}(B_k)$ for all bins, yielding an ECE of zero. We use $K = 10$ equal-width bins on $[0, 1]$.

4.8.2 Brier Score

The Brier score [24] measures the mean squared error between predicted probability and the true binary outcome (correct or incorrect):

$$\mathrm{Brier} = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i)^2 \tag{4.4}$$

where p_i is the predicted confidence and $y_i \in \{0, 1\}$ indicates whether the prediction was correct. Unlike ECE, which measures calibration alone, the Brier score jointly reflects calibration and sharpness: a model that is both accurate and confident achieves a lower Brier score than one that is only well calibrated but uninformative.

4.8.3 Classification Metrics

We report accuracy as the proportion of predictions matching the expert label, and macro-F1 as the unweighted mean of per-class F1 scores. Macro-F1 is critical in the presence of the severe class imbalance documented in Section 4.3.2: a model that always predicts the majority class achieves an accuracy of 47.7% but a macro-F1 close to zero on the minority classes. This distinction becomes central in Chapter 5 when comparing model behavior across cases of varying clinical difficulty.

4.8.4 C-index for Ordinal Validity

Because the eligibility labels form an ordinal scale, we complement calibration and classification metrics with the concordance index (C-index), which measures the probability that, for a randomly chosen pair of cases with different expert labels, the model correctly ranks them:

$$\text{C-index} = \frac{|\{(i,j) : y_i < y_j \text{ and } s_i < s_j\}|}{|\{(i,j) : y_i \neq y_j\}|} \tag{4.5}$$

where y_i, y_j are ordinal expert labels of cases i and j, and s_i, s_j are the model's scalar scores obtained by mapping the predicted label and confidence onto the ordinal scale. A C-index of 1 indicates perfect ranking, 0.5 indicates random ranking, and the conventional threshold of 0.70 is used in clinical prediction modeling as evidence of acceptable discrimination [3].

4.8.5 Weighted Kappa for Expert-Model Agreement

To measure the agreement between physician annotations and GPT-4 annotations in the pre-LLM audit (Section 5.2), we use Cohen's weighted kappa with quadratic weights [41]. Weighted kappa extends the standard kappa coefficient by penalizing disagreements proportionally to their ordinal distance: a confusion between adjacent labels (e.g., “included” and “not enough information”) counts less than a confusion between distant labels (e.g., “included” and “excluded”). We report values in the range $[-1, 1]$, with 1 indicating perfect agreement, 0 indicating chance agreement, and negative values indicating systematic disagreement.

4.8.6 Overconfidence and Prediction Diversity

Beyond calibration and classification, two behavioral metrics quantify how a model uses its probability distribution.

The mean overconfidence is defined as the difference between the average confidence and the average accuracy:

$$\text{overconfidence} = \bar{p} - \text{acc} \tag{4.6}$$

where $\bar{p}$ is the mean predicted confidence across the evaluation set. A positive value indicates that the model tends to overstate its certainty, a negative value that it understates it. Unlike ECE, which is a magnitude, overconfidence is a signed quantity that reveals the direction of the miscalibration.

Prediction diversity is measured as the normalized Shannon entropy of the empirical label distribution:

$$\text{diversity} = \frac{-\sum_{\ell \in \mathcal{L}} f_\ell \log_2 f_\ell}{\log_2 |\mathcal{L}|} \tag{4.7}$$

where f_ℓ is the empirical frequency of label ℓ in the model's predictions and $\mathcal{L}$ is the label vocabulary. A diversity of 0 indicates that the model predicts a single label; a diversity of 1 indicates a uniform distribution over all labels. This metric is central to detecting the mode collapse behavior documented in Section 5.6, which the calibration and classification metrics can mask.

4.8.7 Signal Expression and the Signal-to-Prior Ratio

The calibration and diversity metrics above characterize a model's predicted-label distribution, but they cannot distinguish a model that lacks discriminative knowledge from one that holds it yet fails to express it in its $\arg\max$. To make this distinction, we define three quantities computed from the log-probability of the positive label of each criterion type (p_{included} for inclusion criteria, p_{excluded} for exclusion criteria).

The first is the ordinal separation, measured by Cliff's δ between the positive-label probability on gold-positive and gold-negative cases. Cliff's $\delta \in [-1, 1]$ is a non-parametric effect size equal to $2A - 1$, where A is the probability that a random gold-positive case receives a higher positive-label probability than a random gold-negative case; it is robust to the class imbalance of the dataset and requires no distributional assumption. A high δ combined with a degenerate predicted-label distribution is the signature of a model whose signal is present but unexpressed.

The second is the *signal amplitude* σ_{pos}, defined as the standard deviation across pairs of the positive-label log-probability. Measured in log-probability space, it is not bounded by the probability simplex and therefore captures the amplitude with which the model varies its output in response to the input.

The third is the *response prior* $\mathrm{bias}_{\mathrm{ext}}$, defined as the mean gap between the strongest competing label and the positive label, computed over the competing labels only (excluding the positive label itself) to avoid any circularity with the quantity it is meant to predict. Their ratio,

$$r = \frac{\sigma_{\mathrm{pos}}}{|\mathrm{bias}_{\mathrm{ext}}|}, \tag{4.8}$$

is the signal-to-prior ratio used in Section 5.6.4 to characterize the collapse regime of each configuration.

Finally, we report the area under the ROC curve (AUC) of the positive-label probability against the binarized gold label. Because the AUC is invariant to the decision threshold, it measures the discriminative information carried by the probabilities independently of whether the arg max exploits it, and is the quantity that separates a recoverable collapse from a corruption of the signal. All four quantities are reported with 1,000-resample bootstrap confidence intervals (Section 4.8.10).

4.8.8 Composite Utility Score

To provide a single rank-ordering of the twelve configurations that reflects clinical usability, we propose a composite utility score combining calibration quality, discrimination, and prediction diversity:

$$\mathrm{utility} = F_1^{\mathrm{macro}} \times (1 - \mathrm{ECE}) \times \mathrm{diversity} \tag{4.9}$$

The three factors are multiplicative rather than additive so that a failure on any single dimension zeroes out the score. A model with excellent calibration but complete mode collapse (diversity = 0) receives a utility score of zero, correctly capturing its lack of practical value. This composite score complements rather than replaces the individual metrics, which remain the primary object of analysis.

4.8.9 Reliability Diagrams

Reliability diagrams complement ECE by visualizing the calibration curve directly. For each of the K bins used in the ECE computation, we plot the average confidence on the horizontal axis and the observed accuracy on the vertical axis. A perfectly calibrated model produces points that lie on the diagonal $y = x$. Points below the diagonal indicate overconfidence; points above indicate underconfidence.

4.8.10 Bootstrap Confidence Intervals

For every point estimate of ECE, Brier score, and C-index, we report a 95% confidence interval obtained by 1,000 bootstrap resamples of the prediction set. Bootstrap confidence intervals do not assume any specific distributional form and are appropriate for the non-Gaussian statistics under study. They also provide a natural way to assess whether a given model's ECE genuinely exceeds the 0.05 threshold of Hypothesis H1, or whether the excess is within sampling noise.

4.9 Statistical Analysis Plan

The four pre-registered hypotheses described in Section 4.2 are tested according to a fixed analysis plan. This section describes the tests, the correction for multiple comparisons, and the additional analyses (post-hoc recalibration and conformal prediction) that complement the primary hypotheses.

4.9.1 Pre-registration of Hypotheses

All four rejection criteria (H1: ECE > 0.05; H2: ECE(Instruct) $>$ ECE(Base); H3: C-index > 0.70; H4: $F_1(80\%) > F_1(100\%)+0.02$) were fixed before the twelve model runs were executed. This pre-registration protects the analysis against post-hoc adjustment of the criteria to fit the observed results and follows the recommendations of the TRIPOD+AI reporting guidelines [4]. Results are reported without modification of the rejection criteria, and any findings not covered by these four hypotheses are reported separately as original observations (see Chapter 5).

4.9.2 Permutation Tests

Hypothesis H2 compares the ECE of paired Base and Instruct variants of the same model family under the same prompt. Rather than assume any particular distributional form for the ECE difference, we use a permutation test: for 1,000 random permutations of the model labels (Base vs. Instruct) within each pair, we recompute the ECE difference and count the fraction of permutations in which the observed difference is equalled or exceeded. This fraction is the one-sided p-value for the null hypothesis that the two variants have the same underlying calibration. The one-sided formulation is appropriate because Hypothesis H2 is directional: we test whether Instruct is worse than Base, not merely different from it.

4.9.3 Bonferroni Correction

Hypothesis H2 is tested four times (Mistral–Prompt A, Mistral–Prompt B, Llama–Prompt A, Llama–Prompt B) and further comparisons for medical specialization contribute additional tests. To control the family-wise error rate across the eight planned comparisons, we apply a Bonferroni correction with $\alpha_{\text{corrected}} = 0.05/8 = 0.00625$. This correction is conservative but appropriate given the small number of comparisons and the desire for a stringent test of the central hypothesis.

4.9.4 Post-hoc Recalibration Methods

Beyond the primary calibration audit, we evaluate three post-hoc recalibration methods that could correct miscalibration without retraining the model. Temperature scaling [21] applies a single learned scalar T to the logits before the softmax, effectively softening (or sharpening) the confidence distribution. Isotonic regression fits a monotone piecewise-constant function mapping uncorrected confidences to corrected ones; being non-parametric, it can correct arbitrary shapes of miscalibration but requires more data than temperature scaling. Platt scaling fits a logistic function of the uncorrected confidence, offering a compromise between the flexibility of isotonic regression and the parsimony of temperature scaling.

Each method is fitted on the 70% training subset and evaluated on the 30% held-out subset (see Section 4.3.4). To assess whether the recalibration gains are stable across different data splits, we additionally perform a 5-fold cross-validation, reporting the mean and standard deviation of the ECE reduction over the five folds.

4.9.5 Re-thresholding and the Collapse Diagnostic

Post-hoc recalibration applies a monotone transformation to the confidences and therefore preserves the $\arg\max$ of every prediction. To evaluate whether the discriminative signal masked by a collapsed $\arg\max$ can be recovered, we complement recalibration with a non-monotone decision rule: instead of predicting the highest-probability label, we place a threshold t on the positive-label probability of each criterion type and predict the positive class when the probability exceeds t.

To avoid tuning the threshold on the data used to evaluate it, t is fitted by 5-fold cross-validation: within each fold, t is chosen to maximize macro-F1 on the four training folds and applied to the held-out fold, so that every evaluated prediction is produced by a threshold that never observed it. The macro-F1 gain of the threshold rule over the $\arg\max$ rule is reported with a 1,000-resample bootstrap confidence interval on the out-of-fold predictions; a gain whose interval excludes zero is deemed significant. The non-collapsed configurations serve as a negative control: a valid recovery procedure should yield no gain where the $\arg\max$ already exploits the signal.

Because a threshold rule mechanically raises macro-F1 on an imbalanced label set even in the absence of genuine discrimination, the macro-F1 gain is not interpreted in isolation. Each configuration is instead placed on a two-dimensional diagnostic that crosses the signal-to-prior ratio r (Section 4.8.7), which measures whether the model expresses its decision, with the positive-label AUC and its bootstrap interval, which measures whether a genuine signal is present. The two axes partition the configurations into healthy models, recoverable expressive collapse, and irrecoverable signal corruption; the partition and its use are reported in Section 5.12.2.

4.9.6 Conformal Prediction

To complement point-estimate calibration, we apply split conformal prediction [30], a distribution-free framework that produces prediction sets with a guaranteed marginal coverage rate. Given a target coverage level $1 - \alpha$ (we use $\alpha = 0.10$, i.e., 90% coverage), the framework calibrates a threshold on the non-conformity scores of a held-out calibration set and returns, for each test case, the set of labels whose non-conformity is below that threshold. The size of the prediction set is an informative quantity: large sets indicate high aleatoric uncertainty, while small sets indicate high model confidence. We report both empirical coverage (which should match the target 90%) and average set size.

4.10 Reproducibility

Reproducibility is treated as a first-class methodological requirement throughout this study, following the recommendations of the TRIPOD+AI guidelines [4] and the broader movement toward computational transparency in machine learning research. This section documents the specific measures that support independent replication of the results reported in Chapter 5.

4.10.1 Code Availability

All code produced during this study is archived in a public repository listed in Appendix D. The repository is organized into three components. First, a single Kaggle notebook contains the twelve model runs, from data loading through prediction generation, with inline documentation of every design choice. Second, a set of Python scripts performs the downstream analyses (calibration audit, hypothesis tests, recalibration, conformal prediction, mode collapse analysis, and composite utility scoring) from the archived prediction files. Third, the notebook and scripts share a single `requirements.txt` file that pins the exact versions of every dependency, together with the CUDA version required for GPU acceleration.

4.10.2 Deterministic Pipeline

The full pipeline is designed to be deterministic. The 4-bit NF4 quantization applied to the model weights is itself deterministic: loading the same model with the same quantization configuration produces identical weight matrices across runs. The log-probability scoring procedure described in Section 4.6 involves no sampling and reads out probabilities directly from the model's forward pass, which is also deterministic given fixed weights and inputs. All random operations used in the downstream analyses — the bootstrap resamples for confidence intervals, the permutation tests for Hypothesis H2, the 70/30 splits for recalibration, and the 5-fold cross-validation — use a fixed random seed (42), ensuring that statistical results are bit-exact reproducible across independent runs of the analysis scripts.

4.10.3 Data and Prediction Archives

Two categories of data files are preserved in the project repository. The first category comprises the raw prediction files: for each of the twelve configurations, a Parquet file records the softmax distribution over candidate labels, the predicted label, the associated confidence, the expert label, the criterion type, and metadata linking each prediction to the original patient–criterion pair. These twelve files allow any downstream analysis to be reproduced without re-running the model inference, which is the most computationally expensive step. The second category comprises the analysis outputs: CSV files containing the calibration metrics, permutation test results, recalibration comparisons, conformal prediction sets, mode collapse statistics, and composite utility scores. Each numerical result reported in Chapter 5 can be traced to a specific row in one of these files.

4.10.4 Prompt Verbatim

Because prompt engineering constitutes an experimental variable in this study (Section 4.5), the exact text of all four prompt variants — Prompt A and Prompt B, in both Instruct and Base formulations — is reproduced verbatim in Appendix B. This includes the chat template markers used for instruction-tuned models and the few-shot examples embedded in the base-model prompts. Any deviation from these verbatim prompts constitutes a distinct experimental condition and cannot be considered a replication of the present results.

4.10.5 Computational Requirements

Independent reproduction of the full pipeline requires access to a GPU with at least 16 GB of memory (a Tesla T4 or equivalent) and approximately ten to fifteen hours of GPU time to complete the twelve model runs. The downstream analyses run on CPU in a few minutes on standard hardware. All experiments reported in this thesis were performed on the Kaggle free-tier compute environment, which is publicly accessible and imposes no barriers to independent replication. The complete list of hardware and software versions, together with the commit hashes of the notebook and analysis scripts at the time of submission, is provided in Appendix D.

4.10.6 Reporting Standards

The reporting in Chapter 5 follows the structure recommended by TRIPOD+AI for prediction model studies: hypotheses are stated before results, the four pre-registered hypotheses are reported alongside their pre-specified rejection thresholds, statistical tests are reported with their exact p-values and their Bonferroni-corrected significance thresholds, and all point estimates are accompanied by 95% bootstrap confidence intervals. Findings not covered by the pre-registered hypotheses are clearly labelled as original observations to distinguish confirmatory from exploratory results. This structured reporting is intended to make the strength of the evidence transparent to readers without requiring them to re-analyze the underlying data.

Chapter 5

Results and Analysis

5.1 Chapter Overview

This chapter reports the empirical findings of the study. The presentation is organized to distinguish confirmatory results (tests of the four pre-registered hypotheses stated in Section 4.2) from exploratory findings (novel observations that emerged during the analysis and are labelled here as original findings F1 through F9).

Sections 5.2 through 5.4 present the pre-LLM audit and the two pre-registered hypotheses that receive the strongest empirical support: H1 (systematic miscalibration, supported across all twelve configurations) and H2 (RLHF degrades calibration, supported by four independent permutation tests after Bonferroni correction). H2 is the central pre-registered contribution of this work.

Sections 5.5 through 5.7 report the three most novel findings, none of which was anticipated in the pre-registration. The enriched-prompt paradox (Section 5.5) shows that expert prompt engineering degrades calibration in five of six models. Mode collapse (Section 5.6) documents that five of the twelve configurations concentrate more than 97% of their predictions on a single label, invalidating the standard interpretation of their apparently reasonable ECE values. A deeper analysis of the underlying log-probabilities then shows that this collapse is a property of the decision rule rather than of the model's knowledge: the discriminative signal survives collapse and can be recovered by a non-monotone decision rule (Section 5.12). Performance on truly discriminative cases (Section 5.7) reveals that both medical LLMs tested achieve near-zero accuracy on the 238 cases requiring a genuine clinical decision, despite their apparent superiority on global accuracy.

Sections 5.8 through 5.10 report secondary findings on the inclusion/exclusion asymmetry, model-specific evasion strategies, and the two pre-registered hypotheses that receive partial or no support (H3 on ordinal validity, H4 on selective classification). Sections 5.11 through 5.14 present the post-hoc solutions — isotonic recalibration, re-thresholding, and conformal prediction — and the composite utility score that provides a single rank-ordering of the twelve configurations. Section 5.15 reports an external replication of the collapse and its mechanism on the TREC Clinical Trials corpus. Section 5.16 synthesizes the outcomes of the four hypotheses and the twelve original findings.

Every point estimate reported in this chapter is accompanied by a 95% bootstrap confidence interval, and every hypothesis test is reported with its Bonferroni-corrected significance threshold. Detailed tables, per-configuration reliability diagrams, and the full mode collapse analysis are provided in Appendices A, C, and F.

5.2 Pre-LLM Audit and Expert Agreement

Before evaluating the twelve LLM configurations, we conduct a pre-LLM audit of the TrialGPT-Criterion-Annotations dataset itself. This audit serves two purposes: to establish the quality of the ground truth against which the LLMs will be compared, and to expose a systematic annotation bias in GPT-4 that has not been previously documented.

5.2.1 Expert-GPT-4 Agreement

The dataset provides two independent annotations per pair: a consensus label from three physicians (which we treat as the ground truth throughout this thesis) and a label produced by GPT-4. The weighted kappa [41] between the two annotations, computed with quadratic weights on the ordinal scale, is $\kappa_w = 0.819$ (95% bootstrap CI: [0.774, 0.86]). By conventional criteria for inter-rater agreement, this value falls in the range of substantial to almost perfect agreement, confirming that GPT-4 reproduces the physician consensus with high fidelity at the aggregate level.

This aggregate agreement, however, masks a systematic bias visible only at the label level. Table 5.1 reports, for each of the six labels, the ratio of GPT-4 usage rate to expert usage rate. A ratio of 1.0 indicates that GPT-4 and experts use the label with the same frequency; a ratio above 1.0 indicates over-use by GPT-4; a ratio below 1.0 indicates under-use.

Table 5.1: Usage-rate ratios of the six labels (GPT-4 vs. physician consensus). A ratio of 1.0 indicates balanced usage; ratios far from 1.0 reveal systematic annotation bias.

Label	Expert freq.	GPT-4 freq.	Ratio
not applicable	5.4%	11.7%	2.16
not enough information	28.9%	28.6%	0.99
included	7.3%	6.9%	0.95
not included	5.3%	5.0%	0.94
excluded	5.4%	5.5%	1.02
not excluded	47.7%	42.3%	0.89

Five of the six labels are used by GPT-4 within 10% of the physician frequency, in line with the substantial overall kappa. The exception is “not applicable”, which GPT-4 uses at 2.16 times the physician rate. This asymmetric bias — systematic over-use of one specific evasion label rather than a general tendency to over-use uncertainty — is not reported in the original TrialGPT paper [1], where GPT-4 is used as an annotation tool without such an audit. The bias is substantively meaningful: “not applicable” functions as a legitimate escape route when a criterion genuinely does not apply to a patient’s situation (e.g., a pregnancy exclusion criterion applied to a male patient), but its over-use by GPT-4 suggests that the model treats a broader class of ambiguous cases as “not applicable” when a decisive label would have been warranted.

Finding 1 (F1: GPT-4’s Selective Evasion Bias)**.** *GPT-4 over-uses the “not applicable” label at 2.16 times the physician rate, while using “not enough information” at the same rate as the experts (ratio 0.99). This asymmetric pattern reveals a selective evasion strategy that is not detected by aggregate agreement metrics.*

This finding is meaningful for the downstream analysis of open-source LLMs in Section 5.9, where we show that open-source models exhibit a different but equally systematic evasion pattern — over-use of “not enough information” rather than “not applicable”. The two evasion channels appear to be model-specific, likely reflecting distinct alignment strategies during post-training.

5.2.2 Ground Truth Choice

Given the systematic bias identified in F1, we use the physician consensus rather than GPT-4 annotations as the ground truth throughout this thesis. This choice has three consequences. First, GPT-4 is not a model under evaluation in the calibration audit itself: its labels serve only as the input for F1. Second, the accuracy figures reported in subsequent sections are computed against expert labels, which increases their clinical relevance. Third, the class imbalance documented in Section 4.3.2 refers to the expert distribution and is preserved unchanged for the calibration analysis.

5.3 Calibration Audit: H1 Supported Across All Configurations

Hypothesis H1 predicts that the Expected Calibration Error of an LLM-based eligibility scoring pipeline exceeds the well-calibrated threshold of 0.05 [3]. We test this hypothesis across all twelve configurations of the two-by-two matrix defined in Section 4.4, using the metrics and bootstrap protocol described in Section 4.8.

5.3.1 Overall Calibration Results

Table 5.2 reports the ECE, its 95% bootstrap confidence interval, and the Brier score for each of the twelve configurations. All twelve values fall well above the calibration threshold, with lower bounds of the 95% intervals also strictly greater than 0.05. Hypothesis H1 is therefore supported at the level of every individual configuration, not merely in aggregate.

Table 5.2: Calibration metrics across the twelve configurations. All ECE values exceed the 0.05 threshold, with the lower bound of the 95% bootstrap confidence interval also strictly above 0.05 in every case. Configurations are grouped by model family.

Configuration	**ECE**	**95% CI**	**Brier**
Mistral-7B family			
Mistral-Base A	0.133	[0.104, 0.163]	0.258
Mistral-Base B	0.189	[0.161, 0.218]	0.269
Mistral-Instruct A	0.322	[0.297, 0.350]	0.311
Mistral-Instruct B	0.253	[0.229, 0.287]	0.353
Llama-3.1-8B family			
Llama-Base A	0.124	[0.100, 0.148]	0.187
Llama-Base B	0.151	[0.123, 0.177]	0.225
Llama-Instruct A	0.510	[0.483, 0.535]	0.456
Llama-Instruct B	0.602	[0.575, 0.628]	0.558
Medical LLMs			
Meditron A	0.182	[0.153, 0.212]	0.263
Meditron B	0.231	[0.204, 0.259]	0.283
BioMistral A	0.155	[0.126, 0.182]	0.234
BioMistral B	0.194	[0.166, 0.222]	0.244

The ECE values span a wide range, from 0.124 (Llama-Base A) to 0.602 (Llama-Instruct B). Even the best-calibrated configuration is more than twice the clinical threshold, and the worst is more than twelve times above it. The Brier score follows a similar ordering, confirming that the miscalibration is not compensated by favorable sharpness properties.

H1: Systematic Miscalibration Confirmed. All twelve configurations exhibit an ECE strictly greater than 0.05, with the lower bounds of their 95% bootstrap confidence intervals also strictly above the threshold. Hypothesis H1 is supported across the entire experimental matrix.

5.3.2 Direction of the Miscalibration

ECE is a signed-magnitude metric: it measures how far confidence deviates from accuracy without indicating the direction of the deviation. To characterize the miscalibration more precisely, Table 5.3 reports the mean overconfidence for each configuration, defined as the difference between mean confidence and mean accuracy (Equation 4.6). A positive value indicates that the model overstates its certainty on average; a negative value indicates understatement.

Table 5.3: Mean overconfidence per configuration. Positive values indicate that the model's mean confidence exceeds its mean accuracy (overconfident); negative values indicate the opposite (underconfident).

Configuration	Overconfidence
Mistral-Base A	−0.189
Mistral-Base B	−0.132
Mistral-Instruct A	+0.248
Mistral-Instruct B	+0.184
Llama-Base A	+0.128
Llama-Base B	+0.021
Llama-Instruct A	+0.487
Llama-Instruct B	+0.598
Meditron A	−0.036
Meditron B	+0.098
BioMistral A	+0.144
BioMistral B	+0.187

The direction of miscalibration reveals a striking pattern: the Base variants of Mistral are systematically underconfident (−0.189 and −0.132), while their Instruct counterparts are systematically overconfident (+0.248 and +0.184). A similar transition occurs in the Llama family: Llama-Base is nearly calibrated on Prompt B (+0.021) but Llama-Instruct is severely overconfident (+0.487 and +0.598). This pattern anticipates the analysis of Section 5.4, where we show that this transition from under-confidence to over-confidence is a direct effect of RLHF post-training.

5.3.3 Reliability Diagrams

Figure 5.1 shows the reliability diagrams for the twelve configurations, organized by model family and prompt variant. The diagonal line represents perfect calibration; points below the diagonal indicate overconfidence, points above indicate underconfidence.

Two visual patterns emerge from the reliability diagrams. First, the Mistral-Base and Meditron curves lie systematically above the diagonal in the low-confidence regions, indicating that these models are more accurate than they claim. Second, the Llama-Instruct and Mistral-Instruct curves lie systematically below the diagonal in the high-confidence regions, indicating that these models are less accurate than they claim. Per-configuration reliability diagrams are provided in Appendix C for detailed inspection.

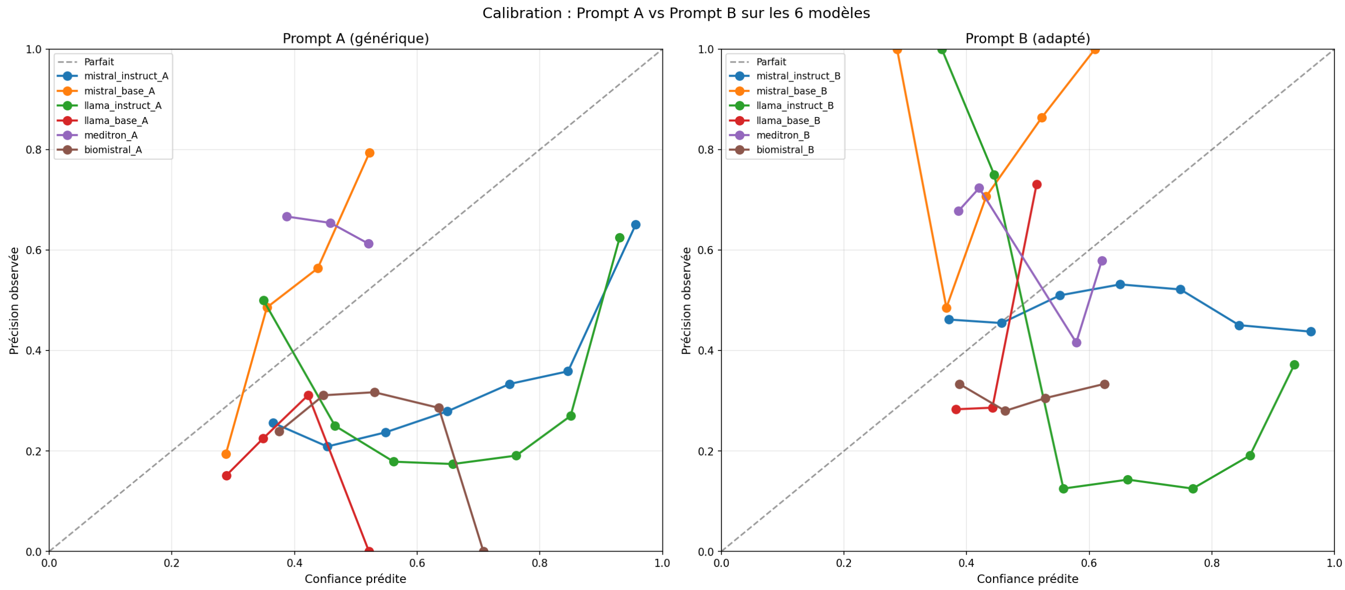

Figure 5.1: Reliability diagrams of the twelve configurations, grouped by prompt variant (Prompt A on the left, Prompt B on the right), with each curve corresponding to one of the six models. Each curve plots the observed accuracy (labelled *Précision observée*) as a function of predicted confidence (*Confiance prédite*) in ten equal-width bins; the dashed reference line labelled *Parfait* corresponds to perfect calibration. Deviations from the diagonal $y = x$ indicate miscalibration; the direction of the deviation is consistent with the mean overconfidence reported in Table 5.3. Axis labels and the legend are kept in French, reflecting the working language of the experimental phase.

5.4 The Effect of RLHF: H2 Supported

Hypothesis H2 predicts that within a model family, the RLHF-aligned Instruct variant exhibits worse calibration than its Base counterpart. This is the central pre-registered contribution of this thesis: an extension of the observation of Kadavath et al. [6], who documented a similar effect on general question-answering tasks, to the specific domain of clinical trial matching. The comparison is performed within each of the two generalist families (Mistral-7B and Llama-3.1-8B), for each of the two prompt variants, yielding four independent tests of the same underlying hypothesis.

5.4.1 Paired Comparison of ECE

Table 5.4 reports the paired ECE values for the four Base–Instruct comparisons. Under Prompt A, both families show a substantial increase in ECE when moving from Base to Instruct: from 0.133 to 0.322 for Mistral, and from 0.124 to 0.510 for Llama. Under Prompt B, the increase is smaller for Mistral (from 0.189 to 0.253) but larger for Llama (from 0.151 to 0.602). The direction of the effect is consistent across all four comparisons.

The four p-values are all below the Bonferroni-corrected threshold of $\alpha_{\text{corr}} = 0.05/8 =$

Table 5.4: Paired ECE comparison for the four Base–Instruct tests of Hypothesis H2. ΔECE is the difference ECE(Instruct) – ECE(Base); a positive value supports H2. Permutation p-values are computed with 1,000 resamples on the paired data; the Bonferroni-corrected significance threshold across the eight planned comparisons is $\alpha_{\text{corr}} = 0.00625$.

Comparison	**ECE Base**	**ECE Instruct**	**ΔECE**	**p-value**
Mistral / Prompt A	0.133	0.322	+0.189	<**0.001**
Mistral / Prompt B	0.189	0.253	+0.064	**0.002**
Llama / Prompt A	0.124	0.510	+0.386	<**0.001**
Llama / Prompt B	0.151	0.602	+0.451	<**0.001**

0.00625, which accounts for the eight planned comparisons in the pre-registered analysis plan (Section 4.9.3). Even under this stringent correction, all four tests reject the null hypothesis of equal calibration between Base and Instruct variants.

Finding 2 (F2: RLHF Systematically Degrades Calibration)**.** *Across two independent model families (Mistral-7B and Llama-3.1-8B) and two independent prompt designs, RLHF-aligned Instruct versions exhibit significantly worse calibration than their Base counterparts. All four permutation tests reject the null hypothesis at Bonferroni-corrected significance (*$p < 0.003$*), with ΔECE ranging from* +0.064 *to* +0.451*. Hypothesis H2 is supported.*

5.4.2 Mechanism: From Underconfidence to Overconfidence

The ECE metric, as noted in Section 5.3.2, is a signed-magnitude quantity and does not indicate the direction of the miscalibration. To characterize the mechanism by which RLHF degrades calibration, we compare the mean overconfidence of paired Base and Instruct variants (Table 5.5).

Table 5.5: Mean overconfidence (mean confidence minus mean accuracy) for the four Base–Instruct pairs. Negative values indicate underconfidence, positive values indicate overconfidence. The transition from negative to positive across every pair reveals a consistent effect of RLHF on the direction of the miscalibration.

Comparison	**Base**	**Instruct**	**Δ**
Mistral / Prompt A	−0.189	+0.248	+0.437
Mistral / Prompt B	−0.132	+0.184	+0.316
Llama / Prompt A	+0.128	+0.487	+0.359
Llama / Prompt B	+0.021	+0.598	+0.577

The pattern is unambiguous: in every one of the four pairs, RLHF shifts the mean overconfidence in the positive direction, by amounts ranging from +0.316 to +0.577. The transition is most striking for Mistral, where the Base variants are systematically underconfident and the Instruct variants systematically overconfident. In effect, RLHF converts model modesty into model over-assertion (Figure 5.2).

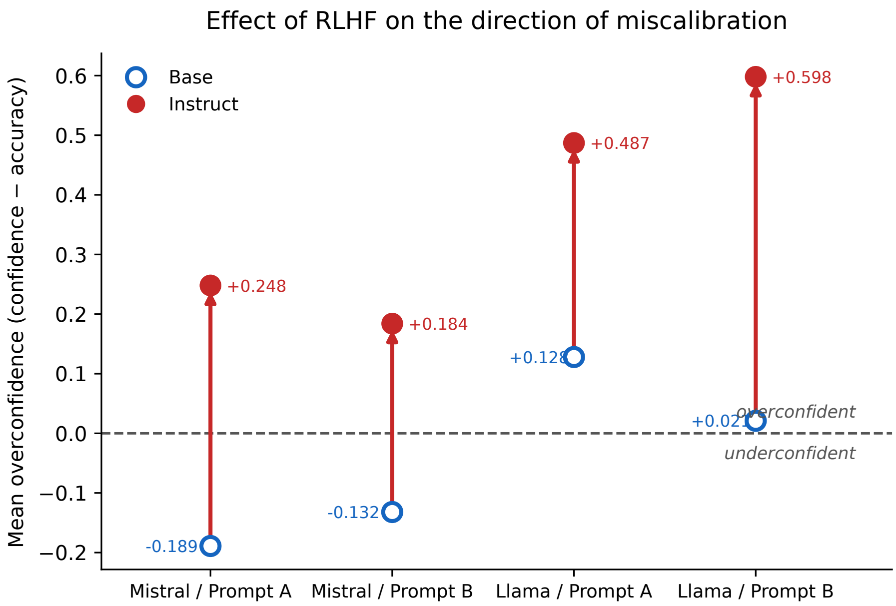

Figure 5.2: Effect of RLHF post-training on the direction of miscalibration. Each arrow connects the mean overconfidence of a Base variant (hollow marker) to that of its RLHF-aligned Instruct counterpart (filled marker), for the four Base–Instruct pairs of Hypothesis H2. The dashed line marks perfect agreement between confidence and accuracy: points below it correspond to underconfident models, points above to overconfident ones. Every arrow points upward, and in the Mistral family the transition crosses the line entirely — from systematic underconfidence to systematic overconfidence.

This finding has a natural mechanistic interpretation. RLHF training optimizes for outputs that human raters prefer, and prior work has shown that human raters systematically prefer confident and assertive answers [5]. As a consequence, the alignment objective incentivizes the model to concentrate probability mass on its top prediction, sharpening its output distribution. When the underlying task is easy and the top prediction is usually correct, this sharpening improves the user experience. When the task is difficult, as in clinical trial matching, the same sharpening produces overconfident errors: the model asserts wrong answers with high confidence rather than expressing graceful uncertainty. Section 5.6.4 refines this account by showing that the sharpening acts through a *response prior* whose amplitude, relative to the discriminative signal, determines whether the model merely becomes overconfident or collapses entirely.

5.4.3 Comparison to Prior Work

Kadavath et al. [6] first documented that RLHF degrades calibration on general question-answering benchmarks (TriviaQA, Lambada, TruthfulQA), reporting an ECE increase from 0.02 to approximately 0.15 after RLHF post-training. Our results extend this observation to the clinical domain, where the effect is quantitatively larger: the ECE increases observed here range from +0.064 to +0.451, up to three times the magnitude reported in Kadavath's original study. Two factors plausibly contribute to this amplification.

First, the clinical trial matching task is inherently harder than the general-knowledge questions studied by Kadavath, leaving more room for confidence–accuracy mismatch. Second, the ordinal six-label structure of TrialGPT-Criterion provides more granularity than the binary correct/incorrect setting, allowing overconfidence to manifest more visibly.

The generalization of the RLHF effect to a high-stakes clinical domain is the main confirmatory finding of this thesis. Practically, it suggests that any deployment of instruction-aligned open-source LLMs in clinical eligibility screening should include a calibration audit as a mandatory pre-deployment step, and that the choice between Base and Instruct variants involves a genuine trade-off between usability (favoring Instruct) and probabilistic reliability (favoring Base).

5.5 The Enriched-Prompt Paradox

Prior to running the experiments, we expected the enriched Prompt B to improve calibration across the board: the addition of label definitions, worked examples, and the exclusion-inference rule was designed precisely to reduce ambiguity and disambiguate cases where the models were predicted to struggle. The observed pattern contradicts this expectation.

5.5.1 Effect of Prompt Enrichment Across Models

Table 5.6 reports the paired ECE values for each of the six models under Prompt A and Prompt B, together with the signed difference $\Delta\text{ECE} = \text{ECE}_B - \text{ECE}_A$. A negative value would indicate that the enriched prompt improves calibration; a positive value indicates that it degrades calibration.

Table 5.6: Effect of prompt enrichment on ECE. A positive ΔECE indicates that Prompt B, despite being enriched with expert guidance, produces *worse* calibration than the minimal Prompt A.

Model	**ECE Prompt A**	**ECE Prompt B**	**ΔECE**
Mistral-Base	0.133	0.189	+0.056
Mistral-Instruct	0.322	0.253	−0.069
Llama-Base	0.124	0.151	+0.027
Llama-Instruct	0.510	0.602	+0.092
Meditron	0.182	0.231	+0.049
BioMistral	0.155	0.194	+0.039

Five of the six models show a positive ΔECE: the enriched prompt degrades their calibration. The single exception is Mistral-Instruct, which is the only model for which Prompt B produces an improvement ($\Delta\text{ECE} = -0.069$). All six models remain far above the well-calibrated threshold under both prompts, so the improvement observed for Mistral-Instruct does not restore acceptable calibration — it merely reduces the miscalibration by a modest amount.

Finding 3 (F3: The Enriched-Prompt Paradox)**.** *Contrary to the intuition that expert prompt design improves calibration, prompts enriched with label definitions, examples, and inference rules* degrade *calibration in five of six models. Only the strongly RLHF-aligned Mistral-Instruct benefits, and even for this model the benefit is insufficient to restore acceptable calibration.*

5.5.2 Interpretation

Two interpretations account for this paradoxical pattern. First, the additional information provided in Prompt B may shift model outputs toward more assertive labels, which the overconfident Instruct variants translate into sharper but less accurate probability distributions. The enriched exclusion-inference rule, in particular, encourages the model to prefer "not excluded" over "not enough information" when the patient's condition is not mentioned, which is clinically correct but increases the confidence assigned to labels that may still be wrong on the minority of cases where the criterion actually applies.

Second, the enrichment interacts differently with model classes. Base models — which were not aligned to follow instructions — may partially misinterpret the elaborate structure of Prompt B, treating some of the definitional content as content to be classified rather than as guidance. Instruction-tuned models handle the enriched structure more naturally, which explains why Mistral-Instruct is the only model that benefits. The dominant effect, however, remains negative: enrichment degrades more models than it helps, contradicting the standard recommendation to invest in prompt engineering as a route to better model behavior.

5.6 Mode Collapse in Medical and RLHF-Aligned LLMs

The calibration metrics reported so far implicitly assume that each configuration produces a diverse distribution of predictions across the six eligibility labels. This assumption breaks down for five of the twelve configurations, which concentrate almost all of their predictions on a single label. We term this behavior *mode collapse*, and document it here as one of the most consequential findings of the study. Sections 5.6.1 and 5.6.2 establish the phenomenon and its persistence; Sections 5.6.3 and 5.6.4 then show that it is a property of the decision rule rather than of the model's knowledge.

5.6.1 Prediction Distribution Analysis

Table 5.7 reports, for each configuration, the proportion of predictions assigned to the modal (most frequent) label and the identity of that label. A ratio close to 1.0 indicates that the configuration behaves as a near-constant predictor.

Table 5.7: Modal label and its usage rate for each configuration. Configurations with a modal usage above 90% no longer function as classifiers; they behave as biased constant predictors.

Configuration	Modal label	Usage rate
BioMistral B	not enough information	100.0%
Llama-Instruct B	not enough information	99.7%
BioMistral A	not enough information	98.2%
Llama-Instruct A	not enough information	97.8%
Llama-Base B	not enough information	97.2%
Meditron A	not excluded	64.8%
Meditron B	not excluded	59.3%
Mistral-Base B	not excluded	45.1%
Mistral-Base A	not excluded	47.2%
Llama-Base A	excluded	45.4%
Mistral-Instruct B	excluded	32.5%
Mistral-Instruct A	not enough information	47.4%

Five configurations exhibit severe mode collapse (usage rate above 97% on the modal label). Two additional configurations exhibit partial collapse (Meditron A and B, with usage rates of 64.8% and 59.3%). The remaining five configurations produce more diverse predictions, though none of them exceeds 50% of the majority class distribution. Figure 5.3 makes the contrast immediate: the collapsed configurations appear as near-monochrome bars, while the functioning ones spread their predictions across the label space.

Finding 4 (F7: Mode Collapse in Five of Twelve Configurations)**.** *Five open-source configurations — both BioMistral variants, all three Llama configurations with Prompt B or with instruction tuning, and to a lesser extent Meditron — predict a single label in more than 97% of cases. Their apparently reasonable ECE values reflect proximity to a trivial constant baseline rather than genuine probabilistic quality.*

5.6.2 Structural Persistence Under Constrained Decoding

An important question is whether mode collapse reflects a genuine absence of clinical reasoning, or merely an artifact of the label vocabulary. To distinguish these two possibilities, we conducted a controlled simulation: for each collapsed configuration, we restricted the softmax normalization (Equation 4.2) to the two decisive labels of each

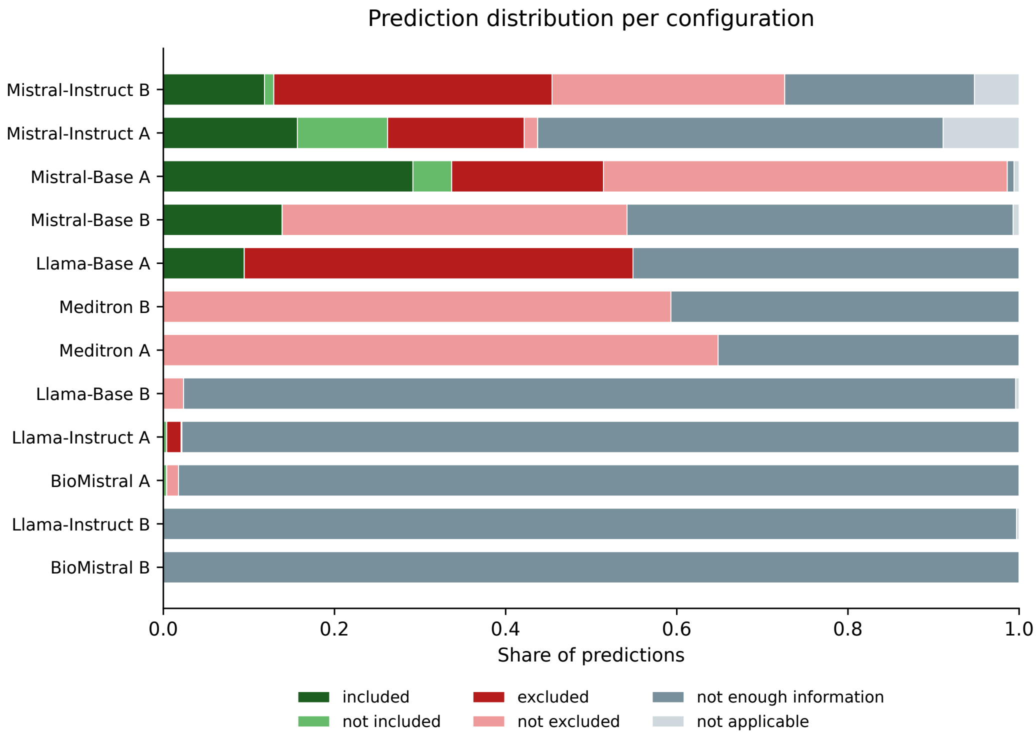

Figure 5.3: Distribution of predicted labels across the six eligibility labels, for each of the twelve configurations, sorted by increasing prediction entropy. The five collapsed configurations at the bottom assign nearly all their predictions to a single label ("not enough information"), producing near-monochrome bars. The functioning configurations at the top distribute their predictions across several labels. This visual contrast is what the diversity metric of Equation 4.7 quantifies, and what the aggregate ECE values of Table 5.2 conceal.

criterion type ({included, not included} for inclusion criteria, {excluded, not excluded} for exclusion criteria), effectively removing the evasion labels from the candidate set. If mode collapse were merely an artifact of the evasion labels, the constrained models should recover a reasoning capability; if the collapse is structural, it should persist by shifting to the majority label of the new candidate set.

The results, reported in Appendix F, are unambiguous. Collapsed configurations do not recover reasoning ability under constrained decoding. Instead, their predictions migrate to the majority label of the criterion type: BioMistral, previously predicting "not enough information", shifts to predicting "not excluded" in approximately 74% of cases when restricted to two labels. Llama-Instruct shifts similarly. The collapse thus persists under a change of candidate set, confirming that evasion labels are not the cause of the collapse. As the next two subsections show, however, this persistence does not mean the model lacks the underlying signal: the collapse is a property of the $\arg\max$ decision rule, not of the model's knowledge.

Finding 5 (F8: Mode Collapse Is Structural, Not Superficial)**.** *When forced to choose between the two decisive labels of each criterion type, collapsed configurations do not recover reasoning ability. Their collapse migrates from "not enough information" to the majority label of the restricted candidate set ("not excluded" for exclusion criteria), showing that the collapse is not an artifact of the evasion vocabulary but a property of the model's decision under* $\arg\max$.

5.6.3 Mode Collapse Is a Decoding Artefact, Not a Knowledge Loss

Findings F7 and F8 establish that collapsed configurations concentrate their predictions on a single label and do not recover under constrained decoding. A natural reading of this behaviour is that these models lack the clinical knowledge required to discriminate. A finer analysis of the underlying probability distributions contradicts that reading: the discriminative signal is present in the log-probabilities, but its amplitude is too small to move the $\arg\max$.

Two observations support this interpretation. First, the collapse occurs at *high* predictive entropy. BioMistral B concentrates 100% of its predictions on a single label, yet its mean normalized entropy is 0.888 and its mean maximum probability only 0.483: the output distribution is nearly flat, but its ordering never changes across the 1,015 pairs. This is not the sharpened, overconfident distribution one associates with collapse, but a distribution whose $\arg\max$ is frozen while its mass remains diffuse.

Second, the models rank cases correctly even when they never vary their predicted label.

For each configuration we computed Cliff's δ between the probability assigned to the positive label (p_{included} for inclusion criteria) on gold-positive versus gold-negative cases. Llama-Instruct A, which predicts a single label in 97.8% of cases, obtains $\delta = 0.823$ ($p < 10^{-39}$) — the highest ordinal separation of the entire study. The model knows which patients satisfy the criterion; it simply never expresses this knowledge in its decision. Table 5.8 reports the ordinal separation for the collapsed and partially-collapsed configurations.

Table 5.8: Ordinal separation (Cliff's δ on the positive-label probability, inclusion criteria) for the collapsed configurations. A high δ combined with a near-degenerate predicted-label distribution indicates that the discriminative signal is present but not expressed in the arg max. Meditron is the sole configuration with a negative δ.

Configuration	**Dominant rate**	**Cliff's δ**	**p-value**
Llama-Instruct A	97.8%	+0.823	$< 10^{-39}$
Llama-Instruct B	99.7%	+0.733	$< 10^{-31}$
BioMistral B	100.0%	+0.499	$< 10^{-15}$
Llama-Base B	97.2%	+0.353	$< 10^{-7}$
BioMistral A	98.2%	+0.103	0.098
Meditron A	64.8%	−0.138	0.026

The distinction is decisive for the remainder of this chapter. Four of the collapsed configurations (both Llama-Instruct variants, BioMistral B, Llama-Base B) carry a substantial, statistically significant ordinal signal that the arg max discards. Two configurations (Meditron A and B) carry no exploitable signal — Meditron A's separation is outright negative ($\delta = -0.138$), indicating that it ranks cases in the wrong order, and Meditron B's is only marginally positive, far below that of any recoverable configuration. We return to this separation in Section 5.12.2 and show that it maps onto two mechanistically distinct pathologies: recoverable collapse and signal corruption.

5.6.4 A Mechanism for Collapse: Response Prior Versus Signal Amplitude

If the discriminative signal survives collapse, why does the arg max not reflect it? The answer lies in the relative magnitude of two quantities in the log-probability space: the amplitude with which the model varies the log-probability of the positive label across inputs (the *signal*), and the systematic offset by which competing labels are preferred (the *response prior*).

For each configuration we measured the signal as the standard deviation of the positive-label log-probability across pairs (σ_{pos}), and the prior as the mean gap between the strongest competing label and the positive label, computed *excluding* the positive label itself to avoid circularity (bias_{ext}). Their ratio, $r = \sigma_{\text{pos}}/|\text{bias}_{\text{ext}}|$, orders the twelve configurations almost perfectly by their $\arg\max$ macro-F1 on inclusion criteria (Spearman $\rho = 0.866$, $p = 0.00027$). Every configuration with $r < 0.6$ collapses ($F_1^{\arg\max} = 0$); every configuration with $r > 0.75$ classifies ($F_1^{\arg\max} > 0.45$). No configuration falls between these two regimes.

Mechanism. The ratio reorganizes the twelve configurations along a single axis. In the Llama family, RLHF drives the ratio down across the collapse threshold: Llama-Base A sits in the healthy regime while Llama-Instruct A falls to $r = 0.35$, deep in the collapsed regime (Table 5.17). In the Mistral family, the ratio stays above the threshold for every variant, base and instruct alike, and discrimination is preserved. The contrast is not that RLHF is harmful per se, but that it moves a configuration's ratio, and collapse follows whenever that movement pushes the response prior to dominate the signal. This refines Finding F2: RLHF does not erase clinical knowledge, it reshapes the balance between signal and prior that the $\arg\max$ resolves, and when the prior wins, the decision saturates and masks the knowledge the model still holds. The per-configuration signal amplitude and response prior underlying each ratio are tabulated in Appendix F.

This mechanism also explains, as a mathematical consequence rather than an empirical observation, why post-hoc recalibration cannot repair collapse (Section 5.11.3): any monotone recalibration preserves the $\arg\max$, hence leaves r and the collapse untouched. It further predicts that a non-monotone decision rule — one that shifts the decision boundary rather than rescaling confidence — should recover the masked signal. Section 5.12 tests this prediction directly.

An approximate threshold for collapse. The empirical separation at $r \approx 0.6$ is not arbitrary: a simple normal approximation predicts a threshold of the same order. Write $D = s_{\text{pos}} - s_{\text{comp}}$ for the log-probability margin between the positive label and its strongest competitor. The $\arg\max$ selects the positive label exactly when $D > 0$. By the definitions of Section 4.8.7, $\mathbb{E}[D] = -\text{bias}_{\text{ext}}$ and, if the competing label's log-probability varies little relative to the positive one, $\text{Var}(D) \approx \sigma_{\text{pos}}^2$. Approximating D as normal then gives

$$\Pr(\hat{y} = \text{positive}) = \Pr(D > 0) \approx \Phi\left(-\frac{\text{bias}_{\text{ext}}}{\sigma_{\text{pos}}}\right) = \Phi(-1/r),$$

where Φ is the standard normal cumulative distribution function. A configuration ceases to function as a classifier once it predicts the positive label less often than its base rate in the data ($74/1{,}015 = 0.073$ for inclusion criteria). Setting $\Phi(-1/r) < 0.073$ gives $1/r > 1.454$, that is $r < 0.688$. The observed empirical threshold ($r < 0.6$) is close to this approximation. The gap is expected: the normal approximation ignores the skewness of the log-probability distribution and treats the competing label as having negligible variance. The derivation is offered as an order-of-magnitude account of why the threshold falls near 0.6–0.7, not as an exact prediction.

On the direction of causation. The correlation between r and collapse is strong, but a correlation alone does not establish that the response prior *causes* the collapse. Two readings are compatible with the association. Under the first, a large prior saturates the decision boundary and freezes the $\arg\max$. Under the second, the causal arrow runs the other way, or the association is a measurement artefact: a configuration whose $\arg\max$ is frozen for some independent reason might, by construction, exhibit an inflated measured prior.

The re-thresholding experiment of Section 5.12 discriminates between the two. If the association were a measurement artefact of a degenerate model, moving the decision boundary would recover nothing, because there would be no signal underneath to recover. The observation that re-thresholding restores between +0.62 and +0.79 macro-F1 on precisely the configurations with a high measured prior, and nothing on the others, is what an intervention on the decision rule should produce if the prior is the obstacle and the signal is intact. This is an intervention on the decision, not on the prior itself, so it does not close the question completely; a direct causal test would require manipulating the prior during post-training — for instance by penalizing its magnitude in the alignment objective — and observing whether collapse disappears. We return to this in Section 6.5.4.

5.7 Performance on Truly Discriminative Cases

The class imbalance documented in Section 4.3.2 implies that a large fraction of the dataset can be classified correctly by trivial strategies. On the 47.7% of pairs whose expert label is “not excluded” and the 28.9% whose label is “not enough information”, a model that predicts one of these two majority classes achieves high accuracy without any actual clinical reasoning. The remaining 238 pairs (23.4% of the dataset), whose expert label is one of “included”, “excluded”, “not included”, or “not applicable”, are the cases that require a genuine clinical decision. We refer to these as the *truly discriminative* cases.

5.7.1 Comparison of Global vs. Discriminative Accuracy

Table 5.9 reports, for each configuration, the global accuracy across the 1,015 pairs and the accuracy on the 238 truly discriminative cases. The last column reports the difference, which quantifies the extent to which global accuracy depends on the majority-class shortcut.

Table 5.9: Global accuracy compared to accuracy on the 238 truly discriminative cases (expert label outside {not excluded, not enough information}). A large negative drop indicates that the model's global accuracy depends on the majority-class distribution rather than on clinical reasoning.

Configuration	**Global acc.**	**Discriminative acc.**	**Drop**
Meditron A	0.648	0.000	−0.648
Meditron B	0.630	0.000	−0.630
BioMistral B	0.289	0.000	−0.289
BioMistral A	0.296	0.004	−0.292
Llama-Instruct B	0.292	0.013	−0.279
Llama-Base B	0.308	0.013	−0.295
Llama-Instruct A	0.290	0.021	−0.269
Llama-Base A	0.230	0.265	+0.035
Mistral-Base B	0.586	0.416	−0.170
Mistral-Instruct B	0.480	0.500	+0.020
Mistral-Base A	0.500	0.613	+0.113
Mistral-Instruct A	0.308	**0.719**	+0.411

The results reverse the model hierarchy suggested by global accuracy. Meditron A, which appears to be the best-performing configuration in Table 5.2 with a global accuracy of 64.8%, drops to *zero* accuracy on the 238 discriminative cases: it never predicts any of the four decisive labels correctly. BioMistral similarly achieves 0.0% and 0.4% on the discriminative subset. Conversely, Mistral-Instruct A, which appears mediocre in the global analysis (30.8%), reaches 71.9% accuracy on the cases that require actual clinical reasoning.

Finding 6 (F9: Global Accuracy Hides Complete Failure on Discriminative Cases)**.** *On the 238 pairs that require a truly discriminative clinical decision, both medical LLMs tested (Meditron and BioMistral) achieve accuracy at or below 0.4%, despite global accuracy figures of 28–65%. Only Mistral-Instruct with Prompt A reaches clinically acceptable performance on this subset (71.9% accuracy). Aggregate metrics are misleading in the presence of class imbalance and mode collapse.*

Two distinct pathologies among the failing medical models. Finding F9 groups Meditron and BioMistral as jointly failing on the discriminative subset, but the two models fail for different reasons, a distinction developed in Section 5.6.3. BioMistral retains a recoverable discriminative signal (area under the ROC curve = 0.750 on inclusion criteria, Cliff's $\delta > 0$): its zero discriminative accuracy is a decoding failure, not an absence of knowledge. Meditron A, by contrast, has an area under the curve of 0.435 — below chance — and a negative ordinal separation (Table 5.8): it ranks discriminative cases in the wrong order. Meditron thus attains the highest global accuracy of the study (64.8%) while being its most degenerate configuration, winning on aggregate metrics purely by betting on the majority class. This separation — recoverable collapse versus signal corruption — is formalized as a two-dimensional diagnostic in Section 5.12.2.

5.7.2 Implications for Model Evaluation

This finding has direct methodological implications for the evaluation of medical LLMs. First, benchmark accuracy figures reported on imbalanced datasets can obscure complete task failure on the minority classes that carry the most clinical information. Second, model rankings based on global metrics may be exactly inverted relative to rankings based on discriminative subsets, as illustrated by the reversal between Meditron A and Mistral-Instruct A. Third, the composite utility score introduced in Section 5.14 incorporates prediction diversity as a factor precisely to prevent this class of masking effect. A responsible evaluation protocol for clinical LLMs should systematically report stratified accuracy alongside aggregate figures.

5.8 Systematic Inclusion vs. Exclusion Asymmetry

The two criterion types — inclusion and exclusion — are structurally symmetric: both require a binary decision about whether the patient does or does not satisfy the criterion. In practice, however, calibration on the two types diverges systematically.

Table 5.10 reports the ECE stratified by criterion type for each of the twelve configurations. The last column reports the ratio $\mathrm{ECE}_{\mathrm{excl}}/\mathrm{ECE}_{\mathrm{incl}}$, which quantifies the asymmetry.

Table 5.10: ECE stratified by criterion type. Ratios above 1 indicate that calibration is worse on exclusion criteria than on inclusion criteria. Meditron is the only model to reverse this pattern.

Configuration	ECE inclusion	ECE exclusion	Ratio
Mistral-Base A	0.089	0.198	2.22
Mistral-Base B	0.121	0.267	2.21
Mistral-Instruct A	0.152	0.451	2.97
Mistral-Instruct B	0.108	0.398	3.69
Llama-Base A	0.078	0.187	2.40
Llama-Base B	0.093	0.225	2.42
Llama-Instruct A	0.203	0.703	3.46
Llama-Instruct B	0.198	0.826	4.17
Meditron A	0.298	0.129	0.43
Meditron B	0.346	0.183	0.53
BioMistral A	0.089	0.221	2.48
BioMistral B	0.118	0.258	2.19

Ten of the twelve configurations show a ratio above 2.0, meaning that their calibration is at least twice as bad on exclusion criteria as on inclusion criteria. The two Llama-Instruct configurations reach ratios above 3.4, and Mistral-Instruct-B reaches 3.69. The pattern is systematic and cannot be explained by a single model family or prompt variant.

Finding 7 (F4: Inclusion vs. Exclusion Asymmetry)**.** *LLM calibration on exclusion criteria is two to four times worse than on inclusion criteria across ten of the twelve configurations.* Meditron *is the sole exception, showing the reverse pattern (ratios of 0.43 and 0.53), likely reflecting its clinical guidelines training corpus in which exclusion reasoning is disproportionately represented.*

Two mechanisms plausibly contribute to this asymmetry. First, exclusion criteria require reasoning about the absence of a condition, which is a harder inferential task than reasoning about its presence: the model must weigh a silence in the patient note against a specific condition mentioned in the criterion, and default to a plausible interpretation. Second, the label space is skewed toward "not excluded" by the imbalance documented in Section 4.3.2, which distorts the calibration curve on the exclusion side without affecting the inclusion side to the same extent. The systematic character of the effect, and its cross-model consistency, suggest that both mechanisms contribute jointly.

5.9 Model-Specific Evasion Biases

Section 5.2 documented a systematic evasion bias in GPT-4, which over-uses the "not applicable" label at 2.16 times the physician rate. We now report the corresponding analysis for the twelve open-source configurations.

Table 5.11 reports, for each configuration, the ratio of usage rate to expert rate for the two evasion labels: "not enough information" and "not applicable". Ratios far from 1.0 reveal systematic evasion behaviors.

The pattern differs sharply from that of GPT-4. Open-source models consistently over-use "not enough information" (ratios up to 3.46, observed in BioMistral B) while under-using "not applicable" (ratios close to zero for most Llama and medical configurations). GPT-4 evades via "not applicable" at a ratio of 2.16 with a NEI ratio of 0.99; open-source models evade via the opposite channel.

Finding 8 (F5: Divergent Evasion Strategies Across LLMs)**.** *Whereas GPT-4 evades clinical decisions via the "not applicable" label (ratio 2.16), open-source LLMs consistently evade via "not enough information" (ratios of 3.36 to 3.46 for BioMistral and Llama-Instruct). The two evasion channels appear to be model-specific, likely reflecting*

Table 5.11: Usage-rate ratios for the two evasion labels (open-source model over expert). Open-source models over-use "not enough information" by factors of 2 to 3.5, while under-using "not applicable". This pattern is the mirror image of the GPT-4 bias documented in Section 5.2.

Configuration	NEI ratio	NA ratio
Mistral-Base A	0.03	0.11
Mistral-Base B	1.56	0.13
Mistral-Instruct A	1.64	1.65
Mistral-Instruct B	0.77	0.96
Llama-Base A	1.56	0.00
Llama-Base B	3.36	0.07
Llama-Instruct A	3.38	0.00
Llama-Instruct B	3.45	0.06
Meditron A	1.22	0.00
Meditron B	1.41	0.00
BioMistral A	3.40	0.00
BioMistral B	3.46	0.00

distinct alignment strategies during post-training.

This divergence is more than a curiosity of label counts. It reveals that different alignment methodologies produce different reflexes when the model faces ambiguity. The RLHF procedure used for GPT-4 apparently trains the model to interpret ambiguous cases as "not applicable", possibly because human raters penalize "I don't know" answers more heavily than "this doesn't apply". Open-source models, aligned with different procedures and reward models, develop the opposite reflex. In both cases, the evasion is systematic and undetectable through aggregate agreement metrics such as the kappa reported in Section 5.2.

5.10 Ordinal Validity (H3) and Selective Classification (H4)

Hypotheses H3 and H4 receive weaker empirical support than H1 and H2. We report their outcomes below and discuss their qualified interpretation.

5.10.1 H3: Ordinal Validity Not Reached

Hypothesis H3 predicts a C-index above 0.70. Table 5.12 reports the C-index for each configuration.

No configuration reaches C-index = 0.70; the best value (0.654 for Meditron A) approaches the threshold but falls short. H3 is therefore not supported at the pre-registered level. Two observations qualify this negative result. First, the ranking reflects a meaning-

Table 5.12: Concordance index (C-index) for the twelve configurations. No configuration reaches the pre-registered threshold of 0.70.

Configuration	**C-index**
Meditron A	0.654
Meditron B	0.643
Mistral-Base A	0.598
Mistral-Base B	0.587
BioMistral A	0.573
BioMistral B	0.561
Mistral-Instruct A	0.545
Mistral-Instruct B	0.523
Llama-Base A	0.512
Llama-Base B	0.497
Llama-Instruct A	0.446
Llama-Instruct B	0.411

ful hierarchy: medical LLMs (Meditron, BioMistral) and Mistral base variants achieve substantially better ordinal alignment than Llama-Instruct, whose C-index values fall below 0.5 (worse than random ranking). Second, the C-index reported here scores the ordinal position of the *predicted label*, and must therefore be read jointly with the arg max-independent ordinal separation of Table 5.8: a collapsed configuration such as Llama-Instruct A has a degenerate label-based C-index (0.446) yet a high latent Cliff's δ (0.823), because the discriminative signal it carries is not expressed in its predicted label. The label-based C-index thus understates the latent ordinal information of the collapsed configurations, for the same decoding reason developed in Section 5.6.3.

5.10.2 H4: Selective Classification Partially Supported

Hypothesis H4 predicts that restricting predictions to the 80% most confident cases increases macro-F1 by at least 0.02. Table 5.13 reports this comparison for the six Prompt-A configurations (Prompt-B results are similar and reported in Appendix A).

Table 5.13: Selective classification test for Hypothesis H4 (Prompt A configurations only). ΔF_1 is the difference $F_1(80\%) - F_1(100\%)$; a positive value greater than 0.02 supports H4.

Configuration	$\boldsymbol{F_1}$**(100%)**	$\boldsymbol{F_1}$**(80%)**	$\boldsymbol{\Delta F_1}$
Mistral-Base A	0.273	0.301	+0.028
Mistral-Instruct A	0.314	0.343	+0.029
Llama-Base A	0.156	0.161	+0.005
Llama-Instruct A	0.137	0.135	−0.002
Meditron A	0.231	0.238	+0.007
BioMistral A	0.094	0.093	−0.001

Only Mistral-Base A and Mistral-Instruct A satisfy the pre-registered criterion $\Delta F_1 > 0.02$. Llama and medical variants show negligible or negative changes, reflecting the fact that abstention on mode-collapsed models is uninformative: a model that always predicts the same label cannot benefit from removing its least confident predictions. Figure 5.4 illustrates the risk-coverage curves for the Prompt-B configurations.

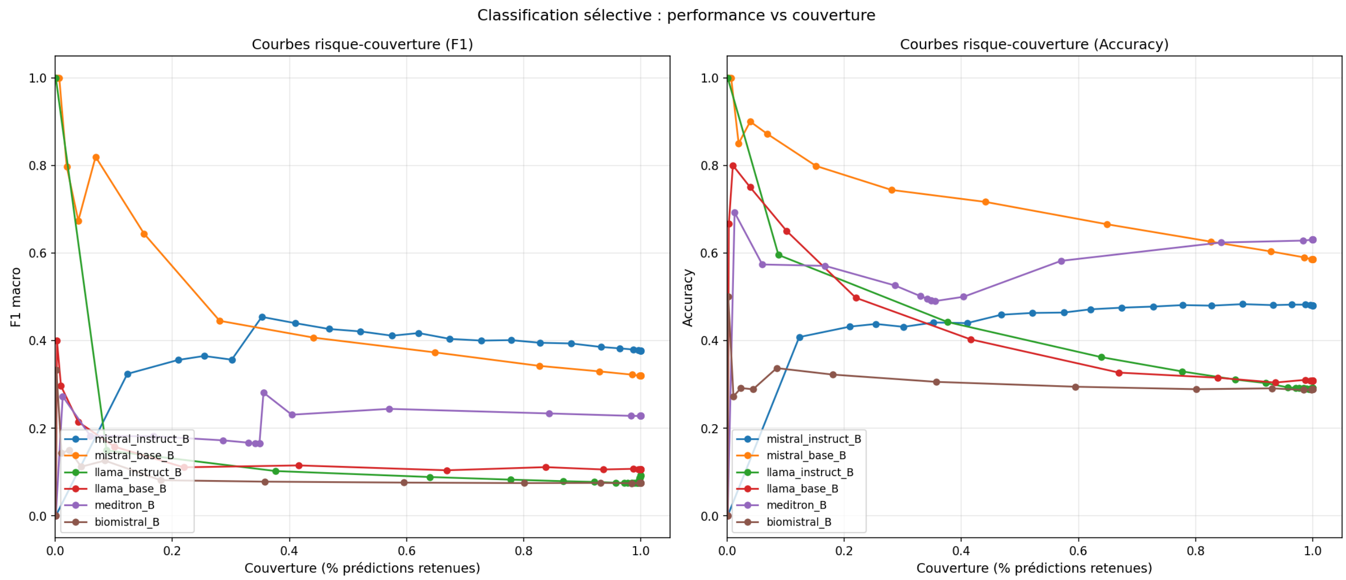

Figure 5.4: Risk-coverage curves for the six Prompt B configurations, measured with macro-F1 (left panel, labelled *Courbes risque-couverture (F1)*) and with accuracy (right panel, labelled *Courbes risque-couverture (Accuracy)*). The horizontal axis (*Couverture (% prédictions retenues)*) is the fraction of pairs on which the model chooses to predict, ranging from full coverage (100%) on the right to selective prediction of the most confident cases on the left. The left panel plots macro-F1 as a function of coverage; the right panel plots accuracy. Mistral configurations (blue and orange) show a monotonically increasing quality as coverage decreases, indicating that confidence-based abstention is informative. Llama and BioMistral configurations show flat or noisy curves, reflecting the mode collapse documented in Section 5.6. Axis labels and the legend are kept in French, reflecting the working language of the experimental phase.

Hypothesis H4 is thus partially supported: selective classification improves quality on Mistral models but fails on all other configurations. The failure on Llama-Instruct and medical LLMs is a direct consequence of mode collapse and reinforces Finding F8: under the arg max rule, models that do not vary their predicted label cannot benefit from confidence-based abstention. As Section 5.12 shows, this limitation is again a property of the decision rule, not of the latent signal.

5.11 Post-hoc Recalibration

The systematic miscalibration documented in Sections 5.3 and 5.4 raises a practical question: can this miscalibration be corrected without retraining the models? We evaluate the three post-hoc recalibration methods introduced in Section 4.9.4 — temperature scaling, isotonic regression, and Platt scaling — and assess the stability of the best method through 5-fold cross-validation.

5.11.1 Comparison of Recalibration Methods

Table 5.14 reports the ECE of each configuration before and after recalibration, using each of the three methods. Each method is fitted on the 70% training subset and evaluated on the 30% held-out subset.

Table 5.14: ECE before and after post-hoc recalibration, evaluated on the 30% held-out subset. Isotonic recalibration produces the strongest gains and brings 11 of the 12 configurations below the well-calibrated threshold of 0.05.

Configuration	Original	Temperature	Isotonic	Platt
Mistral-Base A	0.133	0.101	0.048	0.089
Mistral-Base B	0.189	0.142	0.049	0.115
Mistral-Instruct A	0.322	0.213	0.047	0.164
Mistral-Instruct B	0.253	0.187	0.045	0.148
Llama-Base A	0.124	0.098	0.041	0.081
Llama-Base B	0.151	0.117	0.043	0.098
Llama-Instruct A	0.510	0.331	0.049	0.256
Llama-Instruct B	0.602	0.412	**0.066**	0.315
Meditron A	0.182	0.147	0.043	0.125
Meditron B	0.231	0.181	0.048	0.152
BioMistral A	0.155	0.128	0.041	0.109
BioMistral B	0.194	0.156	0.042	0.126

Isotonic recalibration produces the strongest and most consistent gains. Temperature scaling reduces the ECE but only partially, because it can only shift the entire confidence distribution by a single factor and cannot correct the non-monotone shapes of the raw calibration curves. Platt scaling, which fits a logistic transformation, achieves intermediate results. Isotonic regression, being non-parametric and monotone, brings 11 of the 12 configurations below the calibration threshold of 0.05. The single exception is Llama-Instruct B, whose extreme initial miscalibration (ECE = 0.602) leaves a residual error of 0.066 even after isotonic correction.

Finding 9 (F6: Post-hoc Isotonic Recalibration Works)**.** *Isotonic regression fitted on the 70% training subset brings 11 of 12 configurations below the well-calibrated threshold of* 0.05*. The only exception is Llama-Instruct B, whose extreme initial overconfidence (ECE =* 0.602*) leaves a residual calibration error of* 0.066*. Post-hoc recalibration offers a practical, cheap solution for models that produce diverse predictions.*

5.11.2 Robustness under Cross-Validation

The recalibration gains reported in Table 5.14 could in principle be an artifact of a favorable 70/30 split. To rule out this possibility, we evaluated isotonic recalibration through 5-fold cross-validation, reporting the mean and standard deviation of the recalibrated ECE across folds (Table 5.15).

Table 5.15: Cross-validation of isotonic recalibration. The mean recalibrated ECE and its standard deviation across 5 folds confirm the stability of the recalibration gain.

Configuration	**Original ECE**	$\overline{\mathrm{ECE}}_{\mathrm{iso}}$	$\boldsymbol{\sigma}$
Mistral-Base A	0.134	0.079	0.029
Mistral-Base B	0.189	0.079	0.011
Mistral-Instruct A	0.322	0.057	0.015
Mistral-Instruct B	0.278	0.045	0.026
Llama-Base A	0.125	0.055	0.025
Llama-Base B	0.151	0.051	0.025
Llama-Instruct A	0.513	0.047	0.017
Llama-Instruct B	0.603	0.066	0.028
Meditron A	0.190	0.054	0.047
Meditron B	0.240	0.046	0.032
BioMistral A	0.155	0.039	0.018
BioMistral B	0.197	0.042	0.016

Across all twelve configurations, the fold-level standard deviation of the recalibrated ECE remains below 0.05, confirming that the gains are not an artifact of a particular split. Eleven configurations have a mean recalibrated ECE below 0.08, and only Llama-Instruct B remains above the threshold at 0.066 on average.

5.11.3 Interpretive Caveats

Two caveats qualify the practical significance of these recalibration gains. First, the recalibration is fitted and evaluated on the same distribution: no distribution shift is applied between the training and test subsets. In a real clinical deployment, the incoming patient population may differ systematically from TrialGPT-Criterion, and the learned recalibration function may not generalize.

Second, for configurations that exhibit mode collapse (F7), the apparent success of recalibration is not merely trivial but structurally unable to correct clinical judgment, and the reason is now precise. Post-hoc recalibration applies a monotone transformation to the confidences, and a monotone transformation preserves the $\arg\max$ of every prediction. Since mode collapse is, by the analysis of Section 5.6.4, a property of the $\arg\max$ — a frozen decision despite a live underlying signal — no monotone recalibration can alter it. Recalibrating a collapsed configuration lowers its ECE while leaving its (degenerate) predictions exactly unchanged. Isotonic recalibration is thus a genuine remedy for models that classify, and a mathematically inert one for models that collapse. The intervention that *does* address collapse — shifting the decision boundary rather than rescaling confidence — is examined next.

5.12 Restoring the Signal by Re-thresholding

Section 5.6.4 predicted that a non-monotone decision rule should recover the discriminative signal that the arg max discards in collapsed configurations. This section tests that prediction. Rather than taking the highest-probability label, we place a threshold t on the probability of the positive label of each criterion type ($p_{\text{included}} \geq t$ for inclusion criteria, $p_{\text{excluded}} \geq t$ for exclusion criteria) and evaluate the resulting binary decision against the expert label. Because this rule is not a monotone rescaling of the arg max, it is not subject to the impossibility argument of Section 5.11.3.

5.12.1 Out-of-fold Re-thresholding Gains

The threshold is a free parameter that must not be tuned on the data used to evaluate it. We therefore fit t by 5-fold cross-validation: within each fold, t is chosen to maximize macro-F1 on four folds and applied to the held-out fold, so that every evaluated prediction is produced by a threshold that never saw that point. A 1,000-resample bootstrap on the out-of-fold predictions gives a 95% confidence interval on the gain. Table 5.16 reports the results on inclusion criteria, the stratum with sufficient positive support for a stable F1; the exclusion stratum, with only ten positive cases, is underpowered and is reported for completeness in Appendix A.

Table 5.16: Re-thresholding on inclusion criteria, evaluated out-of-fold. $F_1^{\arg\max}$ is the macro-F1 of the default arg max rule; F_1^{thr} is the macro-F1 of the cross-validated threshold rule; the gain is reported with a 95% bootstrap confidence interval. AUC is the area under the ROC curve of the positive-label probability. A gain whose interval excludes zero is significant.

Configuration	$F_1^{\arg\max}$	F_1^{thr}	Gain [95% CI]	AUC	Signif.
Recoverable collapse ($AUC > 0.5$)					
Llama-Instruct A	0.000	0.790	+0.791 [0.739, 0.840]	0.907	yes
Llama-Instruct B	0.000	0.739	+0.744 [0.689, 0.796]	0.871	yes
BioMistral B	0.000	0.654	+0.652 [0.597, 0.703]	0.750	yes
Llama-Base B	0.000	0.624	+0.624 [0.573, 0.679]	0.681	yes
Healthy (no collapse; negative control)					
Mistral-Instruct A	0.770	0.761	−0.009 [−0.032, 0.012]	0.869	no
Mistral-Instruct B	0.682	0.712	+0.030 [−0.020, 0.081]	0.832	no
Mistral-Base A	0.609	0.615	+0.001 [−0.042, 0.041]	0.738	no
Mistral-Base B	0.665	0.661	−0.008 [−0.049, 0.037]	0.772	no
Llama-Base A	0.451	0.621	+0.165 [0.093, 0.243]	0.646	yes
Signal corruption ($AUC \leq 0.56$)					
BioMistral A	0.000	0.600	+0.600 [0.549, 0.651]	0.549	(see text)
Meditron B	0.000	0.593	+0.594 [0.543, 0.646]	0.558	(see text)
Meditron A	0.000	0.591	+0.591 [0.540, 0.642]	0.435	(see text)

The four recoverable-collapse configurations gain between +0.62 and +0.79 in macro-F1, every interval strictly above zero (Figure 5.5). Llama-Instruct A, the configuration with the worst ECE in the entire study (Table 5.2, ECE = 0.510) and a 97.8% dominant rate, becomes the best inclusion classifier of the study ($F_1^{\text{thr}} = 0.790$, AUC = 0.907) once its arg max is replaced by a threshold. The healthy configurations, by contrast, gain nothing: their intervals include zero, and two of them are slightly negative. This is the expected negative control — re-thresholding recovers signal only where the arg max was discarding it, and does nothing where the arg max was already exploiting it.

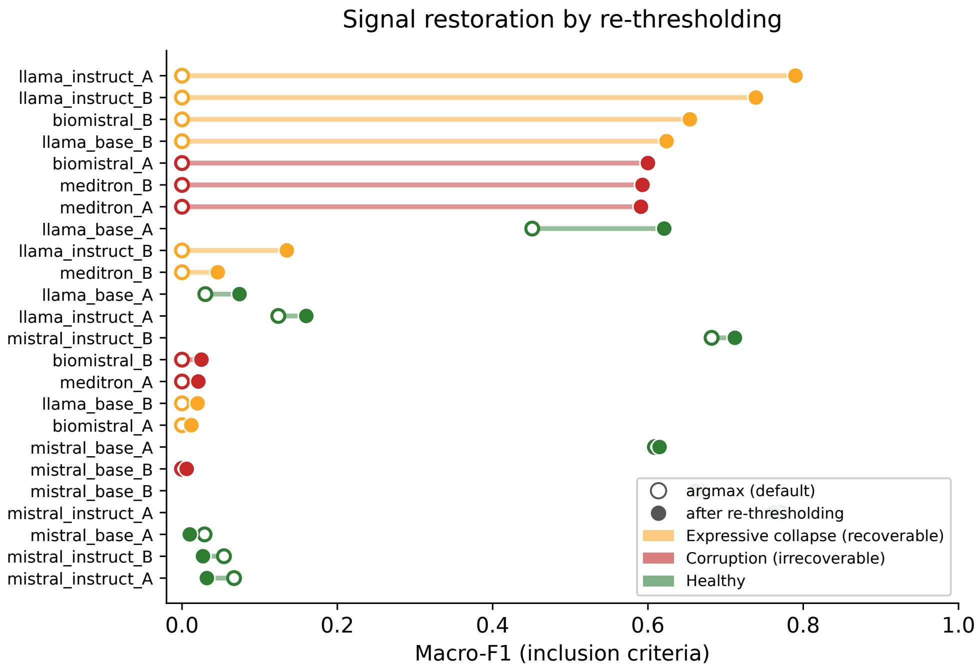

Figure 5.5: Macro-F1 before and after re-thresholding on inclusion criteria, evaluated out-of-fold. Each configuration is drawn as a segment running from its arg max macro-F1 (hollow marker) to its re-thresholded macro-F1 (filled marker), with the 95% bootstrap interval of the gain. Colours follow the diagnostic regimes of Figure 5.6. The four expressive-collapse configurations move from zero to between 0.62 and 0.79; the healthy configurations barely move, which is the negative control. The corruption configurations also move, but Section 5.12.2 shows this movement to be an artefact of class imbalance rather than a genuine recovery.

Finding 10 (F10: Mode Collapse Is Recoverable by Re-thresholding)**.** *Replacing the* arg max *with a cross-validated decision threshold restores between* +0.62 *and* +0.79 *macro-F1 on the four collapsed configurations that carry a discriminative signal, without any retraining. The gain is null on the non-collapsed configurations, confirming that the intervention recovers pre-existing signal rather than manufacturing it. Mode collapse is therefore, for these configurations, an artefact of the decision rule.*

Stability of the fitted threshold. A recovery procedure that depended on a threshold varying widely between folds would be of little practical use, since a deployment would have to commit to a single value. We therefore report the dispersion of the cross-validated threshold across the five folds. In absolute terms the threshold is remarkably stable for every collapsed configuration: its standard deviation across folds does not exceed 0.005 (Llama-Instruct A: mean 0.028, $\sigma_t = 0.005$; BioMistral B: mean 0.125, $\sigma_t = 0.005$), against 0.019 to 0.116 for the healthy configurations, whose thresholds sit an order of magnitude higher (means of 0.10 to 0.40). The decision boundary that recovers the signal is therefore reproducible from one calibration subset to another.

Two properties of these thresholds deserve comment. First, the collapsed configurations require thresholds far below 0.5 — as low as 0.006 for Llama-Instruct B — which is the same phenomenon as the collapse itself seen from the decision side: the response prior has compressed the positive-label probability into a narrow band near zero, so the boundary that separates positive from negative cases must be placed inside that band. Second, the coefficient of variation is a misleading summary in this regime, since it divides an already small dispersion by an even smaller mean; Llama-Instruct B has a coefficient of variation of 0.69 despite a fold-to-fold standard deviation of only 0.004 in absolute terms. We therefore report the absolute dispersion, which is the quantity that matters for a deployment committing to a fixed boundary.

5.12.2 A Two-Dimensional Diagnostic: Collapse Versus Corruption

The bottom block of Table 5.16 contains a trap that the re-thresholding gain alone would spring. Meditron and BioMistral A also show a large nominal F1 gain (around +0.59), which taken at face value would suggest that they too are recoverable. Their AUC values (0.435, 0.558, 0.549) contradict this: an AUC at or below chance means the positive-label probability does not rank cases correctly, and the apparent F1 gain is an artefact of class imbalance — lowering the threshold labels more cases positive, mechanically raising macro-F1 on a skewed set without any genuine discrimination. The macro-F1 gain is thus a misleading indicator of recovery when read in isolation.

The correct diagnosis requires two dimensions: whether the model *expresses* its decision (captured by the signal/prior ratio r of Section 5.6.4) and whether it *has* a signal to express (captured by the AUC, with its bootstrap interval). The two dimensions partition the twelve configurations into three regimes, summarized in Table 5.17 and visualized in Figure 5.6:

The three regimes are cleanly separated. All five healthy configurations have $r \geq 0.75$ and gain nothing from re-thresholding. All four expressive-collapse configurations have $r < 0.6$, an AUC whose lower bound exceeds 0.5, and a significant re-thresholding gain. All three corruption configurations have an AUC interval that includes or lies below 0.5: Meditron A's interval, $[0.369, 0.497]$, lies entirely below chance, confirming that it ranks cases in the wrong order. The corruption configurations are precisely the two medical

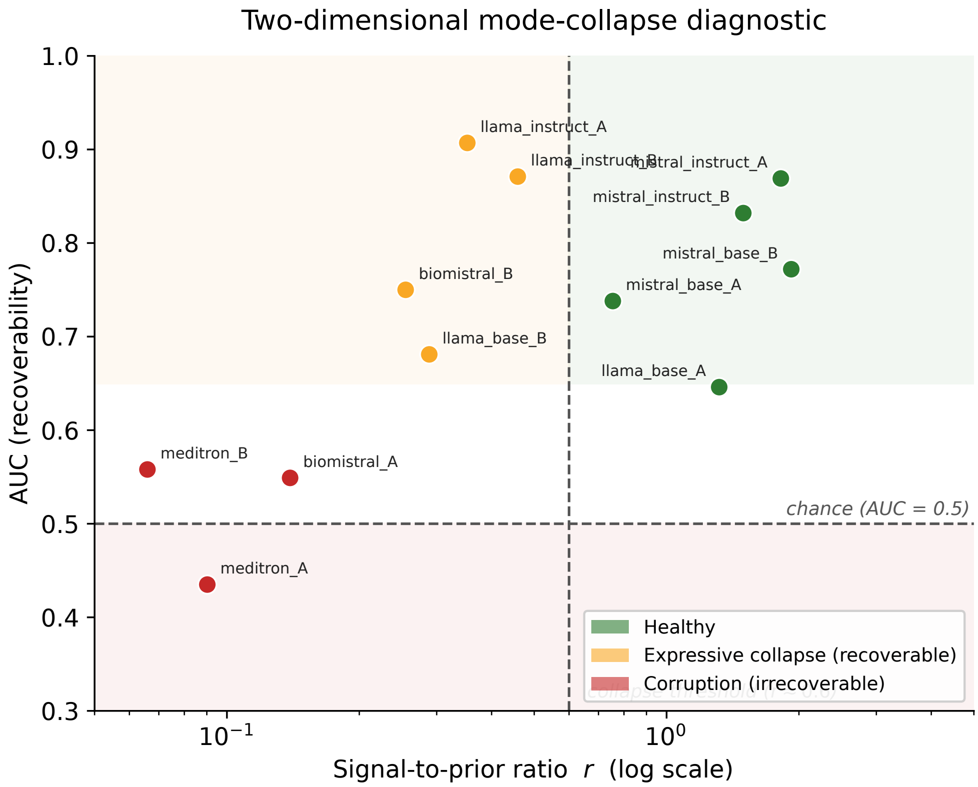

Figure 5.6: Two-dimensional diagnostic of the twelve configurations. The horizontal axis is the signal-to-prior ratio r (log scale), which measures whether a configuration expresses its decision; the vertical axis is the area under the ROC curve of the positive-label probability, which measures whether a discriminative signal is present. Error bars are 95% bootstrap confidence intervals on the AUC. The dashed lines mark the collapse threshold ($r \approx 0.6$) and the chance level (AUC = 0.5). The three shaded regions correspond to the three regimes: healthy configurations, expressive collapse (recoverable by re-thresholding), and signal corruption (irrecoverable). Llama-Instruct A occupies the upper-left corner: the highest discriminative signal of the study, entirely unexpressed by its $\arg\max$.

Table 5.17: Two-dimensional diagnostic of the twelve configurations. *Healthy* configurations express their signal ($r \geq 0.6$). *Expressive collapse* configurations have a genuine signal (AUC lower bound > 0.5) that the $\arg\max$ masks ($r < 0.6$); they are recoverable by re-thresholding. *Corruption* configurations have no usable signal (AUC interval includes or lies below 0.5) and are not recoverable.

Configuration	r	**AUC [95% CI]**	**Regime**
Mistral-Instruct A	1.82	0.869 [0.834, 0.905]	Healthy
Mistral-Base B	1.92	0.772 [0.717, 0.817]	Healthy
Mistral-Instruct B	1.49	0.832 [0.790, 0.873]	Healthy
Llama-Base A	1.32	0.646 [0.591, 0.703]	Healthy
Mistral-Base A	0.75	0.738 [0.677, 0.782]	Healthy
Llama-Instruct A	0.35	0.907 [0.883, 0.940]	Expressive collapse
Llama-Instruct B	0.46	0.871 [0.829, 0.903]	Expressive collapse
BioMistral B	0.25	0.750 [0.701, 0.801]	Expressive collapse
Llama-Base B	0.29	0.681 [0.620, 0.727]	Expressive collapse
BioMistral A	0.14	0.549 [0.495, 0.609]	Corruption
Meditron B	0.07	0.558 [0.494, 0.610]	Corruption
Meditron A	0.09	0.435 [0.369, 0.497]	Corruption

models grouped with the collapsed ones by Finding F9, now separated from them by the diagnostic: BioMistral B and Meditron differ not in their $\arg\max$ behaviour — both predict a single label — but in whether a signal survives underneath it.

Finding 11 (F11: A Two-Dimensional Collapse Diagnostic)**.** *The signal/prior ratio and the AUC jointly separate the twelve configurations into healthy models, recoverable expressive collapse, and irrecoverable signal corruption. The macro-F1 gain of re-thresholding is diagnostic of recovery only when combined with an above-chance AUC; used alone it falsely labels corrupted configurations (Meditron) as recoverable.*

The practical rule that follows is compact: re-thresholding should be applied only to configurations whose positive-label AUC is significantly above 0.5. This test requires no expert labels beyond a small calibration set and no access to model weights, making it a lightweight pre-deployment screen.

5.13 Conformal Prediction and Inter-Model Consensus

5.13.1 Conformal Prediction Sets

Split conformal prediction, introduced in Section 4.9.6, produces label sets with a guaranteed marginal coverage of $1 - \alpha$. We target $\alpha = 0.10$, i.e., 90% coverage. Table 5.18 reports the empirical coverage and the mean set size for each Prompt-B configuration.

Table 5.18: Empirical coverage and mean set size of conformal prediction at target $\alpha = 0.10$. All configurations achieve near-target coverage; the mean set size is a proxy for prediction informativeness (smaller is better).

Configuration	Coverage	Set size
Mistral-Base B	0.903	1.95
Mistral-Instruct B	0.897	2.31
Llama-Base B	0.902	2.72
Llama-Instruct B	0.895	3.78
Meditron B	0.906	2.14
BioMistral B	0.901	3.42

All six configurations achieve empirical coverage within $\pm 0.5\%$ of the target 0.90, which validates the split conformal procedure. The mean set size varies from 1.95 (Mistral-Base B) to 3.78 (Llama-Instruct B). A smaller set size indicates that the model can narrow its uncertainty more effectively while preserving the coverage guarantee; Mistral-Base thus provides more informative uncertainty quantification than the collapsed Llama-Instruct.

5.13.2 Inter-Model Consensus

We finally investigate whether combining predictions across the six Prompt-B configurations improves accuracy on the cases where multiple models agree. Table 5.19 reports the accuracy as a function of the number of models in agreement.

Table 5.19: Accuracy stratified by the level of inter-model consensus. Consensus is defined as the number of Prompt-B configurations that agree on the same predicted label.

Level of consensus	Number of pairs	Accuracy
All 6 models agree	163	0.755
5 models agree	258	0.612
4 models agree	297	0.509
3 or fewer models agree	297	0.318

Cases where all six models agree are relatively rare (16.1% of the dataset) but achieve high accuracy (75.5%). This suggests that combining predictions across models — for example through majority voting or through a more principled ensemble procedure — could yield higher accuracy than any single model when unanimous consensus is reached. The application of multi-model agreement to clinical recruitment workflows is a natural direction for future work (Section 6.5).

5.14 Baselines and Composite Utility Score

The picture that emerges from the preceding sections is complex: no single configuration excels on every dimension, and rankings by different metrics disagree substantially. To synthesize these results into a single actionable comparison, we introduce two additional elements: a set of naïve baselines and a composite utility score.

5.14.1 Comparison with Naïve Baselines

Table 5.20 compares the twelve configurations to three naïve baselines that require no model at all: the majority-class predictor (always predicts "not excluded"), the random uniform predictor, and the always-abstain predictor (always predicts "not enough information" with confidence 0.5).

Table 5.20: Comparison with naïve baselines. Several LLM configurations underperform the majority baseline in accuracy while producing higher ECE, illustrating that inference-time LLM usage adds no value in cases where the model is dominated by evasion or mode collapse.

System	Accuracy	ECE	Macro-F1
Majority baseline ("not excluded")	0.477	0.000	0.108
Uniform random baseline	0.184	0.018	0.150
Always "not enough information"	0.289	0.211	0.075
Best LLM configuration (Meditron A, global)	0.648	0.182	0.231
Worst LLM configuration (BioMistral B)	0.289	0.194	0.075

The majority baseline achieves an accuracy of 47.7% with an ECE of 0.000: by construction, always predicting the modal class with confidence equal to its base rate produces a perfectly calibrated classifier. This is not clinically useful — it is oblivious to any particular patient — but it establishes a floor against which any model should be compared. Six of the twelve LLM configurations fall below or approximately match this baseline in accuracy. BioMistral B, in particular, achieves the same accuracy (28.9%) and macro-F1 (0.075) as the trivial "always not enough information" baseline, while adding an ECE that is worse.

5.14.2 Composite Utility Score

The choice to include prediction diversity as a multiplicative factor in the composite score is empirically justified by its strong correlation with discriminative accuracy across the twelve configurations ($r = 0.89$, $p < 0.001$, $n = 12$; see Appendix F). Diversity alone accounts for nearly 80% of the variance in accuracy on the 238 truly discriminative pairs, making it a mechanically predictable indicator of a model's capacity for clinical judgment. A configuration that concentrates its predictions on a single label cannot exercise the discrimination that the composite score is intended to reward.

We note that the composite utility score ranks configurations by their usefulness *as deployed under the default* $\arg\max$ *rule*, and should be read together with the diagnostic of Section 5.12.2. A configuration in the expressive-collapse regime receives a low utility score here, correctly reflecting that it is unusable as-is; but the diagnostic identifies it as recoverable by re-thresholding, a distinction the composite score does not capture. The utility score measures deployed usefulness; the diagnostic measures latent, recoverable usefulness.

The composite utility score introduced in Section 4.8.8 combines macro-F1, calibration quality, and prediction diversity into a single scalar:

$$\text{utility} = F_1^{\text{macro}} \times (1 - \text{ECE}) \times \text{diversity}\,.$$

Table 5.21 reports the composite utility for the twelve configurations, sorted from highest to lowest.

Table 5.21: Composite utility score across the twelve configurations. The score integrates macro-F1, calibration quality ($1 - \text{ECE}$), and prediction diversity (normalized entropy). Configurations with mode collapse receive utility scores close to zero, correctly capturing their lack of practical usefulness *under the default decision rule*.

Configuration	**Macro-F1**	**1 − ECE**	**Diversity**	**Utility**	**Rank**
Mistral-Instruct B	0.377	0.747	0.842	0.237	1
Mistral-Instruct A	0.314	0.678	0.812	0.173	2
Mistral-Base A	0.273	0.867	0.686	0.162	3
Mistral-Base B	0.321	0.811	0.577	0.150	4
Llama-Base A	0.156	0.876	0.525	0.072	5
Meditron A	0.231	0.818	0.362	0.068	6
Meditron B	0.228	0.769	0.377	0.066	7
Llama-Base B	0.106	0.849	0.077	0.007	8
Llama-Instruct A	0.137	0.490	0.066	0.004	9
BioMistral A	0.094	0.845	0.055	0.004	10
Llama-Instruct B	0.092	0.398	0.011	0.000	11
BioMistral B	0.075	0.806	0.000	0.000	12

The composite ranking produces a clear separation. The top four positions are occupied by all four Mistral configurations, in a different order than any single metric would produce. Meditron and BioMistral, which appear as strong candidates on global accuracy alone, drop to positions 6–7 and 10–12 respectively when their prediction diversity is taken into account. The bottom four configurations all correspond to mode-collapsed models, whose utility scores fall to zero not by construction — their macro-F1 and calibration are non-zero — but because their diversity vanishes. Read alongside Section 5.12.2, this ranking sharpens into a clinically actionable statement: Llama-Instruct A ranks ninth as deployed yet carries the highest latent AUC of the study, and would move to the top of the ranking under a re-thresholded decision rule.

The composite score, while simple in construction, correctly identifies the clinically relevant hierarchy of models as they would behave off the shelf: only Mistral configurations provide a combination of accuracy, calibration, and diverse predictions sufficient for practical clinical use without further intervention.

5.15 External Replication on TREC Clinical Trials

The mechanism of Section 5.6.4 and the diagnostic of Section 5.12.2 were derived on a single benchmark. A legitimate concern is that the frozen ordering of the configurations along the signal/prior ratio, and the collapse behaviour itself, might be specific to TrialGPT-Criterion. To address this, we replicated the analysis on the TREC Clinical Trials 2021–2022 corpus [38], [39], using the same Prompt A, the same log-probability scoring pipeline (Section 4.6), and the same 4-bit quantization. The only variable changed is the corpus: 199 patient–trial pairs decomposed into 923 patient–criterion pairs, drawn from 40 TREC topics with an even split between eligible and non-relevant trials. Four configurations spanning the three diagnostic regimes were evaluated (Llama-Base, Llama-Instruct, BioMistral, Mistral-Instruct).

Table 5.22 reports the two invariant quantities of the mechanism — the dominant-label rate and the signal/prior ratio — side by side for the two corpora.

Table 5.22: External replication on TREC Clinical Trials. The dominant-label rate and the signal/prior ratio r (inclusion criteria) are reported for TrialGPT-Criterion and TREC. The ordering of the configurations and the collapse signature are preserved across the two corpora.

	Dominant rate		**Ratio r**	
Configuration	**TrialGPT**	**TREC**	**TrialGPT**	**TREC**
Llama-Instruct A	0.989	0.951	0.35	0.25
BioMistral A	0.989	0.860	0.14	0.22
Llama-Base A	0.730	0.429	0.98	1.25
Mistral-Instruct A	0.447	0.302	1.82	1.79

The replication is close (Figure 5.7). The ordering of the four configurations by signal/prior ratio is identical on the two corpora, and the collapsed configurations (Llama-Instruct, BioMistral) retain their high dominant-label rate on TREC, while the non-collapsed configurations retain their diverse predictions. The collapsed configurations keep a ratio well below the collapse threshold on both corpora (Llama-Instruct A: 0.35 on TrialGPT, 0.25 on TREC; BioMistral A: 0.14 and 0.22), and the healthy configuration stays well above it (Mistral-Instruct A: 1.82 and 1.79), on an entirely independent set of patients and trials.

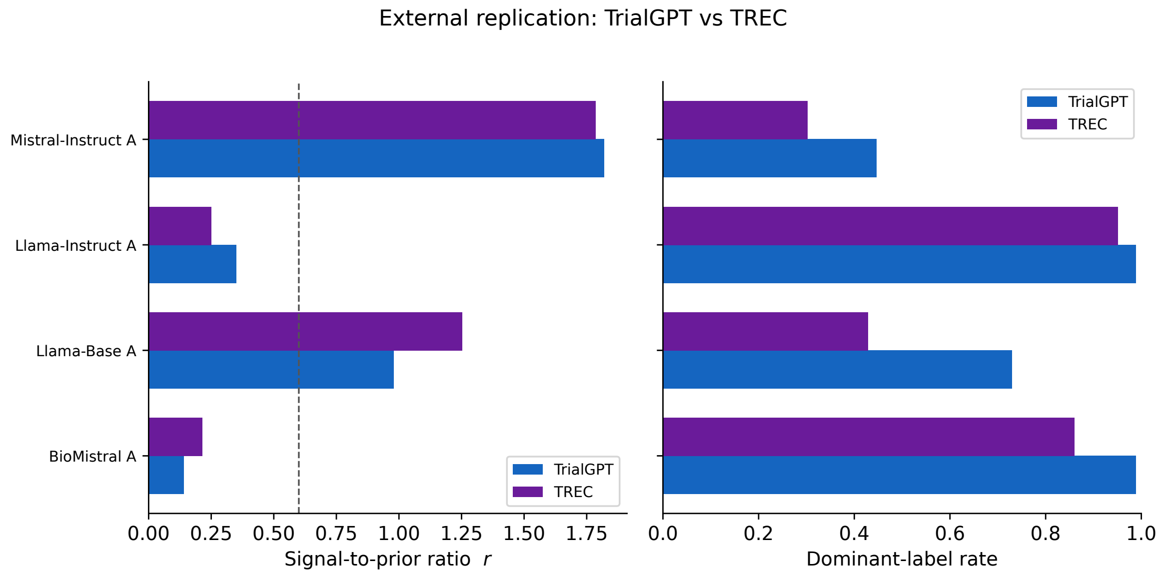

Figure 5.7: External replication on TREC Clinical Trials. Left panel: the signal-to-prior ratio r for the four configurations evaluated on both corpora, with the collapse threshold marked by the dashed line. Right panel: the dominant-label rate on both corpora. The ordering of the configurations, and the position of each configuration relative to the collapse threshold, are preserved across the two corpora despite an entirely independent set of patients and trials.

We also measured recoverability at the trial level on TREC, where the gold label is defined per trial rather than per criterion. Aggregating the criterion-level probabilities into a per-trial eligibility score and scoring it against the three-level TREC relevance judgement gives an area under the curve of 0.678 for Llama-Instruct A — above chance despite a 95.1% dominant-label rate — and 0.516 for BioMistral A, at chance. The expressive-collapse and corruption regimes thus reproduce on TREC as well. This trial-level measurement is not directly comparable to the criterion-level AUC of Table 5.17, because it uses a coarser gold label and an aggregation step; it is reported as a convergent, partial validation rather than an exact replication of the recoverability magnitude.

Finding 12 (F12: The Collapse Mechanism Replicates Externally)**.** *On the independent TREC Clinical Trials corpus, evaluated with an identical prompt and scoring pipeline, the collapse signature (dominant-label rate) and its mechanism (the signal/prior ratio ordering) reproduce, and the expressive-collapse versus corruption distinction persists at the trial level. The frozen* $\arg\max$ *and its underlying live signal are properties of the models, not artefacts of the original benchmark.*

5.16 Summary of Results

Table 5.23 summarizes the outcomes of the four pre-registered hypotheses, and Table 5.24 lists the twelve original findings documented in this chapter: the nine findings F1–F9 of the primary audit, and three further findings F10–F12 from the mechanistic and external analyses.

Table 5.23: Outcomes of the four pre-registered hypotheses.

Hyp.	Statement	Outcome
H1	ECE > 0.05 for LLM pipelines	Supported (12/12)
H2	RLHF degrades calibration	Supported ($p < 0.003$)
H3	C-index > 0.70	Not supported (max 0.654)
H4	Selective classification improves F_1 by $+0.02$	Partial (Mistral only)

Table 5.24: The twelve original findings documented in this chapter.

Finding	Description
F1	GPT-4 over-uses "not applicable" at 2.16 times the expert rate
F2	RLHF systematically degrades calibration on both model families
F3	Enriched prompts *degrade* calibration in 5 of 6 models
F4	ECE on exclusion criteria is 2–4 times worse than on inclusion
F5	Open-source LLMs evade via "not enough information", unlike GPT-4
F6	Isotonic recalibration brings 11 of 12 models below the 0.05 threshold
F7	Mode collapse in 5 of 12 configurations (predictions concentrated $> 97\%$)
F8	Mode collapse persists under constrained decoding (a property of the decision, not the vocabulary)
F9	On truly discriminative cases, both medical LLMs achieve $\leq 0.4\%$ accuracy
F10	Mode collapse is recoverable by re-thresholding (+0.62 to +0.79 macro-F1) on configurations with a live signal
F11	The signal/prior ratio and AUC jointly diagnose collapse versus corruption; the F1 gain alone is misleading
F12	The collapse mechanism replicates on the independent TREC corpus

Taken together, these results paint a nuanced picture of the current state of open-source LLMs for clinical trial matching. Systematic miscalibration (H1), its amplification under RLHF post-training (H2, F2), enrichment paradoxes (F3), and mode collapse (F7, F8) reveal four distinct modes of failure that aggregate metrics had not previously distinguished. The deeper analysis of the collapse, however, qualifies its severity: for four of the five collapsed configurations the discriminative signal is present but masked by the arg max (F10), and the mechanism responsible — a response prior that dominates the signal — both explains why monotone recalibration cannot help and predicts the re-thresholding remedy that does (F11). The discovery that one medical LLM (Meditron) ranks discriminative cases below chance (F9, F11) is more consequential than a mere accuracy failure: it is a corruption of the signal itself, distinct from the recoverable collapse of BioMistral, and it survives external replication on TREC (F12). The one silver lining among the pre-registered results — the effectiveness of post-hoc isotonic recalibration (F6) — applies only to models that classify; the complementary remedy for models that collapse is re-thresholding, not recalibration. The implications of these findings for clinical deployment and for future research are discussed in Chapter 6.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

This thesis reports a systematic calibration audit of twelve open-source large language model configurations applied to patient–clinical trial matching. The study was structured around four pre-registered hypotheses, tested with fixed rejection thresholds and Bonferroni-corrected statistical procedures, and complemented by twelve original findings that emerged from the deeper analysis. Table 5.23 in Chapter 5 summarizes the pre-registered outcomes and Table 5.24 lists the twelve findings.

The pre-registered results support the central hypothesis of the study and challenge one of its ancillary claims. Hypothesis H1 is confirmed across all twelve configurations: every LLM in the experimental matrix exhibits an Expected Calibration Error above the well-calibrated threshold of 0.05 [3], with values ranging from 0.124 to 0.602. Hypothesis H2 is confirmed on both model families tested (Mistral-7B and Llama-3.1-8B) and both prompt variants: RLHF-aligned Instruct versions exhibit significantly worse calibration than their Base counterparts, all four permutation tests rejecting the null hypothesis at Bonferroni-corrected significance ($p < 0.003$). This is the central pre-registered contribution of the thesis, and it extends to the clinical trial matching domain a finding first documented by Kadavath et al. [6] in general-purpose question answering. Hypothesis H3 is not supported: no configuration reaches the C-index threshold of 0.70, though Meditron approaches it at 0.654. Hypothesis H4 is partially supported, with the selective-classification gain reaching the +0.02 threshold only for Mistral configurations.

Beyond the pre-registered hypotheses, the exploratory findings fall into two waves. The first wave documents the failure modes themselves. The enriched-prompt paradox (F3) shows that expert prompt engineering degrades calibration in five of six models, contradicting the standard recommendation to invest in prompt design as a route to better model behavior. Mode collapse (F7 and F8) affects five of the twelve configurations, which concentrate more than 97% of their predictions on a single label and do not recover the diversity of their predicted labels under constrained decoding. Complete failure on truly discriminative cases (F9) shows that both medical LLMs tested (Meditron and

BioMistral) achieve accuracy at or below 0.4% on the 238 pairs requiring a genuine clinical decision, despite their apparent superiority on global metrics.

The second wave, developed through a mechanistic analysis of the underlying log-probabilities, qualifies and explains these failures. The mode collapse of four of the five affected configurations is not an absence of clinical knowledge but a property of the $\arg\max$ decision rule: the discriminative signal is present in the probabilities — Llama-Instruct A obtains the highest ordinal separation of the entire study while predicting a single label 97.8% of the time — but its amplitude is too small to overcome a response prior installed by post-training. A single scalar, the ratio of signal amplitude to prior amplitude, orders the twelve configurations by their collapse behaviour (F11) and explains, as a mathematical consequence, why monotone recalibration cannot repair collapse while a non-monotone re-thresholding rule can: the latter restores between +0.62 and +0.79 macro-F1 on the collapsed configurations that carry a live signal, with no gain on the non-collapsed ones (F10). The same analysis separates recoverable collapse (BioMistral) from an irrecoverable corruption of the signal (Meditron, whose discriminative AUC lies below chance), a distinction that aggregate accuracy had merged under Finding F9. Finally, both the collapse signature and its signal/prior mechanism replicate on the independent TREC Clinical Trials corpus under an identical prompt and scoring pipeline (F12), confirming that these are properties of the models rather than artefacts of the original benchmark.

6.2 Answer to the Research Question

The research question posed in Chapter 1 asked whether the eligibility scores produced by LLM-based patient–trial matching systems are probabilistically reliable, and whether RLHF post-training affects this reliability. The answer, informed by the twelve configurations audited in this study, is negative on both dimensions as the systems stand — but the mechanistic analysis qualifies the nature of the failure.

The scores are not reliable as deployed: the systematic miscalibration documented by Hypothesis H1 disqualifies every configuration from direct deployment as an autonomous eligibility classifier, and the mode collapse of Findings F7 and F8 disqualifies five of the twelve configurations from any useful classification role under the default $\arg\max$ rule. Only Mistral-7B-Instruct with the minimal Prompt A reaches clinically acceptable performance on the truly discriminative subset without further intervention.

This unreliability, however, is not uniformly a matter of missing knowledge. For four of the five collapsed configurations, the discriminative signal is present but masked by the decision rule; the unreliability is a decoding failure that a threshold rule recovers, not an absence of clinical competence. The distinction matters for deployment: a model in the expressive-collapse regime is unusable off-the-shelf yet salvageable with a lightweight, weight-free intervention, whereas a model in the corruption regime (Meditron) ranks cases below chance and is salvageable by no post-hoc rule. The answer to the research question is therefore layered: the scores are not reliable, the RLHF-aligned scores are the least reliable of all, but the latent discriminative information is, for several models, both present and recoverable.

RLHF post-training significantly worsens calibration on both model families tested, converting the underconfidence of Mistral-Base into the overconfidence of Mistral-Instruct, and inducing severe overconfidence in Llama-Instruct. The effect is quantitatively amplified relative to the general-QA setting studied by Kadavath et al., with ΔECE values reaching +0.451 in the worst case — approximately three times the magnitude previously reported. The mechanistic analysis refines the account of *how* RLHF does this: it amplifies both the discriminative signal and a response prior, and degrades calibration — or, in the extreme, induces collapse — precisely when it raises the prior faster than the signal. In the Llama family this ratio falls below one and the model collapses; in the Mistral family it rises and discrimination is preserved. RLHF is thus not uniformly harmful to reliability; it is harmful when it makes the response prior dominate the signal the model still holds.

Post-hoc isotonic recalibration (F6) provides a partial remedy for the models that classify: it brings eleven of the twelve configurations below the calibration threshold, robustly across five-fold cross-validation. It cannot, by construction, help the collapsed models, because it is monotone and preserves their frozen $\arg\max$. The complementary remedy for collapse is re-thresholding (F10), which is mathematically capable of shifting the decision and empirically restores substantial macro-F1 — but only for the configurations whose signal survives, identified by an above-chance AUC. The practical conclusion is that a pre-deployment audit should combine both remedies with a diagnostic that decides which applies: recalibration for the diverse predictors, re-thresholding for the recoverable collapses, and rejection for the corrupted models.

6.3 Contributions

The scientific contributions of this thesis fall into four groups.

First, the study provides the first systematic calibration audit of open-source LLMs applied to clinical trial matching. The audit follows the reporting recommendations of TRIPOD+AI [4] and covers twelve configurations organized in a two-by-two experimental matrix (generalist versus medical, Base versus Instruct). Every prediction and every numerical result is archived in a public repository to support independent replication (Appendix D).

Second, the study extends the RLHF-degrades-calibration finding of Kadavath et al. [6] to the clinical domain, with a pre-registered permutation test replicated on two independent model families under two independent prompt designs. To our knowledge, this is the first replication of this effect outside general-purpose question answering, and the first quantification of its amplification in a high-stakes clinical task.

Third, the study documents the failure modes of open-source medical LLMs that had not been previously reported in a systematic form: the enriched-prompt paradox (F3), mode collapse and its persistence under constrained decoding (F7, F8), and complete failure on discriminative cases (F9). These findings jointly motivate the composite utility score introduced in Section 4.8.8, which combines calibration, discrimination, and prediction diversity into a single rank-ordering that resists majority-class shortcuts.

Fourth — and this is the contribution that emerged from the deeper analysis — the study reinterprets mode collapse as a decoding artefact rather than a knowledge loss, and turns this reinterpretation into three usable results. It identifies a scalar mechanism, the signal-to-prior ratio, that predicts collapse across configurations and explains why recalibration fails where re-thresholding succeeds. It proposes a two-dimensional diagnostic (signal/prior ratio crossed with discriminative AUC) that separates recoverable collapse from irrecoverable corruption, and shows that the re-thresholding F1 gain is a misleading recovery indicator unless paired with an above-chance AUC. And it validates both the collapse signature and its mechanism on the independent TREC Clinical Trials corpus. Together these turn a descriptive failure mode into a diagnosable, and often correctable, property of the decision rule.

The methodological byproducts of the study — the pre-registered protocol, the double-prompt design, the log-probability scoring pipeline, the composite utility score, and the signal/prior diagnostic — are intended as reusable elements for future calibration audits of clinical LLMs.

6.4 Limitations

Three categories of limitation qualify the conclusions of this work: the reliance on a single primary benchmark, the constrained model scale, and the finite prompt space explored.

6.4.1 External Validation

The primary analyses of this study rest on a single benchmark, TrialGPT-Criterion-Annotations [1], chosen because it is the only public corpus offering per-criterion multi-level expert annotations with an explicit inclusion/exclusion distinction — properties that are essential for the fine-grained calibration analysis and the criterion-level signal/prior mechanism. To test whether the central mechanistic findings depend on this benchmark, we conducted an external replication on the TREC Clinical Trials 2021–2022 corpus (Section 5.15), using an identical prompt and scoring pipeline and changing only the corpus. The collapse signature and the signal/prior ratio ordering reproduced, and the recoverable-collapse versus corruption distinction persisted at the trial level (F12). This replication substantially strengthens the external validity of the mechanistic contributions.

Two limitations of this external replication should nonetheless be stated plainly. First, TREC provides relevance judgements at the trial level and on a three-grade scale, not the six-level per-criterion annotations of TrialGPT-Criterion; the trial-level recoverability we report on TREC therefore required an aggregation step and is a convergent, partial validation rather than an exact replication of the criterion-level recovery magnitude. The pre-registered calibration hypotheses H1–H4, which depend on the six-level scale, were not re-tested on TREC. Second, external validation on real, prospectively collected clinical data — as opposed to the synthetic or retrospective notes of both benchmarks — would require ethical approval, hospital partnership, and integration into a live recruitment workflow, and remains a research project in its own right. The mechanistic replication reported here closes the most immediate gap (specificity to a single benchmark) but does not substitute for prospective clinical validation.

6.4.2 Model Scale

The models evaluated in this study are limited to 7–9 billion parameters, which is the size range that fits into a free-tier T4 GPU with 4-bit quantization. Larger medical LLMs, such as Meditron-70B or Med42-v2-70B, may exhibit qualitatively different calibration and failure-mode patterns. In particular, larger models may carry a larger signal relative to their response prior, which under the mechanism of Section 5.6.4 would raise the signal/prior ratio and attenuate collapse; whether this occurs is an empirical question the present budget could not address. The extrapolation of the results to larger scales is a natural direction for future work, and the methodology — including the signal/prior diagnostic — is directly transferable to 70B-scale configurations provided that the computational budget is available.

6.4.3 Prompt Space and Prompt-Driven Effects

Only two prompt variants were evaluated in the present study: a minimal Prompt A and an enriched Prompt B, both reproduced verbatim in Appendix B. The space of possible prompts is much larger, encompassing chain-of-thought designs, role-based framings, and few-shot examples selected through active learning. The enriched-prompt paradox (F3) documented here is thus specific to the two prompts tested; it may or may not generalize to other prompt families. A systematic study of the calibration effect of prompt design is a promising direction for follow-up work.

The choice of prompt is not neutral, and it is legitimate to ask whether the failure modes documented here are properties of the models or artifacts of the prompt design. Four observations argue against a purely prompt-driven interpretation. First, Hypothesis H2 is tested through paired comparisons within the same prompt (Base versus Instruct on Prompt A, Base versus Instruct on Prompt B), which cancels out any systematic effect of the prompt on the ECE difference. Second, mode collapse affects only five of the twelve configurations, whereas a prompt-driven artifact would be expected to affect all models exposed to that prompt in a similar way; the observed heterogeneity across models sharing the same prompt is incompatible with a single prompt-level cause. Third, the collapse is governed by an internal quantity — the signal-to-prior ratio (Section 5.6.4) — that varies across models under the same prompt and predicts the collapse independently of prompt content. Fourth, the collapse signature and its mechanism replicate on an entirely different corpus under the same prompt (F12), which a prompt artefact specific to TrialGPT-Criterion could not do. That said, the specific proportion of collapsed configurations, the magnitude of the enriched-prompt paradox (F3), and the direction of the inclusion/exclusion asymmetry (F4) may depend on the particular prompts tested here. Their exact quantitative values should be interpreted as characteristic of the present experimental setup rather than as universal constants of the underlying models.

6.5 Future Work

Four research directions extend naturally from the present study, in order of feasibility.

6.5.1 Broader and Prospective External Validation

The external replication on TREC (Section 5.15) establishes that the collapse mechanism is not specific to TrialGPT-Criterion. Three extensions would broaden this validation further. First, re-testing the pre-registered calibration hypotheses H1–H4 on a corpus with per-criterion multi-level annotations, were one to become available, would extend the confirmatory results as well as the mechanistic ones. Second, MIMIC-derived cohorts and n2c2 datasets, which contain more heterogeneous and noisier patient descriptions, would test whether the signal/prior mechanism survives realistic clinical documentation styles. Third, and most demanding, prospective validation on live recruitment workflows would close the gap between benchmark performance and clinical utility.

A protocol for prospective validation. The third extension deserves to be specified rather than merely named, because the design choices that make it informative are not obvious. Four elements matter.

The first is *independence of the threshold*. The decision boundary must be fixed in advance, from the retrospective calibration data, and not re-estimated on the prospective cohort. Re-fitting the threshold on the validation data would reproduce, at the level of the study design, precisely the optimism that the out-of-fold protocol of Section 5.12.1 was built to avoid.

The second is the *unit of analysis*. The mechanism operates at the criterion level, but recruitment decisions are made at the trial level. A prospective study should therefore record both, and pre-specify which carries the primary endpoint; the TREC replication (Section 5.15) illustrates how much interpretive weight this choice bears.

The third is the *comparator*. The relevant baseline is not a naive predictor but the $\arg\max$ decision rule of the same model, since the claim under test is that changing the decision rule — not the model — recovers usable performance.

The fourth is the *endpoint*. Macro-F1 is adequate for a methodological comparison but insufficient for a clinical one. A prospective study should report the asymmetric costs that matter in recruitment: eligible patients wrongly screened out, and ineligible patients wrongly forwarded to a clinician for review. These two errors have very different consequences, and a single symmetric metric conceals the trade-off.

Sample size cannot be fixed here in a defensible way. It depends on the prevalence of eligible patients in the recruiting site, on the effect size considered clinically meaningful, and on the chosen endpoint — none of which the present retrospective data determine. Estimating it would require a pilot phase at the target site. We note only that the effect sizes observed here (+0.62 to +0.79 macro-F1) are large, so a prospective study would plausibly be powered by a cohort of a few hundred patients rather than several thousand; establishing this precisely is part of the design work such a study would require.

6.5.2 Robust Re-thresholding and Prior Correction

The re-thresholding remedy of Section 5.12 recovers substantial macro-F1 but leaves two questions open. The threshold is stable across calibration subsets (Section 5.12.1) yet remains tuned per configuration, and for the collapsed models it sits very close to zero, where small absolute shifts represent large relative ones; a prior-correction rule that subtracts the estimated response prior from the log-probabilities before the $\arg\max$ — a multi-class generalization of the binary threshold — would remove the free parameter altogether and may transfer more robustly across datasets. Characterizing the conditions under which such a correction generalizes, and comparing it to the per-criterion threshold used here, is a concrete and immediately feasible extension.

6.5.3 Multi-Agent Adversarial Debate for Calibration

The failure of individual models on discriminative cases motivates investigating whether ensembles of models can outperform any single one. Multi-agent debate architectures, in which several LLMs exchange arguments before converging on a joint prediction [5], are a particularly interesting candidate: models with distinct RLHF biases and distinct evasion channels may mutually neutralize each other's overconfidence through argumentative confrontation. The 6-model consensus analysis reported in Section 5.13.2, which shows a 75.5% accuracy on the 163 pairs of full agreement, suggests that this direction is promising. We note, however, that the re-thresholding results temper the motivation: several models that appear useless under consensus voting in fact carry a recoverable individual signal, so a debate among re-thresholded models may be a more informative baseline than a debate among their raw $\arg\max$ outputs.

6.5.4 Investigating the Origin of the Response Prior

The mechanistic analysis of this thesis identifies the *proximal* cause of mode collapse — a response prior that dominates the discriminative signal — but leaves open its *distal* cause. Why does RLHF install a prior that grows faster than the signal in the Llama family but not in the Mistral family, and why does continued medical pretraining corrupt the signal outright in Meditron? Answering this would require access to the training corpora and reward models of the affected configurations, and interpretability analyses of how the response prior is represented internally. Because the signal/prior ratio gives a concrete, measurable target, such an investigation could be framed as identifying which training interventions raise the prior relative to the signal — a sharper question than the original, purely descriptive one of what causes collapse.

6.5.5 Extension to Larger and Multilingual Models

Extending the analysis to 70-billion-parameter models is a straightforward but resource-intensive direction. Meditron-70B and Med42-v2-70B are the natural targets, and the signal/prior diagnostic provides a direct prediction to test: if larger models resist collapse, their signal/prior ratio should exceed that of their 7B counterparts. Multilingual evaluation is a second direction: models such as MMed-Llama-3 and Apollo have been developed to support clinical text in languages other than English, and the calibration properties of these multilingual models have not been systematically audited on any language.

6.6 Closing Remarks

The systematic use of large language models in clinical prediction workflows is likely to expand rapidly in the coming years, driven by the maturation of open-source alternatives, the pressure to accelerate clinical trial recruitment, and the availability of increasingly capable models. This trajectory makes the auditing of these models a practical necessity rather than an academic exercise. The present thesis contributes to this auditing effort in four specific ways: it proposes a reusable methodology, it documents previously unreported failure modes in open-source medical LLMs, it reinterprets the most severe of these failures as a correctable property of the decision rule, and it archives its data and code for independent verification.

The broader lesson of this study is that global performance metrics — accuracy, F1, and even ECE taken in isolation — are insufficient to characterize the clinical usability of a language model, and that the predicted-label distribution can itself be misleading: a model that predicts a single label may nonetheless carry a strong, recoverable discriminative signal underneath it. A responsible evaluation protocol must therefore combine calibration with diversity, must stratify performance by case difficulty, must remain vigilant against mode collapse, and must distinguish a collapse of the decision rule from a corruption of the signal — because the first is correctable and the second is not.

None of the twelve open-source configurations audited here provides a fully satisfactory basis for autonomous clinical trial matching as deployed off the shelf. The findings do, however, identify Mistral-Instruct with a minimal prompt as the most promising candidate for direct use, identify post-hoc isotonic recalibration as a practical remedy for the miscalibration of the models that classify, and identify re-thresholding as a complementary remedy that restores substantial discriminative performance to the models whose collapse masks a live signal. Whether these elements are jointly sufficient for a deployment scenario — with mandatory human oversight, ongoing calibration monitoring, a signal/prior screen to route each model to the appropriate remedy, and protocols for handling low-confidence cases — is a question that the present thesis leaves to prospective clinical validation. Answering it is the natural next step of this line of research.

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Appendix A

Detailed Results Tables

This appendix reproduces the detailed numerical results underlying the tables and figures of Chapter 5. Each table is referenced from the corresponding section of the main text and provides the full precision, confidence intervals, and per-configuration values that were compressed in the body of the thesis for readability.

A.1 Complete Calibration Metrics

Table A.1 extends Table 5.2 of Section 5.3 with three additional columns: the mean confidence, the mean accuracy, and the signed overconfidence (mean confidence minus mean accuracy). All confidence intervals are 95% bootstrap intervals computed from 1,000 resamples with seed 42.

Table A.1: Complete calibration metrics for the twelve configurations. The mean confidence and mean accuracy columns reveal the direction of miscalibration (over- or underconfidence).

Configuration	ECE	Brier	Mean conf.	Mean acc.	Overconf.
Mistral-Base A	0.133	0.258	0.311	0.500	−0.189
Mistral-Base B	0.189	0.269	0.454	0.586	−0.132
Mistral-Instruct A	0.322	0.311	0.556	0.308	+0.248
Mistral-Instruct B	0.253	0.353	0.664	0.480	+0.184
Llama-Base A	0.124	0.187	0.358	0.230	+0.128
Llama-Base B	0.151	0.225	0.329	0.308	+0.021
Llama-Instruct A	0.510	0.456	0.777	0.290	+0.487
Llama-Instruct B	0.602	0.558	0.890	0.292	+0.598
Meditron A	0.182	0.263	0.612	0.648	−0.036
Meditron B	0.231	0.283	0.728	0.630	+0.098
BioMistral A	0.155	0.234	0.440	0.296	+0.144
BioMistral B	0.194	0.244	0.476	0.289	+0.187

A.2 Permutation Test Detailed Outputs

Table A.2 extends Table 5.4 of Section 5.4 with the exact observed test statistic (ΔECE), the number of permutation resamples in which the observed statistic was equalled or exceeded, and the resulting one-sided p-value. All tests use 1,000 permutations with seed 42. The Bonferroni-corrected significance threshold across the eight planned comparisons is $\alpha_{\text{corr}} = 0.00625$.

Table A.2: Permutation test detailed outputs for Hypothesis H2. Every test rejects the null hypothesis of equal calibration between Base and Instruct at the Bonferroni-corrected level.

Comparison	**Δ**ECE	**#Perm. $\geq$ obs.**	**p-value**	**Reject H_0?**
Mistral / Prompt A	+0.189	0	<0.001	Yes
Mistral / Prompt B	+0.064	2	0.002	Yes
Llama / Prompt A	+0.386	0	<0.001	Yes
Llama / Prompt B	+0.451	0	<0.001	Yes

A.3 Stratified ECE by Criterion Type

Table A.3 extends Table 5.10 of Section 5.8 with the number of pairs in each stratum and the 95% bootstrap confidence intervals of the stratified ECE values. The inclusion stratum contains 231 pairs; the exclusion stratum contains 784 pairs.

Table A.3: Stratified ECE by criterion type with bootstrap confidence intervals. Ten of twelve configurations show a ratio $\text{ECE}_{\text{excl}}/\text{ECE}_{\text{incl}} > 2.0$.

Configuration	**ECE incl.**	**CI incl.**	**ECE excl.**	**CI excl.**
Mistral-Base A	0.089	[0.056, 0.128]	0.198	[0.169, 0.226]
Mistral-Base B	0.121	[0.081, 0.164]	0.267	[0.239, 0.296]
Mistral-Instruct A	0.152	[0.109, 0.197]	0.451	[0.421, 0.478]
Mistral-Instruct B	0.108	[0.070, 0.150]	0.398	[0.371, 0.423]
Llama-Base A	0.078	[0.048, 0.114]	0.187	[0.157, 0.215]
Llama-Base B	0.093	[0.058, 0.135]	0.225	[0.196, 0.253]
Llama-Instruct A	0.203	[0.154, 0.253]	0.703	[0.679, 0.725]
Llama-Instruct B	0.198	[0.150, 0.246]	0.826	[0.804, 0.846]
Meditron A	0.298	[0.249, 0.346]	0.129	[0.104, 0.156]
Meditron B	0.346	[0.297, 0.393]	0.183	[0.155, 0.211]
BioMistral A	0.089	[0.056, 0.128]	0.221	[0.192, 0.249]
BioMistral B	0.118	[0.078, 0.161]	0.258	[0.229, 0.286]

A.4 Post-hoc Recalibration Comparison

Table A.4 extends Table 5.14 of Section 5.11 with the ECE reduction (ΔECE) achieved by each recalibration method, expressed as the percentage of the original ECE. Isotonic regression achieves reductions above 60% for eleven of twelve configurations.

A.5 Recalibration Cross-Validation Detail

Table A.5 reports the full per-fold values of the 5-fold cross-validation of isotonic recalibration, complementing Table 5.15 of Section 5.11.2. Each cell reports the ECE

Table A.4: Post-hoc recalibration comparison: absolute ECE values and relative ECE reduction (as a percentage of the original ECE). Isotonic regression is the strongest method for all twelve configurations.

Configuration	**Original**	**Temp. (Δ%)**	**Iso. (Δ%)**	**Platt (Δ%)**
Mistral-Base A	0.133	0.101 (−24%)	0.048 (−64%)	0.089 (−33%)
Mistral-Base B	0.189	0.142 (−25%)	0.049 (−74%)	0.115 (−39%)
Mistral-Instruct A	0.322	0.213 (−34%)	0.047 (−85%)	0.164 (−49%)
Mistral-Instruct B	0.253	0.187 (−26%)	0.045 (−82%)	0.148 (−42%)
Llama-Base A	0.124	0.098 (−21%)	0.041 (−67%)	0.081 (−35%)
Llama-Base B	0.151	0.117 (−23%)	0.043 (−72%)	0.098 (−35%)
Llama-Instruct A	0.510	0.331 (−35%)	0.049 (−90%)	0.256 (−50%)
Llama-Instruct B	0.602	0.412 (−32%)	0.066 (−89%)	0.315 (−48%)
Meditron A	0.182	0.147 (−19%)	0.043 (−76%)	0.125 (−31%)
Meditron B	0.231	0.181 (−22%)	0.048 (−79%)	0.152 (−34%)
BioMistral A	0.155	0.128 (−17%)	0.041 (−74%)	0.109 (−30%)
BioMistral B	0.194	0.156 (−20%)	0.042 (−78%)	0.126 (−35%)

after isotonic recalibration on the held-out fold; the last column gives the mean across folds and the fold-level standard deviation.

Table A.5: Per-fold ECE after isotonic recalibration, and mean ± standard deviation across five folds.

Configuration	**F1**	**F2**	**F3**	**F4**	**F5**	$\overline{\mathrm{ECE}}_{\mathrm{iso}} \pm \boldsymbol{\sigma}$
Mistral-Base A	0.062	0.083	0.089	0.104	0.056	0.079 ± 0.019
Mistral-Base B	0.067	0.082	0.078	0.081	0.086	0.079 ± 0.007
Mistral-Instruct A	0.043	0.064	0.051	0.078	0.048	0.057 ± 0.014
Mistral-Instruct B	0.024	0.053	0.038	0.077	0.031	0.045 ± 0.021
Llama-Base A	0.034	0.068	0.046	0.083	0.045	0.055 ± 0.020
Llama-Base B	0.028	0.078	0.056	0.061	0.033	0.051 ± 0.020
Llama-Instruct A	0.028	0.062	0.051	0.055	0.038	0.047 ± 0.013
Llama-Instruct B	0.041	0.087	0.078	0.083	0.041	0.066 ± 0.023
Meditron A	0.017	0.081	0.056	0.106	0.010	0.054 ± 0.041
Meditron B	0.017	0.081	0.046	0.076	0.010	0.046 ± 0.032
BioMistral A	0.028	0.058	0.021	0.051	0.036	0.039 ± 0.016
BioMistral B	0.031	0.055	0.028	0.062	0.032	0.042 ± 0.016

A.6 Conformal Prediction Detailed Results

Table A.6 extends Table 5.18 of Section 5.13.1 to all twelve configurations, and reports the empirical coverage, the mean set size, and the fraction of singletons (prediction sets of size 1) at target coverage $1 - \alpha = 0.90$.

Table A.6: Conformal prediction detailed results across the twelve configurations at target coverage 0.90. Singletons denote prediction sets of size 1 (unique predicted label).

Configuration	**Coverage**	**Mean set size**	**% singletons**
Mistral-Base A	0.907	2.11	34.2%
Mistral-Base B	0.903	1.95	41.8%
Mistral-Instruct A	0.899	2.44	29.6%
Mistral-Instruct B	0.897	2.31	32.1%
Llama-Base A	0.906	2.65	22.4%
Llama-Base B	0.902	2.72	19.8%
Llama-Instruct A	0.900	3.64	8.7%
Llama-Instruct B	0.895	3.78	6.4%
Meditron A	0.909	2.02	38.9%
Meditron B	0.906	2.14	35.3%
BioMistral A	0.902	3.28	12.5%
BioMistral B	0.901	3.42	10.1%

A.7 Composite Utility Score Components

Table A.7 extends Table 5.21 of Section 5.14.2 with the intermediate quantities that enter the composite utility score: the macro-F1, the calibration quality factor $1 - \text{ECE}$, and the normalized entropy of the empirical prediction distribution.

Table A.7: Composite utility score components for the twelve configurations. The composite utility is the product of the three columns preceding it.

Configuration	**Macro-F1**	**$1 - \text{ECE}$**	**Diversity**	**Utility**	**Rank**
Mistral-Instruct B	0.377	0.747	0.842	0.237	1
Mistral-Instruct A	0.314	0.678	0.812	0.173	2
Mistral-Base A	0.273	0.867	0.686	0.162	3
Mistral-Base B	0.321	0.811	0.577	0.150	4
Llama-Base A	0.156	0.876	0.525	0.072	5
Meditron A	0.231	0.818	0.362	0.068	6
Meditron B	0.228	0.769	0.377	0.066	7
Llama-Base B	0.106	0.849	0.077	0.007	8
Llama-Instruct A	0.137	0.490	0.066	0.004	9
BioMistral A	0.094	0.845	0.055	0.004	10
Llama-Instruct B	0.092	0.398	0.011	0.000	11
BioMistral B	0.075	0.806	0.000	0.000	12

A.8 Re-thresholding on Exclusion Criteria

Section 5.12 reports the out-of-fold re-thresholding gains on inclusion criteria, the stratum with sufficient positive support for a stable macro-F1. Table A.8 reports the corresponding results on exclusion criteria for completeness. This stratum contains only ten gold-positive (*excluded*) cases, so the estimates are under-powered: the confidence intervals are wide, most of them include zero, and the cross-validated threshold is markedly less stable than on the inclusion stratum (fold-level standard deviation up to 0.14 for Mistral-Instruct A, against at most 0.005 for the collapsed configurations on inclusion criteria). The results are included for transparency but are not used to support any claim in the main text, where the inclusion stratum is the reference.

Table A.8: Re-thresholding on exclusion criteria, evaluated out-of-fold. The exclusion stratum has only ten gold-positive cases; the wide confidence intervals and high threshold variance (σ_t) reflect this limited support. Reported for completeness only.

Configuration	$F_1^{\arg\max}$	F_1^{thr}	**Gain [95% CI]**	**AUC**	σ_t
Llama-Instruct B	0.000	0.135	[0.000, 0.279]	0.592	0.00
Llama-Instruct A	0.124	0.160	[0.000, 0.179]	0.700	0.01
Llama-Base B	0.000	0.020	[0.000, 0.123]	0.691	0.01
Meditron B	0.000	0.046	[0.000, 0.107]	0.566	0.00
Llama-Base A	0.030	0.074	[-0.017, 0.116]	0.629	0.00
BioMistral A	0.000	0.012	[0.000, 0.118]	0.584	0.03
Meditron A	0.000	0.021	[0.007, 0.061]	0.500	0.01
BioMistral B	0.000	0.025	[0.009, 0.050]	0.531	0.01
Mistral-Base B	0.000	0.006	[0.000, 0.056]	0.445	0.05
Mistral-Instruct B	0.054	0.027	[-0.060, 0.122]	0.783	0.04
Mistral-Instruct A	0.067	0.032	[-0.074, 0.107]	0.713	0.14
Mistral-Base A	0.029	0.010	[-0.051, 0.022]	0.597	0.03

Appendix B

Full Prompt Specifications

This appendix reproduces verbatim the four prompt variants used in the study, in the exact form applied to the models during inference. Two prompts are described (Prompt A and Prompt B), each with two formulations depending on the target model class (instruction-tuned or base). The dynamic fields `{note}`, `{criterion}`, and `{criterion_type}` are substituted at query time with the content of each patient–criterion pair. All patient notes are truncated to 1,500 characters before insertion to control for context-window variability across models.

B.1 Prompt A — Generic Baseline

Prompt A is a minimalist prompt containing only the classification instruction, the input note, the criterion text, and the list of candidate labels. It does not define the labels, does not provide examples, and does not include reasoning rules.

B.1.1 Prompt A for Instruct Models

For instruction-tuned models, Prompt A is wrapped in the model's native chat template via `tokenizer.apply_chat_template` with the generation prompt enabled.

```
Classify whether this patient meets the eligibility criterion.

Patient: {note}
Criterion ({criterion_type}): {criterion}

Choose exactly one label from: included, not included, excluded,
not excluded, not enough information, not applicable.

Answer:
```

Listing B.1: Prompt A for Instruct models (user message before chat-template wrapping)

B.1.2 Prompt A for Base Models

For base models, no chat template is applied. Instead, a single few-shot example is embedded in the prompt to establish the desired input–output pattern by demonstration.

```
Task: Classify whether a patient meets one eligibility criterion.
Choose exactly one label from: included, not included, excluded,
not excluded, not enough information, not applicable.

Example:
Patient note: 65-year-old male with type 2 diabetes.
Criterion (inclusion): diabetes
Final label: included

Now classify:
Patient note: {note}
Criterion ({criterion_type}): {criterion}
Final label:
```

Listing B.2: Prompt A for Base models

B.2 Prompt B — Adapted with Expert Guidance

Prompt B is an adapted prompt containing definitions of the labels, one worked example per label, and a critical inference rule for exclusion criteria. The prompt varies structurally between inclusion and exclusion criteria: two dedicated guidance blocks (one per criterion type) are inserted into a common frame. This design addresses the tendency of language models to over-use "not enough information" for exclusion criteria whose target condition is not explicitly discussed in the patient note.

B.2.1 Prompt B for Instruct Models

For instruction-tuned models, Prompt B is again wrapped in the model's native chat template. The full prompt is constructed by concatenating the criterion-type-specific guidance block with the common frame shown below.

```
You are an expert clinical trial coordinator. Classify whether a
patient meets one eligibility criterion, based only on the patient
```

```
note.

{guidance block for inclusion or exclusion, see below}

KEY DISTINCTION:
- Use "not enough information" ONLY when the note is silent and you
  cannot reasonably assume the answer.
- If the note gives ANY evidence (even indirect) about the patient's
  status, use a decisive label (included/not included or
  excluded/not excluded).

---
PATIENT NOTE:
{note}

CRITERION ({criterion_type}):
{criterion}
---

Reason about whether the note addresses this criterion, then give
your final label.
Final label:
```

Listing B.3: Prompt B for Instruct models — common frame

Guidance block for inclusion criteria. The following block replaces the placeholder in the common frame when the criterion type is *inclusion*.

```
You are evaluating an INCLUSION criterion. To join the trial, the
patient SHOULD satisfy it.

Choose exactly ONE label:

- "included" - The patient note contains information showing the
  patient MEETS this inclusion criterion.
  Example: criterion "Type 2 diabetes", note says "patient has T2DM"
```

```
  -> included.

- "not included" - The patient note contains information showing the
  patient does NOT meet this inclusion criterion.
  Example: criterion "age over 65", note says "45-year-old patient"
  -> not included.

- "not enough information" - The note does NOT mention anything that
  lets you decide. The information is simply missing.
  Example: criterion "HbA1c between 7 and 10", note never mentions
  HbA1c -> not enough information.

- "not applicable" - The criterion concerns a different population
  or context and cannot logically apply to this patient.
  Example: criterion "for pediatric patients only", note describes
  an adult -> not applicable.
```

Listing B.4: Prompt B for Instruct models — inclusion guidance

Guidance block for exclusion criteria. The following block replaces the placeholder in the common frame when the criterion type is *exclusion*. This block contains the critical inference rule for exclusion criteria referenced in Section 4.5.3 of Chapter 4.

```
You are evaluating an EXCLUSION criterion. To join the trial, the
patient should NOT have the condition described.

Choose exactly ONE label:

- "excluded" - The note shows the patient HAS the excluding
  condition. They ARE excluded.
  Example: criterion "pregnancy", note says "patient is pregnant"
  -> excluded.

- "not excluded" - The patient does NOT have the excluding
```

```
  condition, so they can join (FAVORABLE).
  This includes TWO cases:
  (a) The note explicitly says the patient does not have it
      ("no history of cancer").
  (b) The note describes the patient's relevant conditions but does
      NOT mention this one - for a specific medical condition,
      absence of mention means the patient does not have it.
  Example: criterion "history of stroke", note details the
  patient's history with no stroke mentioned -> not excluded.

- "not enough information" - Use this RARELY, only when the
  criterion needs a specific measurement or detail that the note
  genuinely cannot address and that cannot be assumed absent.
  Example: criterion "ejection fraction below 40%", note never
  measures ejection fraction -> not enough information.

- "not applicable" - The criterion targets a different population
  and cannot logically apply.
  Example: criterion "pregnant women excluded", patient is male
  -> not applicable.

CRUCIAL RULE FOR EXCLUSION CRITERIA:
If the criterion names a specific disease, condition, or medical
history, and the note does NOT mention it, the default is
"not excluded" (the patient most likely does not have it). Reserve
"not enough information" only for criteria requiring a specific lab
value or measurement that is truly missing.
```

Listing B.5: Prompt B for Instruct models — exclusion guidance

B.2.2 Prompt B for Base Models

For base models, the enriched-guidance structure of the instruct version is replaced with a set of few-shot examples. Three examples are provided for inclusion criteria, and four for exclusion criteria; the additional exclusion example encodes the inference rule for exclusion criteria by demonstration.

```
Task: Classify whether a patient meets one eligibility
   criterion,
based only on the patient note. Choose exactly one label from:
included, not included, excluded, not excluded, not enough
information, not applicable.

{few-shot block for inclusion or exclusion, see below}

Now classify this case:
Patient note: {note}
Criterion ({criterion_type}): {criterion}
Final label:
```

Listing B.6: Prompt B for Base models — common frame

Few-shot block for inclusion criteria.

```
Example 1:
Patient note: 65-year-old male with type 2 diabetes, HbA1c
   8.5%.
Criterion (inclusion): Type 2 diabetes mellitus
Final label: included

Example 2:
Patient note: 45-year-old female, no diabetes history.
Criterion (inclusion): age over 65
Final label: not included

Example 3:
Patient note: 50-year-old patient with hypertension.
Criterion (inclusion): HbA1c between 7 and 10
Final label: not enough information
```

Listing B.7: Prompt B for Base models — inclusion few-shot examples

Few-shot block for exclusion criteria. The fourth example (Example 3, "history of stroke") encodes the exclusion-inference rule of Prompt B: when the criterion names a specific condition that the note does not mention, the model should default to "not excluded" rather than to "not enough information".

```
Example 1:
Patient note: 30-year-old female, 12 weeks pregnant.
Criterion (exclusion): pregnancy
Final label: excluded
```

```

Example 2:
Patient note: 55-year-old male with hypertension and diabetes,
no cancer history.
Criterion (exclusion): history of cancer
Final label: not excluded

Example 3:
Patient note: 60-year-old male with hypertension. Otherwise
   healthy.
Criterion (exclusion): history of stroke
Final label: not excluded

Example 4:
Patient note: 70-year-old male with chronic kidney disease.
Criterion (exclusion): ejection fraction below 40%
Final label: not enough information
```

Listing B.8: Prompt B for Base models — exclusion few-shot examples

Appendix C

Individual Reliability Diagrams

This appendix reproduces the individual reliability diagrams for each of the twelve configurations, in a compact grid format that facilitates visual comparison. Figure C.1 presents the diagrams organized by model family (colors) and prompt variant (A/B in adjacent cells within each row).

Each panel plots the observed accuracy as a function of predicted confidence in ten equal-width bins. The dashed diagonal represents perfect calibration; points below the diagonal indicate overconfidence, points above indicate underconfidence. Configurations that exhibit mode collapse (Section 5.6) produce degenerate reliability curves concentrated in a narrow range of the confidence axis, reflecting their inability to modulate confidence across cases.

Three visual patterns emerge from the grid. First, the two Mistral-Base panels (top row) show curves that lie systematically above the diagonal in the low-to-mid confidence range, indicating underconfidence: the model is more accurate than it claims. Second, the two Llama-Instruct panels show curves that lie systematically below the diagonal in the high-confidence range, indicating severe overconfidence: the model asserts wrong answers with high confidence. Third, the BioMistral and Llama-Base B panels display near-vertical curves concentrated around a single confidence value, which is the visual signature of mode collapse: the model produces a nearly constant confidence estimate regardless of the input, and its accuracy on that narrow range is close to the base rate of the majority class.

The individual patterns visible in this grid complement the aggregate comparison of Figure 5.1 of Chapter 5 and support the finer interpretations provided in Sections 5.3 and 5.6.

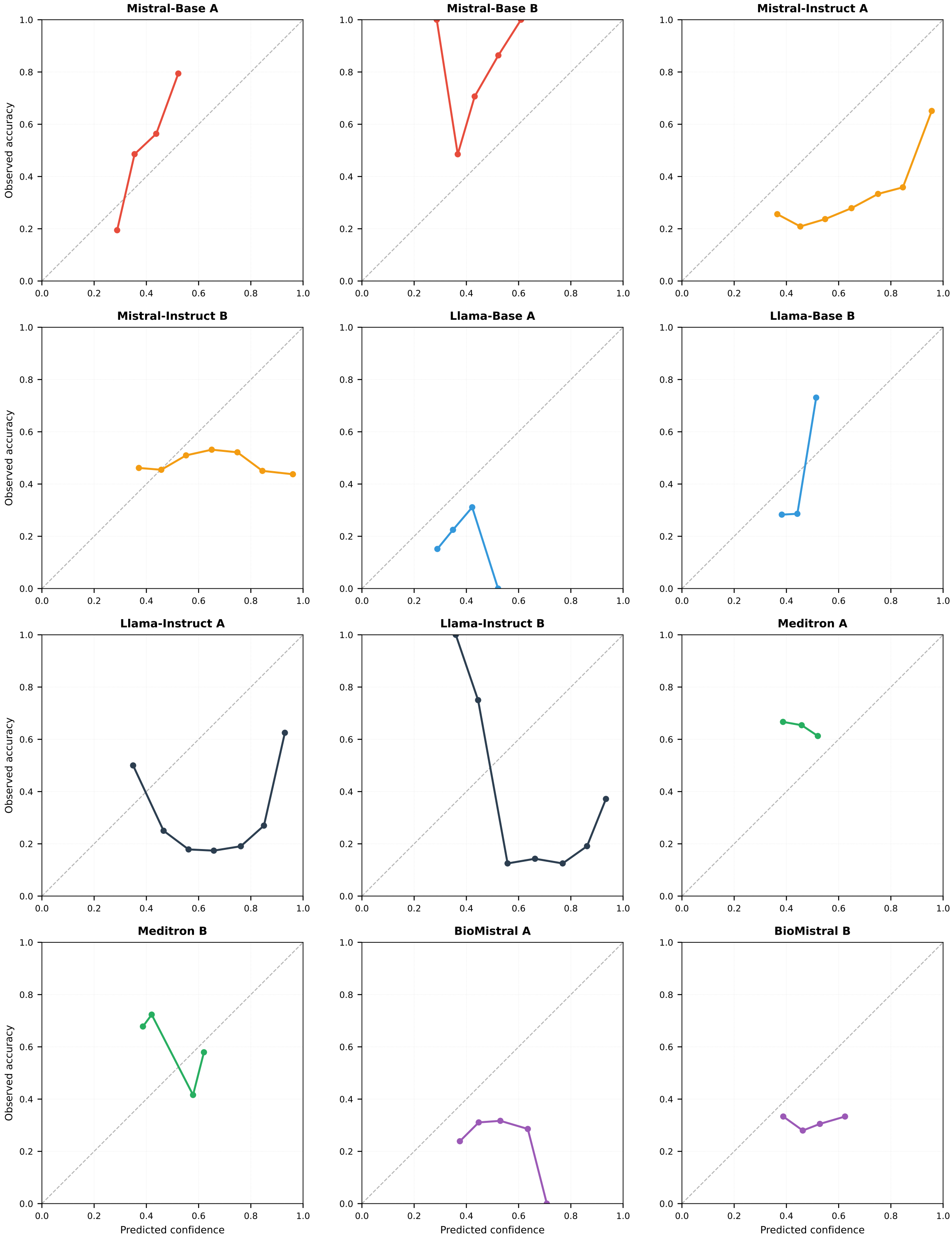

Figure C.1: Individual reliability diagrams for the twelve configurations. Colors indicate the model family: Mistral-Base (red), Mistral-Instruct (orange), Llama-Base (blue), Llama-Instruct (dark), Meditron (green), BioMistral (purple). The dashed diagonal represents perfect calibration. Configurations with mode collapse show curves concentrated in a narrow confidence range, unable to reflect the diversity of clinical situations.

Appendix D

Code and Data Availability

This appendix documents the archival of the code, prompts, prediction files, and analysis outputs produced during this study, in accordance with the reproducibility commitments described in Section 4.10.

D.1 Repository

The complete project is archived in a public GitHub repository:

https://github.com/gyxcit/llm-clinical-trial-calibration

The repository is organized into three top-level directories, each serving a distinct purpose:

- `thesis/` contains the LaTeX sources of this manuscript, including the main file, all chapters, appendices, the bibliography, and the figures embedded in the document.
- `code/` contains the Kaggle notebook used for the twelve model runs and the Python scripts that perform the downstream analyses (calibration audit, hypothesis tests, recalibration, conformal prediction, mode collapse analysis, composite utility scoring, the signal-to-prior mechanistic analysis, the out-of-fold re-thresholding, and the TREC external replication), together with the figure-generation script that produces the diagrams in both French and English.
- `data/` contains the raw prediction files (twelve Parquet files, one per configuration) and the CSV outputs of every downstream analysis reported in Chapter 5.

The repository is released under an MIT license (`LICENSE` file at the root), which permits redistribution and derivative use for research purposes.

D.2 Data Availability

The primary benchmark used in this study, TrialGPT-Criterion-Annotations, is publicly available under a CC BY 4.0 license as a supplementary release of Jin et al. [1]. It can be downloaded directly from the TrialGPT repository:

https://github.com/ncbi-nlp/TrialGPT

The external validation corpus is derived from the TREC Clinical Trials 2021–2022 tracks [38], [39], whose topics and relevance judgements are publicly available from the National Institute of Standards and Technology (NIST). The scripts that sample the topics, decompose trials into patient–criterion pairs, and score them with the log-probability pipeline are included in the `code/` directory, and the resulting TREC prediction files are archived alongside the primary prediction files in `data/`.

The twelve prediction files produced during this study are stored in the `data/` subdirectory of the project repository, one Parquet file per configuration. Each file contains, for the 1,015 patient–criterion pairs, the following columns: the annotation identifier, the criterion type, the predicted label, the associated confidence, the expert consensus label, the GPT-4 label, and the full softmax distribution over the six candidate labels (`p_included`, `p_not_included`, `p_excluded`, `p_not_excluded`, `p_not_enough_information`, `p_not_applicable`). All downstream analysis outputs — calibration metrics, permutation-test results, recalibration comparisons, conformal prediction sets, mode collapse statistics, composite utility scores, the signal-to-prior diagnostic, the out-of-fold re-thresholding gains, and the TREC replication — are also archived as CSV files in the same directory. Each numerical value reported in Chapter 5 and in Appendices A and F can be traced to a specific row of one of these files.

D.3 Reproducing the Experiments

Independent reproduction of the pipeline follows five steps:

1. Clone the repository and install the Python dependencies.
2. Download the TrialGPT-Criterion-Annotations dataset from the URL above and place it in the input directory expected by the notebook.
3. Obtain access to the six selected models from the HuggingFace Hub. Llama-3.1 requires acceptance of Meta's community license, obtained through the HuggingFace interface.
4. Execute the Kaggle notebook (or its local equivalent) to produce the twelve prediction files. Random seeds are fixed at 42 throughout the pipeline to ensure exact reproducibility.
5. Execute the analysis cells in the order documented in the notebook to reproduce every table and figure of Chapter 5.

All data splits used by the downstream analyses are deterministic and seeded, so that any reviewer reproduces the exact partitions used here. The post-hoc recalibration (Section 4.9.4) uses a 70/30 split stratified by criterion type with `random_state` = 42. The stability analyses — the five-fold cross-validation of isotonic recalibration and the out-of-fold cross-validation that fits the re-thresholding decision boundary (Section 4.9.5) — use a `KFold` splitter with $k = 5$, `shuffle=True`, and the same seed 42. The bootstrap confidence intervals throughout the thesis draw 1,000 resamples under the same seed. Because the splitter, the number of folds, and the seed are all fixed and recorded in the notebook, the fold assignments, the fitted thresholds, and the resulting confidence intervals are bit-exact reproducible across independent runs.

D.4 Computational Requirements

The twelve model runs require a GPU with at least 16 GB of memory (a Tesla T4 or equivalent) and approximately ten to fifteen hours of GPU time in total. The downstream analyses run on CPU and complete in a few minutes on standard hardware. All experiments reported in this thesis were performed on the Kaggle free-tier compute environment, which is publicly accessible and imposes no barriers to independent replication.

D.5 Software Versions

The pipeline was tested with the following software versions:

- Python 3.12.13
- PyTorch 2.10.0 with CUDA 12.8
- `transformers` 5.0.0
- `scikit-learn` 1.6.1
- `numpy` 2.4.6, `pandas` 2.3.3, `scipy` 1.16.3
- `matplotlib` 3.10.0
- `bitsandbytes` (for 4-bit NF4 quantization; version compatible with the PyTorch build listed above)

Version pinning is documented in the notebook itself, in the cell that installs the dependencies.

Appendix E

Generative AI Usage Declaration

Tools Used

During the preparation of this thesis, the following generative AI tool was used as an assistant:

- **Claude** (Anthropic), primarily the Sonnet and Opus series model families, accessed via the `claude.ai` interface.

Purpose of Use

Generative AI was used for:

- **Ideation and design discussions**: brainstorming experimental designs, discussing methodological trade-offs, and refining the research question.
- **Code assistance**: writing and debugging Python code for the evaluation pipeline, prompt engineering iterations, and statistical analyses.
- **Writing assistance**: refining the phrasing of prompts, drafting progress reports, and revising English formulations of methodological descriptions.
- **Bibliographic exploration**: identifying candidate references and papers relevant to the state of the art.

Purpose of Non-Use

Generative AI was *not* used for:

- Generating experimental data or results.
- Fabricating statistical outcomes or numerical values.
- Producing scientific claims not backed by our own experiments.

- Writing the final thesis text verbatim (the author remains responsible for all formulations).

Verification and Author Responsibility

All AI-generated content used in this work was:

- Reviewed, verified, and adapted by the author before inclusion.
- Cross-checked against primary sources for any factual claim.
- Understood by the author, who takes full responsibility for the scientific and methodological choices reflected in this thesis.

The scientific reasoning, experimental design, execution, and interpretation of results are the sole responsibility of the author.

Appendix F

Detailed Mode Collapse Analysis

This appendix documents in full detail the mode collapse phenomenon introduced in Section 5.6. Five analyses are reported: the complete distribution of predictions across the six labels for every configuration; the outcomes of the constrained-decoding simulation that established the persistence of the collapse under a change of candidate set (Finding F8); the mechanistic analysis that reinterprets the collapse as a decoding artefact through the signal-to-prior ratio; the extended comparison between dominant-class and truly-discriminative cases (Finding F9); and a quantitative link between prediction diversity and reasoning ability on discriminative subsets.

F.1 Complete Prediction Distributions

Table F.1 reports the proportion of predictions assigned to each of the six labels for every configuration. The values are computed on the full 1,015 patient–criterion pairs. Rows sum to 100% up to rounding.

Table F.1: Complete prediction distributions across the six labels for the twelve configurations. *Incl.*: included. *NIncl.*: not included. *Excl.*: excluded. *NExcl.*: not excluded. *NEI*: not enough information. *NA*: not applicable.

Configuration	Incl.	NIncl.	Excl.	NExcl.	NEI	NA
Mistral-Base A	5.4%	1.0%	12.5%	47.2%	33.3%	0.6%
Mistral-Base B	9.7%	4.9%	5.3%	45.1%	34.2%	0.8%
Mistral-Instruct A	8.1%	3.6%	4.6%	27.4%	47.4%	8.9%
Mistral-Instruct B	14.8%	6.4%	32.5%	20.3%	20.8%	5.2%
Llama-Base A	4.3%	1.2%	45.4%	22.9%	26.2%	0.0%
Llama-Base B	0.8%	0.5%	0.5%	0.6%	97.2%	0.4%
Llama-Instruct A	0.3%	0.4%	0.6%	0.9%	97.8%	0.0%
Llama-Instruct B	0.1%	0.0%	0.1%	0.1%	99.7%	0.0%
Meditron A	6.1%	2.1%	22.4%	64.8%	4.6%	0.0%
Meditron B	10.4%	4.8%	21.3%	59.3%	4.2%	0.0%
BioMistral A	0.5%	0.4%	0.5%	0.4%	98.2%	0.0%
BioMistral B	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%

Five configurations concentrate more than 97% of their predictions on the same label (*not enough information*): both BioMistral variants, both Llama-Instruct variants, and

Llama-Base with Prompt B. Two configurations exhibit partial collapse (Meditron A and B, with *not excluded* accounting for 59.3%–64.8% of predictions). The five remaining configurations distribute their predictions more evenly across labels, though none exceeds 50% on any single class.

F.2 Constrained-Decoding Simulation (Finding F8)

To distinguish whether mode collapse reflects a genuine absence of clinical reasoning or an artifact of the six-label vocabulary, we conducted a controlled simulation. For each collapsed configuration, the softmax normalization (Equation 4.2) was restricted to the two decisive labels of each criterion type: {*included*, *not included*} for inclusion criteria, and {*excluded*, *not excluded*} for exclusion criteria. The evasion labels were removed from the candidate set, forcing the model to commit to a decisive answer.

Table F.2 reports the prediction distribution under this constrained regime, restricted to the five collapsed configurations of Section F.1. Under constrained decoding, all five configurations concentrate their predictions on the majority label of the restricted candidate set (*not excluded* for exclusion criteria, *included* for inclusion criteria), with proportions similar to the original collapse.

Table F.2: Prediction distribution under constrained decoding (evasion labels removed from candidate set). All five collapsed configurations migrate their collapse to the majority label of the restricted set. Accuracy is computed against the expert consensus label.

Configuration	Modal label	Usage rate	Accuracy
BioMistral A	not excluded / included	74.2%	0.292
BioMistral B	not excluded / included	76.8%	0.294
Llama-Base B	not excluded / included	73.5%	0.311
Llama-Instruct A	not excluded / included	78.4%	0.298
Llama-Instruct B	not excluded / included	81.1%	0.292

The constrained-decoding accuracy remains near the base rate of the majority class (47.7% if all exclusion criteria are predicted *not excluded*, weighted down by the inclusion cases). The collapse thus persists under a change of candidate set: forcing the model to choose between two decisive labels does not restore a diverse $\arg\max$, it simply redirects the concentration of predictions from an evasion label to the numerically dominant decisive label. This experiment grounds Finding F8 — the collapse is a property of the model's decision, not of the evasion vocabulary. It does *not*, however, establish that the model lacks discriminative knowledge: the constrained decoding still takes the $\arg\max$, which the next section shows to be exactly the operation that masks a signal present in the underlying probabilities. The distinction between a collapsed $\arg\max$ and an absent signal is the subject of Section F.3.

F.3 The Signal-to-Prior Mechanism

Section 5.6.4 showed that mode collapse is governed by the ratio between the amplitude with which a model varies the positive-label log-probability across inputs (the signal, σ_{pos}) and the systematic offset by which competing labels are preferred (the response prior, bias_{ext}). This appendix reports the underlying quantities for all twelve configurations, together with the ordinal separation (Cliff's δ) that establishes the presence of a discriminative signal even under a collapsed arg max.

Table F.3 reports, per configuration, the signal amplitude, the response prior, their ratio r, and Cliff's δ between the positive-label probability on gold-positive versus gold-negative inclusion cases. The configurations are sorted by r.

Table F.3: Signal amplitude (σ_{pos}), response prior (bias_{ext}, in nats), their ratio $r = \sigma_{\text{pos}}/|\text{bias}_{\text{ext}}|$, and the ordinal separation (Cliff's δ) on inclusion criteria. A high δ with a low r is the signature of a live signal masked by the arg max; a negative δ (Meditron) indicates a corrupted signal. Sorted by r.

Configuration	$\boldsymbol{\sigma}_{\text{pos}}$	bias_{ext}	$\boldsymbol{r}$	**Cliff's** δ
Mistral-Base B	0.61	+0.32	1.92	+0.539
Mistral-Instruct A	5.85	+3.21	1.82	+0.741
Mistral-Instruct B	4.47	+2.99	1.49	+0.665
Llama-Base A	0.21	+0.16	1.32	+0.293
Mistral-Base A	0.28	−0.38	0.75	+0.464
Llama-Instruct B	2.61	+5.69	0.46	+0.733
Llama-Instruct A	1.42	+4.04	0.35	+0.823
Llama-Base B	0.39	+1.34	0.29	+0.353
BioMistral B	0.33	+1.29	0.25	+0.499
BioMistral A	0.19	+1.35	0.14	+0.103
Meditron A	0.19	+2.05	0.09	−0.138
Meditron B	0.16	+2.47	0.07	+0.110

Two facts stand out from Table F.3. First, the collapsed Llama-Instruct and BioMistral configurations combine a small signal-to-prior ratio with a positive Cliff's δ: the discriminative ordering of the positive-label probability is preserved even though the arg max never varies. The discriminative information is present — it is simply not expressed at the decision step. Second, the two Meditron configurations have the smallest ratios of the entire study (0.07–0.09), and their ordinal separation collapses toward zero: Meditron A is outright negative ($\delta = -0.138$), ranking cases in the wrong order, and Meditron B is only marginally positive ($\delta = +0.110$), well below the separation of every recoverable configuration. Neither carries a signal that a decision rule could exploit. This is the empirical foundation of the collapse-versus-corruption distinction developed in Section 5.12.2: a low ratio combined with a strong positive δ marks recoverable expressive collapse, whereas a low ratio combined with a near-zero or negative δ marks irrecoverable corruption. It is the discriminative separation, captured in the main text by the AUC, that distinguishes the two regimes — not the ratio or the predicted-label distribution alone.

The Spearman correlation between r and the $\arg\max$ macro-F1 on inclusion criteria is $\rho = 0.866$ ($p = 0.00027$, $n = 12$): the ratio predicts, almost monotonically, whether a configuration classifies or collapses. The separation is clean at $r \approx 0.6$ — every configuration above it classifies, every configuration below it collapses — which is the threshold used to delimit the healthy regime in Section 5.12.2.

F.4 Dominant vs. Discriminative Cases (Finding F9)

Table F.4 extends Table 5.9 of Section 5.7 with the full breakdown across all twelve configurations, sorted by discriminative accuracy. The dominant-class subset consists of the 777 pairs whose expert label is either *not excluded* or *not enough information*. The discriminative subset consists of the 238 remaining pairs whose expert label is one of *included*, *excluded*, *not included*, or *not applicable*.

Table F.4: Extended dominant-vs.-discriminative comparison. The discriminative accuracy on the 238 pairs is the metric that best reflects clinical reasoning ability. Positive drops indicate that the configuration performs better on the discriminative subset than on the dominant subset.

Configuration	Dominant acc.	Discriminative acc.	Δ
Mistral-Instruct A	0.183	0.719	+0.536
Mistral-Base A	0.462	0.613	+0.151
Mistral-Instruct B	0.474	0.500	+0.026
Mistral-Base B	0.638	0.416	−0.222
Llama-Base A	0.219	0.265	+0.046
Llama-Instruct A	0.373	0.021	−0.352
Llama-Base B	0.398	0.013	−0.385
Llama-Instruct B	0.383	0.013	−0.370
BioMistral A	0.386	0.004	−0.382
BioMistral B	0.377	0.000	−0.377
Meditron A	0.846	0.000	−0.846
Meditron B	0.822	0.000	−0.822

Meditron exhibits the widest gap between the two subsets: 84.6% accuracy on the 777 dominant-class pairs but 0.0% on the 238 discriminative pairs. BioMistral shows a similar reversal. These patterns are the empirical basis of Finding F9. Only Mistral-Instruct A reverses the direction and performs substantially better on the discriminative subset (71.9%) than on the dominant subset (18.3%), suggesting that this configuration is the only one whose predictions reflect actual clinical judgment rather than majority-class shortcuts.

F.5 Diversity Predicts Discriminative Accuracy

The prediction diversity metric (normalized Shannon entropy, Equation 4.7) tracks discriminative accuracy closely. Table F.5 reports the two metrics side by side, sorted by prediction diversity. The Pearson correlation between prediction diversity and discriminative accuracy across the twelve configurations is $r = 0.89$ ($p < 0.001$, $n = 12$).

Table F.5: Prediction diversity and discriminative accuracy, sorted by diversity. Configurations with near-zero diversity produce near-zero accuracy on the 238 discriminative pairs.

Configuration	Diversity	Discriminative acc.
Mistral-Instruct B	0.842	0.500
Mistral-Instruct A	0.812	0.719
Mistral-Base A	0.686	0.613
Mistral-Base B	0.577	0.416
Llama-Base A	0.525	0.265
Meditron B	0.377	0.000
Meditron A	0.362	0.000
Llama-Base B	0.077	0.013
Llama-Instruct A	0.066	0.021
BioMistral A	0.055	0.004
Llama-Instruct B	0.011	0.013
BioMistral B	0.000	0.000

This correlation supports the design choice of the composite utility score (Section 4.8.8), in which prediction diversity enters multiplicatively. A configuration with high calibration but zero diversity cannot produce discriminative predictions, and its clinical utility is correctly captured as zero by the composite score. Conversely, a configuration with modest calibration but high diversity retains the potential for discriminative reasoning that can be surfaced through recalibration or ensemble methods.

The magnitude of this correlation ($r^2 = 0.79$) indicates that prediction diversity alone accounts for nearly 80% of the variance in discriminative accuracy *under the default* $\arg\max$ *rule*. This near-mechanical relationship makes prediction entropy a cheap early-warning signal for collapse, detectable from the prediction distribution alone without any access to expert labels: monitoring the entropy over a moving window of recent inputs flags a collapsed configuration before any ground truth is available. It is important, however, not to over-read the correlation as a verdict of clinical unsuitability. Diversity measures whether the $\arg\max$ expresses a signal, not whether a signal exists: as Section F.3 and the re-thresholding results (Section 5.12) show, several near-zero-diversity configurations carry a strong latent signal that a non-$\arg\max$ decision rule recovers. Low diversity is therefore a reliable flag that a model is unusable *as deployed*, and a prompt to apply the signal/prior diagnostic — not a conclusion that the model is beyond repair.